About additional resources

This topic describes how to get even more information about macros or the software.

For more information about macros

This software provides additional resources that contain helpful information about macros. These additional resources, located in the **Data** folder for the installed software, are described in the following table.

Resource	Description and filename
Macro Programming Guide	Provides a streamlined instructional approach to programming macros for the software **Macro Programming Guide.pdf**
Object-model diagram for CorelDRAW	Provides a hierarchical representation of the CorelDRAW object model **CorelDRAW Object Model Diagram.pdf**

- For a more basic introduction to macros, please see the topic “Working with macros” in the main Help file for the application. You can access the main Help from within an application by clicking **Help** } **Help topics**.

For more information about the software

A variety of additional resources for the software are also available to you.

For comprehensive information about the features in the software, you can consult its installed documentation:

- The program group for the software (on the Windows **Start** menu) includes a **Documentation** folder, which provides easy access to various installed resources.
- Each program offers in-product Help, displayed by clicking **Help** } **Help topics**.

For even more information about the software, see the following Web-based resources.

Resource	Description and URL
CorelDRAW website	Provides the latest news, tips and tricks, and information about upgrades www.corel.com/coreldraw
Corel Support Services website	Provides prompt and accurate information about product features, specifications, pricing, availability, services, and technical support www.corel.com/support
Corel Knowledge Base	Provides a repository of articles written by the Corel Technical Support Services team in response to questions by users www.corel.com/knowledgebase
CorelDRAW (and Corel DESIGNER) online community	Provides interaction with other users through sharing experiences, asking questions, and receiving help and suggestions www.coreldraw.com

You can submit any comments or suggestions about the software by using the contact information provided at www.corel.com/contact.

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About this resource

This resource assumes that the reader has experience with at least one procedural programming language, such as BASIC, Microsoft Visual Basic (VB), C, C++, Java, Pascal, Cobol, or Fortran.

This resource does not describe the basics of procedural programming (such as functions, conditional branching, and looping). Therefore, before using this documentation, non-programmers are strongly advised to learn basic programming in a language such as Microsoft Visual Basic for Applications (VBA).

- Most of the code examples provided in this documentation are written in VBA.

- For more detailed instruction on the VB programming environment and on VBA, see Microsoft Visual Basic Help, which is available from the **Help** menu in the Macro Editor.
- For a more basic introduction to macros, please see the topic “Working with macros” in the main Help file for the application. You can access the main Help from within an application by clicking **Help** } **Help topics**.

Documentation conventions

The following table explains the documentation conventions used in this resource.

Wherever you see this	You’ll find
	A note — describes required conditions for performing a procedure or presents other essential information
	A tip — describes helpful information such as shortcuts, alternate methods, or benefits that are related to a procedure
bold text	The name of a control or other element on the user interface
<text in italics and between angle brackets>	A placeholder for user-specified information, such as a path or filename
`monospace text`	A reference to coding

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Activating documents

Each open document is a member of the **Application.Documents** collection. The documents in that collection appear in the order in which they were created or opened.

- To reflect the actual stacking order of the documents, you must use the **Application.Windows** collection.

The **Application.ActiveDocument** property provides direct access to the active document — that is, the document that is in front of all the other documents in the application window. **ActiveDocument** is an object of type **Document** and, therefore, has the same members — properties, methods, and objects — as the **Document** class.

If no documents are open, **ActiveDocument** returns nothing. You can check for open documents by using the following VBA code:

```
If Documents.Count = 0 Then
MsgBox "There aren't any open documents.", vbOK, "No Docs"
Exit Sub
End If
```

The **Document.Activate** method activates a document so that it can be referenced by **ActiveDocument**. The following VBA code activates the third open document (if three or more documents are open):

```
Documents(3).Activate
```

- Using the **Document.Activate** method on the **Application.ActiveDocument** property has no effect.

If you want, you can specify which open document to activate by referencing the one of the following properties:

- **Document.FilePath** — checks only the file path (for example, **C:\My Documents**)
- **Document.FileName** — checks only the filename (for example, **Graphic1.cdr**)
- **Document.FullFileName** — checks both the file path and the filename (for example, **C:\My Documents\Graphic1.cdr**)

You can check the filename of each open document by using the following VBA code:

```
Public Function findDocument(filename As String) As Document
Dim doc As Document
For Each doc In Documents
If doc.FileName = filename Then Exit For
Set doc = Nothing
Next doc
Set findDocument = doc
End Function
```

You can then activate the returned document by using the **Document.Activate** method.

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Activating layers

Each layer is a member of the **Page.Layers** collection for the page on which it appears. The layers in a **Page.Layers** collection appear in the order in which they appear on that page — the first layer is the one at the top of the “stack,” and the last layer is the one at the bottom. If layers are added, reordered, or deleted, the affected **Page.Layers** collection is immediately updated to reflect the new layer order of that page.

The **Document.ActiveLayer** property provides direct access to the active layer for a document, while the **Page.ActiveLayer** property provides direct access to the active layer for a page.

You can activate a layer by using the **Layer.Activate** method:

```
ActivePage.Layers("Layer 1").Activate
```

- Activating a locked layer does not unlock it. Similarly, activating a hidden layer does not make it visible. For information on unlocking and displaying layers, see Locking and hiding layers.

If you want, you can use event handlers to respond to events that are triggered by activating a layer:

- **Document.LayerActivate**

You can also use event handlers to respond to events that are triggered by deactivating a layer:

- **Document.LayerChange**

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Activating pages

Each page is a member of the **Document.Pages** collection for the document in which it appears. The pages in a **Document.Pages** collection appear in the order in which they appear in that document — for example, the fifth page in the active document is `ActiveDocument.Pages.Item(5)`. If pages are added, reordered, or deleted, the affected **Pages** collection is immediately updated to reflect the new page order of that document.

You can access the active page of the active document by using the **Application.ActivePage** property (or `ActiveDocument.ActivePage`, or simply `ActivePage`). A reference to the active page in the active document, of type **Page**, is returned.

```
Dim pg As Page
Set pg = ActivePage
```

You can access the active page of a document, regardless of whether that document is active, by using **ActivePage** property for that document:

```
Public Function getDocsActivePage(doc As Document) As Page
Set getDocsActivePage = doc.ActivePage
End Function
```

You can switch pages by finding the desired page and then invoking its **Activate** method. The following VBA code activates page 3 in a document:

```
ActiveDocument.Pages(3).Activate
```

If you want, you can use event handlers to respond to events that are triggered by activating a page:

- **Document.PageActivate**

You can also use event handlers to respond to events that are triggered by deactivating a page:

- **Document.PageChange**

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Applying effects to shapes

The object model provides a number of methods for applying effects to shapes. For information on these methods, see the following subtopics:

- Applying blends
- Applying contours
- Applying customized effects
- Applying distortions
- Applying drop shadows
- Applying envelopes
- Applying extrusions
- Applying lenses
- Applying perspective

- Applying an effect returns an **Effect** object, which lets you access various properties and methods for the created effect. For example, you can use the **Effect.Separate** method to separate the shapes that are generated by an effect from the shape to which that effect is applied. In addition, you can use the **Effect.Clear** method to remove an effect from a shape.

Applying blends

The **Shape.CreateBlend** method creates a blend between the current shape and the shape that is specified as a parameter. This method provides optional parameters for various blend settings, such as the acceleration of the blend and the path along which the blend is created.

The following VBA code creates a basic ten-step blend:

```
Dim sh As Shapes, eff As Effect
Set sh = ActiveSelection.Shapes
Set eff = sh(1).CreateBlend(sh(2), 10)
```

- In the preceding example, the number of shapes in the blend is twelve: the start and end shapes, plus the ten blend steps that are created.

- The **Shape.CreateBlend** method returns an **Effect** object, the **Effect.Blend** property for which you can use to modify the created blend.

Applying contours

The **Shape.CreateContour** method applies a contour to a shape. This method provides optional parameters for various contour settings, such as the colors and acceleration of the contour.

The following VBA code creates a three-step contour at a five-millimeter spacing:

```
Dim eff As Effect
ActiveDocument.Unit = cdrMillimeter
Set eff = ActiveShape.CreateContour(cdrContourOutside, 5, 3)
```

- The **Shape.CreateContour** method returns an **Effect** object, the **Effect.Contour** property for which you can use to modify the created contour.

Applying customized effects

The **Shape.CreateCustomEffect** method applies a customized effect to a shape. This method provides parameters for various effect settings.

- The **Shape.CreateCustomEffect** method returns an **Effect** object, the **Effect.Custom** property for which you can use to modify the created effect.

Applying distortions

The following methods apply a distortion to a shape:

- **Shape.CreatePushPullDistortion** — applies a Push-and-pull distortion
- **Shape.CreateTwisterDistortion** — applies a Twister distortion
- **Shape.CreateZipperDistortion** — applies a Zipper distortion
- **Shape.CreateCustomDistortion** — applies a customized distortion

These methods provide parameters for various distortion settings.

- The distortion-creation methods return an **Effect** object, the **Effect.Distortion** property for which you can use to modify the created distortion.

If you want, you can use event handlers to respond to events that are triggered by distorting a shape:

- **Document.ShapeDistort**

Applying drop shadows

The **Shape.CreateDropShadow** method applies a drop shadow to a shape. This method provides optional parameters for various drop-shadow settings, such as the feathering and offset of the drop shadow.

- The **Shape.CreateDropShadow** method returns an **Effect** object, the **Effect.DropShadow** property for which you can use to modify the created drop shadow.

Applying envelopes

The following methods apply an envelope to a shape:

- **Shape.CreateEnvelope** — applies a basic envelope
- **Shape.CreateEnvelopeFromCurve** — applies an envelope by using the specified curve as a template
- **Shape.CreateEnvelopeFromShape** — applies an envelope by using the specified shape as a template

These methods provide parameters for various envelope settings.

- The envelope-creation methods return an **Effect** object, the **Effect.Envelope** property for which you can use to modify the created envelope.

Applying extrusions

The **Shape.CreateExtrude** method applies an extrusion to a shape. This method provides optional parameters for various extrusion settings, such as the angle and color of the extrusion.

- The **Shape.CreateExtrude** method returns an **Effect** object, the **Effect.Extrude** property for which you can use to modify the created extrusion.

Applying lenses

The **Shape.CreateLens** method applies a lens to a shape. This method provides optional parameters for various lens settings, such as the color and magnitude of the lens.

- The **Shape.CreateLens** method returns an **Effect** object, the **Effect.Lens** property for which you can use to modify the created lens.

Applying perspective

The **Shape.CreatePerspective** method applies perspective to a shape. This method provides optional parameters for specifying horizontal and vertical vanishing points.

- The **Shape.CreatePerspective** method returns an **Effect** object, the **Effect.Perspective** property for which you can use to modify the created perspective effect.

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Capturing coordinates

When capturing mouse actions, or when developing a complex macro solution, you may want to convert between screen coordinates and document coordinates. This conversion is done with the methods **ScreenToDocument** and **DocumentToScreen** of the **Window** class.

The following VBA example converts a set of screen coordinates into a point in the document that is visible in the active window:

```
Dim docX As Double, docY As Double
ActiveDocument.Unit = cdrMillimeter
ActiveWindow.ScreenToDocument 440, 500, docX, docY
```

The following VBA example returns the screen coordinates of a point in the document as it appears on the screen:

```
Dim screenX As Long, screenY As Long
ActiveDocument.Unit = cdrMillimeter
ActiveWindow.DocumentToScreen 40, 60, screenX, screenY
```

In both examples, the converted coordinates are returned in the last two parameters.

- Screen coordinates start from the upper-left corner of the screen, so positive y-values are down the screen, whereas negative y-values are up the screen.

You can test whether a set of coordinates (that is, a point) is inside, outside, or on the outline of a curve by using the **Shape.IsOnShape** method. For a set of document coordinates, this method returns one of the following:

- **cdrInsideShape** — if the coordinate is inside the shape
- **cdrOutsideShape** — if the coordinate is outside the shape
- **cdrOnMarginOfShape** — if the coordinate is on or near the outline of the shape

For example, the following VBA code tests where the point **(4, 6)** is in relation to the active shape:

```
Dim onShape As Long
ActiveDocument.Unit = cdrInch
onShape = ActiveShape.IsOnShape(4, 6)
```

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Capturing mouse clicks

To get the position of a single mouse click, you can use the **GetUserClick** method of the **Document** class. This method pauses the macro until the specified period of time elapses, or until the user clicks in the document or presses **Escape**. Here is a VBA example that uses the **Document.GetUserClick** method:

```
Dim doc As Document, retval As Long
Dim x As Double, y As Double, shift As Long
Set doc = ActiveDocument
doc.Unit = cdrCentimeter
retval = doc.GetUserClick(x, y, shift, 10, True, cdrCursorPick)
```

The following parameters for the **Document.GetUserClick** method are coded into the preceding example:

- The variable `x` returns the horizontal position of the mouse click.
- The variable `y` returns the vertical position of the mouse click.
- The parameter `shift` returns the combination of the **Shift**, **Ctrl**, and **Alt** keys that is held down by the user when clicking the mouse. The **Shift**, **Ctrl**, and **Alt** keys are assigned values of **1**, **2**, and **4** (respectively), the sum of which is the returned value.
- The value `10` specifies the number of seconds for the user to click in the document.
- The value `True` specifies that the `SnapToObjects` parameter is enabled.
- The value `cdrCursorPick` specifies that the icon for the **Pick** tool is used for the cursor icon. (You cannot use a custom icon.)

One of the following values is returned:

- **0** — The user successfully completes the click.
- **1** — The user cancels by pressing **Escape**.
- **2** — The operation times out.

- The returned coordinates are relative to the origin of the page and, unless explicity specified, are in document units.

To get the shapes under the returned click point, you can use the method **SelectShapesAtPoint** (which is a member of **Page**), as in the following VBA example:

```
doc.ActivePage.SelectShapesAtPoint x, y, True
```

A value of `True` selects unfilled objects, while `False` does not select unfilled objects.

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Capturing mouse drags

To get the position of a mouse drag (or an area or rectangle), you can use the **GetUserArea** method of the **Document** class. This method pauses the macro until the specified period of time elapses, or until the user clicks, drags, and releases in the document or presses **Escape**. Here is a VBA example that uses the **Document.GetUserArea** method:

```
Dim doc As Document, retval As Long, shift As Long
Dim x1 As Double, y1 As Double, x2 As Double, y2 As Double
Set doc = ActiveDocument
doc.Unit = cdrCentimeter
retval = doc.GetUserArea(x1, y1, x2, y2, shift, 10, True, cdrCursorExtPick)
ActivePage.SelectShapesFromRectangle x1, y1, x2, y2, False
```

The following parameters for the **Document.GetUserArea** method are coded into the preceding example:

- The variables `x1` and `y1` return the horizontal and vertical positions (respectively) of the upper-left corner of the area.
- The variables `x2` and `y2` return the horizontal and vertical positions (respectively) of the lower-right corner of the area.
- The parameter `shift` returns the combination of the **Shift**, **Ctrl**, and **Alt** keys that is held down by the user when dragging the mouse. The **Shift**, **Ctrl**, and **Alt** keys are assigned values of **1**, **2**, and **4** (respectively), the sum of which is the returned value.
- The value `10` specifies the number of seconds for the user to click in the document.
- The value `True` specifies that the `SnapToObjects` parameter is enabled.
- The value `cdrCursorExtPick` specifies the icon to use for the cursor.

In the preceding example, the code ends by selecting the shapes that lie completely within the area by using the **Page.SelectShapesFromRectangle** method.

One of the following values is returned:

- **0** — The user successfully completes the selection.
- **1** — The user cancels by pressing **Escape**.
- **2** — The operation times out.

- This method returns two points that are interpreted as the corners of a rectangle. However, the two points can also be used as the start point and end point of a mouse drag.
- The returned coordinates are relative to the origin of the page and, unless explicity specified, are in document units.

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cdrColorMultiChannel

The following enumerations have this constant:

- cdrColorType

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cdrColorUserInk

The following enumerations have this constant:

- cdrColorType

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A

The classes in this section begin with the letter *A*:

- ActiveView
- Application
- AppStatus
- AppWindow
- ArrowHead
- ArrowHeadOptions
- ArrowHeads

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B

The classes in this section begin with the letter *B*:

- BBoxSnapPoint
- Bitmap
- BSpline
- BSplineControlPoint
- BSplineControlPoints

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C

The classes in this section begin with the letter *C*:

- Clipboard
- CloneLink
- Color
- ColorContext
- ColorManagementPolicy
- ColorManager
- ColorProfile
- ColorProfiles
- Colors
- Component
- Components
- Connector
- CorelScriptTools
- CrossPoint
- CrossPoints
- Curve
- CustomEffect
- CustomShape

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D

The classes in this section begin with the letter *D*:

- DataField
- DataFields
- DataItem
- DataItems
- Dimension
- DimensionAngular
- DimensionLinear
- DockHost
- DockHosts
- DockItem
- DockItems
- Document
- Documents
- Duotone
- DuotoneInk
- DuotoneOverprint

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E

The classes in this section begin with the letter *E*:

- EdgeSnapPoint
- Effect
- EffectBlend
- EffectContour
- EffectControlPath
- EffectCustomDistortion
- EffectDistortion
- EffectDropShadow
- EffectEnvelope
- EffectExtrude
- EffectLens
- EffectPerspective
- EffectPushPullDistortion
- Effects
- EffectTextOnPath
- EffectTwisterDistortion
- EffectZipperDistortion
- Ellipse
- EPS
- ExportFilter
- ExtrudeVanishingPoint

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F

The classes in this section begin with the letter *F*:

- Fill
- FillMetadata
- FontList
- FountainColor
- FountainColors
- FountainFill
- FrameWindow
- FrameWindows
- FrameWork

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G

The classes in this section begin with the letter *G*:

- GlobalMacroStorage
- GMSManager
- Grid
- Guide

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H

The classes in this section begin with the letter *H*:

- HatchFill
- HatchFills
- HatchLibraries
- HatchLibrary
- HatchPattern
- HatchPatterns

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I

The classes in this section begin with the letter *I*:

- ImageList
- ImportFilter

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L

The classes in this section begin with the letter *L*:

- Layer
- Layers

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M

The classes in this section begin with the letter *M*:

- Metadata

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Classes

The CorelDRAW object model contains several classes, which are arranged alphabetically in this section.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

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N

The classes in this section begin with the letter *N*:

- Node
- NodeRange
- Nodes

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O

The classes in this section begin with the letter *O*:

- ObjectSnapPoint
- OLE
- Outline
- OutlineStyle
- OutlineStyles

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P

The classes in this section begin with the letter *P*:

- Page
- Pages
- PageSize
- PageSizes
- Palette
- PaletteManager
- Palettes
- PatternCanvas
- PatternCanvases
- PatternFill
- PDFVBASettings
- Polygon
- PostScriptFill
- PowerClip
- PrintDocuments
- Printer
- PrintJob
- PrintLayout
- PrintOptions
- PrintPages
- PrintPostScript
- PrintPrepress
- PrintSeparations
- PrintSettings
- PrintTrapping
- ProofColorSettings
- Properties
- PSScreenOptions

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R

The classes in this section begin with the letter *R*:

- RecentFile
- RecentFiles
- Rect
- Rectangle
- Rulers

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S

The classes in this section begin with the letter *S*:

- ScreenRect
- Segment
- SegmentRange
- Segments
- SelectionInfo
- SeparationPlate
- SeparationPlates
- Shape
- ShapeRange
- Shapes
- SnapPoint
- SnapPointRange
- SnapPoints
- Spread
- Spreads
- StructAlignProperties
- StructColorConversionOptions
- StructCreateOptions
- StructExportOptions
- StructFontProperties
- StructHyphenationSettings
- StructImportCropOptions
- StructImportOptions
- StructImportResampleOptions
- StructOpenOptions
- StructPaletteOptions
- StructPasteOptions
- StructSaveAsOptions
- StructSpaceProperties
- Style
- StyleCharacter
- StyleFill
- StyleFrame
- StyleOutline
- StyleParagraph
- Styles
- StyleSheet
- SubPath
- SubPaths
- Symbol
- SymbolDefinition
- SymbolDefinitions
- SymbolLibraries
- SymbolLibrary
- SystemPrinters

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T

The classes in this section begin with the letter *T*:

- TableBorders
- TableCell
- TableCellRange
- TableCells
- TableColumn
- TableColumns
- TableRow
- TableRows
- TableShape
- Text
- TextCharacters
- TextColumns
- TextFrame
- TextFrames
- TextLines
- TextParagraphs
- TextRange
- TextRanges
- TextTabPosition
- TextTabPositions
- TextureFill
- TextureFillProperties
- TextureFillProperty
- TextWords
- TraceSettings
- Transparency
- TrapLayer
- TrapLayers
- TreeManager
- TreeNode
- TreeNodes

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U

The classes in this section begin with the letter *U*:

- URL
- UserSnapPoint

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V

The classes in this section begin with the letter *V*:

- View
- Views
- ViewHost
- ViewHosts
- ViewWindow
- ViewWindows

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W

The classes in this section begin with the letter *W*:

- Window
- Windows
- Workspace
- Workspaces

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Closing documents

You can close a document by calling the **Document.Close** method.

The following VBA code closes the active document and activates the document behind it:

```
ActiveDocument.Close
```

- If the code closes a document that is not active, the document referenced by the **Application.ActiveDocument** property does not change.
- You must explicitly test the **Dirty** property for a document and take appropriate action if that document has been modified.

- You can also close a document by using the **Close** method of the **Document** object itself (as in `doc.Close`).

If you want, you can use event handlers to respond to events that are triggered by closing a document:

- **Application.DocumentClose**
- **Document.Close**
- **GlobalMacroStorage.DocumentClose**

You can also use event handlers to respond to events that are triggered when the user responds to a request to close a document:

- **Application.QueryDocumentClose**
- **Document.QueryClose**
- **GlobalMacroStorage.QueryDocumentClose**

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Coding dialog boxes

After setting up a dialog box, you can develop the VBA code for displaying it. You can also develop the code for providing its text boxes, combination boxes and list boxes, buttons, and images.

Displaying dialog boxes

The **Show** method for a dialog box lets you determine how the dialog box is displayed.

For example, the following code uses the **Show** method to display the dialog box **frmFooForm**:

```
frmFooForm.Show
```

In addition, the **Show** method provides a **Modal** parameter, which lets you specify whether the dialog box is modal or modeless. A value of **vbModal** (or **1**) for this parameter creates a modal dialog box, while a value of **vbModeless** (or **0**) creates a modeless dialog box. The following VBA example creates a modeless dialog box:

```
frmFooForm.Show vbModeless
```

To open a dialog box from a macro that is available from within the application itself, you must create a public subroutine within a code module. However, a subroutine cannot be made available from within the application if the subroutine exists either within the code for a dialog box or within a class module. In addition, the subroutine cannot take any parameters.

The following VBA example subroutine opens **frmFooForm** as a modeless dialog box:

```
Public Sub showFooForm()
frmFooForm.Show vbModeless
End Sub
```

- When a dialog box loads, it triggers its own **UserForm_Initialize** event. From this event handler, you must initialize all the relevant controls on the dialog box. For more information, see Providing event handlers.

Finally, you can also use the **Show** method to open additional dialog boxes from within the current one, as in the following VBA example:

```
UserForm2.Show vbModal
```

However, control is not returned to you until all open dialog boxes are unloaded.

Providing text boxes in dialog boxes

Text boxes (that is, **TextBox** controls) are the mainstay of user input. They are simple to use and quick to program, and they are suitable for a number of purposes.

To set the text in a text box when initializing it, set the **Text** (default or implicit) property for the **TextBox** control, as in the following VBA example:

```
txtWidth.Text = "3"
txtHeight = "1"
```

To get the value of a **TextBox** control, get its **Text** property in the **Properties** window, as in the following VBA example:

```
Call SetSize(txtWidth.Text, txtHeight.Text)
```

Providing combination boxes and list boxes in dialog boxes

In a combination box (that is, a **ComboBox** control), the user can either choose an item from the list or type a value into the text box. You can prevent users from being able to type into a **ComboBox** control by setting its **Style** property (in the **Properties** window) to **fmStyleDropDownList**.

In a list box (that is, a **ListBox** control), the user can choose one or more items (typically, from between three and ten items) from the list.

To populate a list of any type, you must call the member function **AddItem** for the list. This function takes two parameters: the string or numerical value, and the position in the list. The position parameter is optional, so omitting it inserts the item at the last position in the list. For example, the following VBA code populates the list **ComboBox1** with four items:

```
ComboBox1.AddItem 1
ComboBox1.AddItem 2
ComboBox1.AddItem 3
ComboBox1.AddItem 0, 0
```

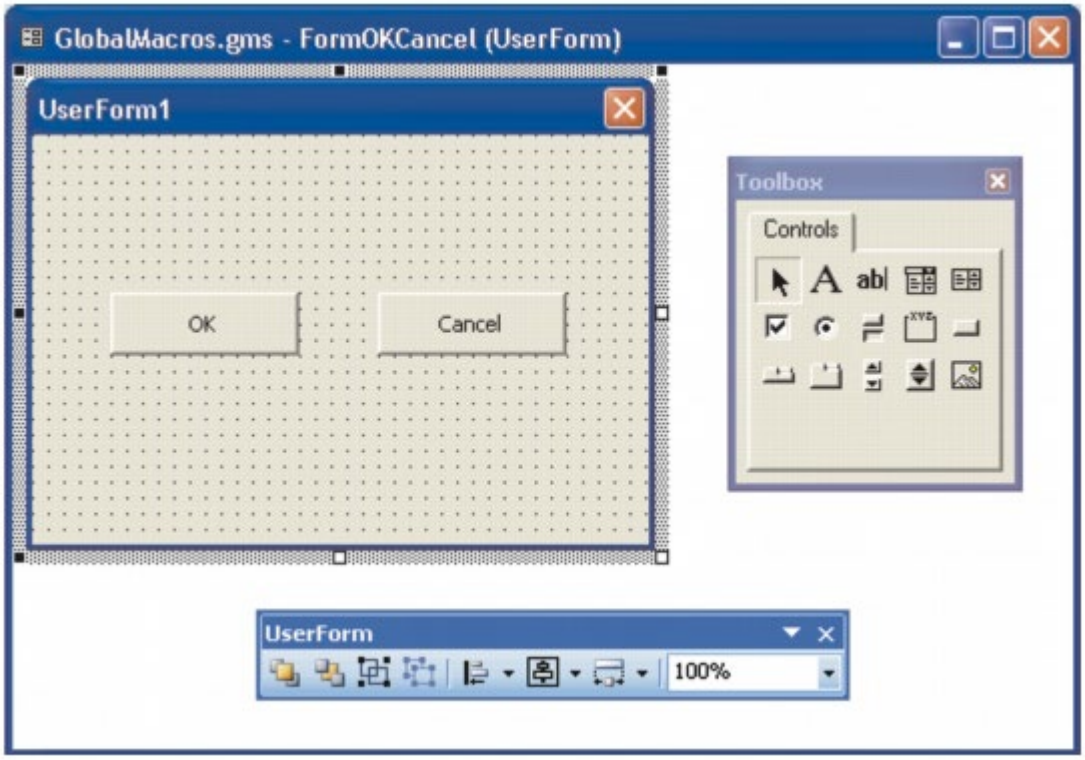
To test which item is selected when the **OK** button is clicked, test the **ListIndex** property for the list.

To get the value of the caption for a selected combination box or list box, test the **Text** property for the item, as in the following VBA example:

```
Dim retList As String
retList = ComboBox1.Text
```

Providing buttons in dialog boxes

You can add a button to a dialog box by using the **CommandButton** control. Click the dialog box to add a default-sized button, or drag to create a custom-sized one. Click the caption to edit it, or select the button and edit its **Caption** property in the **Properties** window. You might also want to change the name of the button to something more descriptive, such as **buttonOK** or **buttonCancel**.



Designing buttons in the Form Designer

Most dialog boxes have an **OK** button and a **Cancel** button. However, no button functions until its dialog box has code for handling the click event for the button. (This is because dialog boxes in VBA and VSTA are event-driven.)

For an **OK** button, you can set its **Default** property to **True** so that the event handler for the button is called when the user presses **Enter** to activate the dialog box. That way, the click-event handler for the button performs the functionality of the dialog box and then unloads that dialog box.

If the dialog box is used to set the size of the selected shapes by setting their width and height, then the click-event handler for the **OK** button could resemble the following VBA code sample (which assumes you have already created two text boxes called **txtWidth** and **txtHeight**):

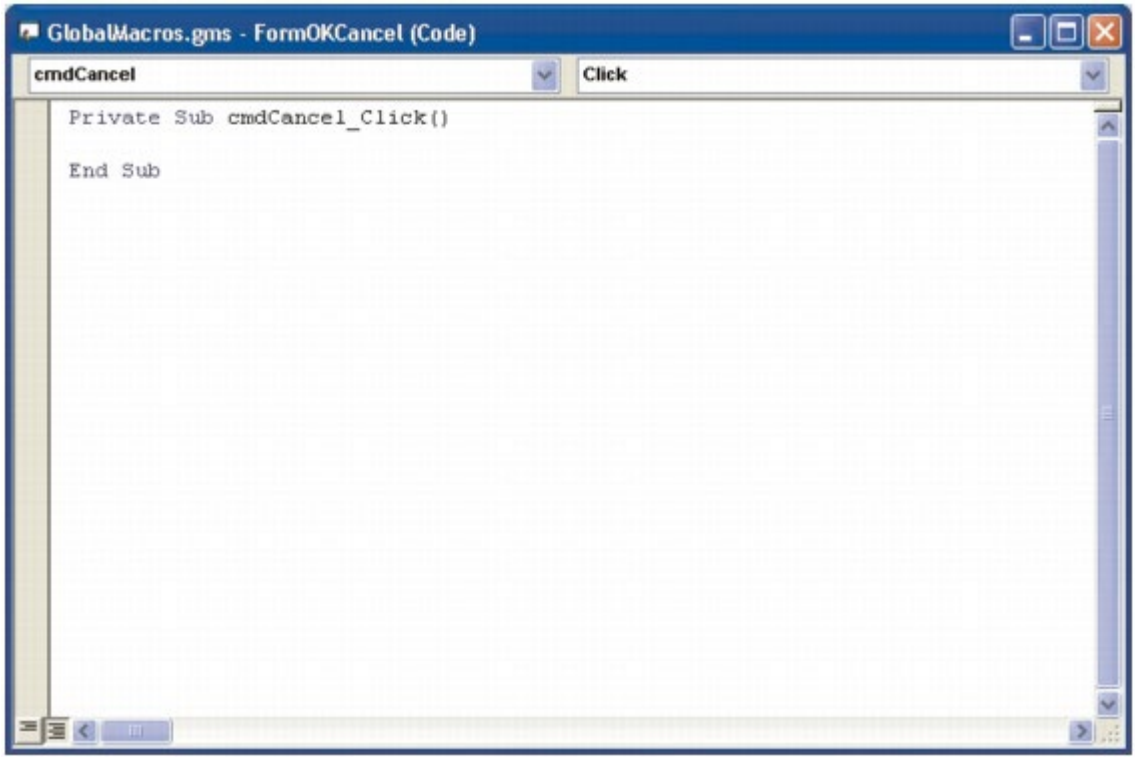
```
Private Sub buttonOK_Click()
Me.Hide
Call SetSize(txtWidth.Text, txtHeight.Text)
Unload Me
End Sub
```

Similarly, the size-setting subroutine could resemble the following:

```
Private Sub SetSize(width As String, height As String)
ActiveDocument.Unit = cdrInch
ActiveSelection.SetSize CDbl(width), CDbl(height)
End Sub
```

From inside the code module for the dialog box, the dialog-box object is implicit, so all the controls can be simply accessed by name. From other modules, the controls must be accessed through their full name (as in **UserForm1.buttonOK**).

The **Cancel** button is the simplest control: it must dismiss the form without doing anything else. To add a cancel action to a **Cancel** button, double-click the button from within the Form Designer to display its code in the **Code** window. This creates a new subroutine called **cmdCancel_Click**:



*The **Code** window with code for a **Cancel** button*

The following VBA code, if applied to a **Cancel** button, dismisses the dialog box when the button is clicked:

```
Private Sub cmdCancel_Click()
Unload Me
End Sub
```

If you continue by setting the **Cancel** property for the dialog box to **True**, then when the user presses **Escape**, the **cmdCancel_Click** event is triggered and the provided code unloads the form.

Providing images in dialog boxes

The **Image** control is used to place graphics on a dialog box. The image (a bitmap) is contained in the **Picture** property, so you can either load an RGB image from a file (such as a GIF, JPEG, or Windows Bitmap BMP file) or paste one into the property.

At run time, you can change the **Picture** property if you want to load a new image into the **Image** control. To change the **Picture** property, use the function **LoadPicture** and provide a path to the new image file, as in the following VBA example:

```
Image1.Picture = LoadPicture("C:\Images\NewImage.gif")
```

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Coloring shapes

You can add color to a shape by applying a fill (or **Fill** object) to it. The fill type for a shape is recorded by the **Fill.Type** property as one of the following constants for the **cdrFillType** enumeration:

- **cdrUniformFill** — uniform fill
- **cdrFountainFill** — fountain fill
- **cdrPatternFill** — pattern fill
- **cdrTextureFill** — texture fill
- **cdrPostScriptFill** — PostScript fill
- **cdrHatchFill** — hatch fill
- **cdrNoFill** — no fill

The following VBA code returns the type of fill that is applied to the active shape:

```
Dim fillType As cdrFillType
fillType = ActiveShape.Fill.Type
```

- You cannot change the fill type for a shape by modifying its **Fill.Type** property. Instead, you must use the appropriate **Fill.Apply...Fill** method, as described in the subsections that follow.

You can also add color to a shape by applying an outline (or **Outline** object) to it.

In addition, the object model provides a variety of properties and methods for working with the colors (or **Color** objects) that you apply to shapes.

For information on applying fills and outlines and on working with colors, see the following subtopics:

- Applying uniform fills
- Applying fountain fills
- Applying pattern fills
- Applying texture fills
- Applying PostScript fills
- Applying hatch fills
- Applying outlines
- Working with color

- In your macros, you can include queries that search for shapes that have specific fill properties, outline properties, or color properties. For information, see “Including queries in macros.”

Applying uniform fills

Uniform fills consist of a single, solid color. A uniform fill is represented by the **Fill.UniformColor** property as a **Color** object.

You can apply a uniform fill to a shape by using the **Fill.ApplyUniformFill** method. The following VBA example applies a red uniform fill to the active shape:

```
ActiveShape.Fill.ApplyUniformFill CreateRGBColor(255, 0, 0)
```

You can change the color of a uniform fill by modifying its **Fill.UniformColor** property. The following VBA example changes the uniform fill of the active shape to deep navy blue:

```
ActiveShape.Fill.UniformColor.RGBAssign 0, 0, 102
```

- You can remove the uniform fill from a shape by using the **Fill.ApplyNoFill** method.

Applying fountain fills

Fountain fills display a progression between two colors. A fountain fill is represented by the **Fill.Fountain** property as a **FountainFill** object, which specifies the various properties for the fountain fill: start color, end color, angle, blend type, and so on. The colors in a fountain fill are represented by a **FountainColors** collection.

You can apply a fountain fill to a shape by using the **Fill.ApplyFountainFill** method. This method provides optional parameters for various fountain-fill settings, such as the midpoint and offset of the blend. The following VBA example creates a simple linear fountain fill, from red to yellow, at 30 degrees to the horizontal:

```
Dim startCol As New Color, endCol As New Color
startCol.RGBAssign 255, 0, 0
endCol.RGBAssign 255, 255, 0
ActiveShape.Fill.ApplyFountainFill startCol, endCol, cdrLinearFountainFill, 30
```

You can add a color to a fountain fill by using the **FountainColors.Add** method. Color positions are integer values in percent, where 0% is the start-color position and 100% is the end-color position. The following VBA example adds a green color to the fountain fill at a position about one-third (33%) of the way from the existing red color:

```
Dim fFill As FountainFill
Set fFill = ActiveShape.Fill.Fountain
fFill.Colors.Add CreateRGBColor(0, 102, 0), 33
```

You can move a color in a fountain fill by using the **FountainColor.Move** method. The following VBA code moves the green color from the previous example to a position that is 60% of the way from the red (that is, more towards the yellow):

```
ActiveShape.Fill.Fountain.Colors(1).Move 60
```

You can use the **FountainColors.Count** property to determine the number of colors between the start color and end color of a fountain fill. (For the preceding example, this value is **1**.) The first color in the collection is that start color, and its index number is **0**; this color cannot be moved, but its color can be changed. The last color in the collection is the end color, and its index number is **(Count + 1)**; this color cannot be moved, but its color can be changed. The following VBA code changes the end color from yellow to blue:

```
Dim cols As FountainColors
Set cols = ActiveShape.Fill.Fountain.Colors
cols(cols.Count + 1).Color.RGBAssign 0, 0, 102
```

- You can remove the fountain fill from a shape by using the **Fill.ApplyNoFill** method.

Applying pattern fills

Pattern fills display a series of repeating vector objects or bitmap images. A pattern fill is represented by the **Fill.Pattern** property as a **PatternFill** object, which specifies the various properties for the pattern fill: foreground color, background color, tile offset, and so on.

- The collection of available pattern fills is stored in the **PatternCanvases** collection.

You can apply a pattern fill to a shape by using the **Fill.ApplyPatternFill** method.

- You can remove the pattern fill from a shape by using the **Fill.ApplyNoFill** method.

Applying texture fills

Texture fills are fractally generated and fill a shape with one image rather than a series of repeating images. A texture fill is represented by the **Fill.Texture** property as a **TextureFill** object, which specifies the various properties for the texture fill: origin, resolution, tile offset, and so on.

- The properties for a texture fill are stored in a **TextureFillProperties** collection.

You can apply a texture fill to a shape by using the **Fill.ApplyTextureFill** method.

- You can remove the texture fill from a shape by using the **Fill.ApplyNoFill** method.

Applying PostScript fills

PostScript fills are texture fills that are designed by using the PostScript language. A PostScript fill is represented by the **Fill.PostScript** property as a **PostScriptFill** object, which specifies the various properties for the PostScript fill.

You can apply a PostScript fill to a shape by using the **Fill.ApplyPostScriptFill** method.

- You can remove the PostScript fill from a shape by using the **Fill.ApplyNoFill** method.

Applying hatch fills

Hatch fills are composed of vector-based lines and can be used to clearly distinguish the materials or object relationships in a drawing. A hatch fill is represented by the **Fill.Hatch** property as a **HatchFill** object, which specifies the various properties for the hatch fill.

- The collection of available hatch-fill patterns is stored in the **HatchPatterns** collection, and each document stores its own library of hatch-fill patterns in a **HatchLibraries** collection.

You can apply a hatch fill to a shape by using the **Fill.ApplyHatchFill** method.

- You can remove the hatch fill from a shape by using the **Fill.ApplyNoFill** method.

Applying outlines

You can use the various properties and methods of the **Outline** class to define the outline of a shape.

The **Outline.Type** property uses the following constants of the **cdrOutlineType** enumeration to record whether the specified shape has an outline:

- **cdrOutline** — indicates that the shape has an outline
- **cdrNoOutline** — indicates that the shape does not have an outline

- If a shape has no outline, setting its **Outline.Type** property to **cdrOutline** applies the document-default outline style.
- If a shape has an outline, setting its **Outline.Type** property to **cdrNoOutline** removes that outline.

The **Outline.Width** property for an outline sets its width in document units. In the following VBA example, the outline of the selected shapes is set to 1 millimeter:

```
ActiveDocument.Unit = cdrMillimeter
ActiveSelection.Outline.Width = 1
```

- If a shape does not have an outline, its **Outline.Width** value is **0**. Changing this value applies an outline and automatically changes the value for the **Outline.Type** property from **cdrNoOutline** to **cdrOutline**.
- Similarly, if a shape has an outline, its **Outline.Width** value is greater than **0**. Changing this value to **0** removes the outline and automatically changes the value for the **Outline.Type** property from **cdrOutline** to **cdrNoOutline**.

The **Outline.Color** property for an outline defines its color, as in the following VBA example:

```
ActiveSelection.Outline.Color.GrayAssign 0 ' Set to black
```

- Setting the color of an outline automatically sets the **Outline.Type** property of that outline to **cdrOutline** and applies the default outline width.

The **Outline.Style** property for an outline specifies the dash settings of that outline. These dash settings are defined by the following properties of the **OutlineStyle** class:

- **DashCount** — represents the number of pairs of dashes and gaps in an outline. This value ranges from **1** to **5**.
- **DashLength** — represents the length of each dash in an outline. This value is calculated as a multiple of the outline width, which is measured in document units. For example, if `DashLength(1)` is `5` and the outline is 0.2" wide, the length of the dash is 1"; however, if the width of the line is changed to 0.1", the length of the dash becomes 0.5".
- **GapLength** — represents the length ofeach gap in an outline. This value is calculated as a multiple of the outline width, which is measured in document units.
- **Index** — represents the index number of a predefined outline style in the **OutlineStyles** collection for the application. The **OutlineStyles** collection is customizable, so the index number that is associated with each outline style in the collection may vary from user to user; however, the expression `OutlineStyles.Item(0)` always specifies a solid line.

- **Outline** objects have many other properties, including the following:
  - **StartArrow** and **EndArrow** — specify the arrowhead on each end of an open curve
  - **LineCaps** and **LineJoin** — respectively, specify the type of line caps (butt, round, or square) and line joins (bevel, miter, or round)
  - **NibAngle** and **NibStretch** — specify the shape of the nib used to draw the outline
  - **BehindFill** and **ScaleWithShape** — respectively, draw the outline behind the fill and scale the outline with the shape
- **Outline** objects also have methods, including the following:
  - **ConvertToObject** — converts the outline to an object
  - **SetProperties** — sets most of the available outline properties in a single call

Working with color

The **Color** class defines the fill colors and outline colors that you apply to shapes. This class provides a number of properties and methods for working with color.

You can determine the color model of a color by accessing its **Color.Type** property, as in the following VBA example:

```
Dim colType As cdrColorType
colType = ActiveShape.Outline.Color.Type
```

The **Color.Type** property is defined by the **cdrColorType** enumeration, which provides the following constants (among many others) for supported color models:

- **cdrColorCMYK** — specifies the CMYK color model
- **cdrColorRGB** — specifies the RGB color model
- **cdrColorGray** — specifies the grayscale color model

- The color components for each supported color model are defined by additional properties of the **Color** class, as demostrated by the following VBA examples:
  - CMYK color model — is defined by the **Color.CMYKCyan**, **Color.CMYKMagenta**, **Color.CMYKYellow**, and **Color.CMYKBlack** properties
  - RGB color model — is defined by the **Color.RGBRed**, **Color.RGBGreen**, and **Color.RGBBlue** properties
  - grayscale color model — is defined by the **Color.Gray** property
- The range of values that is supported by a color component depends on the color model for that component.

- To create a color, you can use the automation keyword **New**, as in `Dim col As New Color`.
- To assign a color model to a new color, you can use the desired **...Assign** method (such as **Color.CMYKAssign**, **Color.RGBAssign**, or **Color.GrayAssign**). Each of these methods provides one parameter for each color component in its respective color model. For example, `col.RGBAssign 0, 0, 102` assigns a deep-blue RGB color to the new color that was created in the previous tip.
- To use the application's color-management settings to change the color model that is assigned to a color, you can use the desired **ConvertTo...** method (such as **Color.ConvertToCMYK**, **Color.ConvertToRGB**, or **Color.ConvertToGray**). For example, `ActiveShape.Fill.UniformColor.ConvertToRGB` converts the fill of the active shape to the RGB color model.

You can copy the properties of one color to another color by using the **Color.CopyAssign** method, as in the following VBA example:

```
Dim sh As Shape
Set sh = ActiveShape
sh.Outline.Color.CopyAssign sh.Fill.UniformColor
```

- The color "none" does not exist. To set a fill color or outline color to "none," you must instead set the fill type or outline type to "none."

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cdr

The constants in this section begin with the letters *cdr*.

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clr

The constants in this section begin with the letters *clr*.

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Constants

The CorelDRAW object model contains several constants, which are arranged alphabetically in this section.

cdr clr pdf prn

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pdf

The constants in this section begin with the letters *pdf*.

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prn

The constants in this section begin with the letters *prn*.

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Creating command groups for documents

Two very useful methods of the **Document** class combine to create a “command group,” which can reduce a series of programmed, document-related actions to a single, undoable step. These methods — **BeginCommandGroup** and **EndCommandGroup** — are demonstrated in the following VBA example:

```
Dim sh As Shape
ActiveDocument.BeginCommandGroup "CreateCurveEllipse"
Set sh = ActiveLayer.CreateEllipse(0, 1, 1, 0)
sh.ConvertToCurves
ActiveDocument.EndCommandGroup
```

The preceding code sets the **Undo** string in the **Edit** menu as **Undo CreateCurveEllipse**. Clicking this command undoes not only the **ConvertToCurves** operation but also the **CreateEllipse** operation.

A command group can contain many hundreds of commands, if required. Creating command groups can make your macros appear to be fully integrated into the application.

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Creating documents

The **Application** object has two methods for creating documents: **CreateDocument** and **CreateDocumentFromTemplate**.

The **Application.CreateDocument** method creates an empty document based on the default page size, orientation, and styles:

```
Application.CreateDocument() As Document
```

The **Application.CreateDocumentFromTemplate** method creates an untitled document from a specified template (CDT) file:

```
Application.CreateDocumentFromTemplate(Template As String, _
[IncludeGraphics As Boolean = True])As Document
```

Both of these functions return a reference to the new document, so they are typically used in the following manner:

```
Dim newDoc as Document
Set newDoc = CreateDocument
```

The new document becomes active immediately and can be referenced by using the **Application.ActiveDocument** property. For more information on this property, see Activating documents.

If you want, you can use event handlers to respond to events that are triggered by creating a document:

- **Application.DocumentNew**
- **GlobalMacroStorage.DocumentNew**

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Creating layers

You can create a layer by using the **Page.CreateLayer** method. Creating a layer inserts a new layer at the top of the list of non-master layers.

The following VBA code creates a new layer called “My New Layer”:

```
ActivePage.CreateLayer "My New Layer"
```

If you want, you can use event handlers to respond to events that are triggered by activating a layer:

- **Document.LayerCreate**

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Creating macro projects

The process of creating a macro begins with creating a macro project. A macro project can be created in one of two ways:

- as a Global Macro Storage (GMS), or “project,” file
- in a document

Using GMS files

For best results in storing and distributing a macro project, it is highly recommended that you use a GMS file. GMS files are stored in the **GMS** folder for the application, the location of which depends on the type of macro project.

Macro-project type	GMS folder
Default projects installed with the software	**X:\Program Files\Corel**<suite>\<program>**\GMS** Legend: • **X:** is the drive and **Program Files\Corel**<*suite*> is the path where the software suite is installed • **<program>** is the program subfolder
User-created projects on Windows 7 and Windows 8	**X:\Users**<username>**\AppData\Roaming\Corel**<suite>\<program>**\GMS** Legend: • **X:** is the drive where the software is installed • **<username>** is the name of the user • ***<suite>*** is the folder where the software suite is installed • **<program>** is the application subfolder

Understanding macro projects

The **Macro Manager** docker provides basic tools for working with macro projects. For access to more advanced tools, you can use the Macro Editor (for VBA macro projects) or the VSTA Editor (for VSTA macro projects).

In the Macro Editor, a VBA macro project is broken into four types of components, which are displayed as the following folders in the Project Explorer (see Using the Project Explorer):

- **<application> Objects** — contains a single item that is used mostly for event handling: **ThisMacroStorage** for GMS-based macro projects, or **ThisDocument** for document-based macro projects. For normal code, this module is not used.
- **Forms** — contains customized dialog boxes and user interfaces, plus the code to control them
- **Modules** — contains code modules, for storing general code and macros
- **Class Modules** — contains object-oriented Visual Basic class modules (which are not discussed in this documentation)

- In the Macro Editor, you cannot move a component from one folder to another within the same project. However, you can drag a component to another project to make a copy of it there.

Related procedures

For details on creating macro projects, see the following procedures:

- To create a macro project
- To add a dialog box to a macro project
- To add a code module to a macro project
- To add a class module to a macro project

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Creating macros

Now that you are familiar with the concept of automation and with the macro-related tools and features of CorelDRAW, you are ready to create macros.

In this section

This section contains the following topics:

- Creating macro projects
- Writing macros
- Recording macros
- Running macros
- Debugging macros

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Creating pages

The methods for creating pages belong to the **Document** class.

Both the **Document.AddPages** method and the **Document.AddPagesEx** method add the specified number of pages to the end of a document. The difference between these methods is that **AddPages** uses the default page size, while **AddPagesEx** uses a specified size.

Similarly, both the **Document.InsertPages** method and the **Document.InsertPagesEx** method insert the specified number of pages at the specified location in a document. The difference between these methods is that **InsertPages** uses the default page size, while **InsertPagesEx** uses a specified size.

As an example, the following VBA code uses the **AddPages** method to add three default-sized pages to the end of the document:

```
Public Function AddSomeSimplePages() as Page
Set AddSomeSimplePages = ActiveDocument.AddPages(3)
End Function
```

The following VBA example uses the **AddPagesEx** method to add to the end of the document three pages that are 8.5 inches wide by 11 inches high:

```
Public Function AddSomeSpecifiedPages() as Page
Dim doc as Document
Set doc = ActiveDocument
doc.Unit = cdrInch
Set AddSomeSpecifiedPages = doc.AddPagesEx(3, 8.5, 11)
End Function
```

The preceding examples return the first page that was added; all other added pages follow this page. You can therefore reference any of the added pages by incrementing the **Index** property of the returned page:

```
Dim firstNewPage As Page, secondNewPage As Page
Set firstNewPage = AddSomeSimplePages
Set secondNewPage = ActiveDocument.Pages(firstNewPage.Index + 1)
```

If you want, you can use event handlers to respond to events that are triggered by creating a page:

- **Document.PageCreate**

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Creating shapes

Every document is made up of shapes, or **Shape** objects, which are created by using the drawing tools. The shapes on a document page are stored on layers, so the various shape-creation methods belong to the **Layer** class.

For information on creating specific types of shapes, see the following subtopics:

- Creating rectangles
- Creating ellipses
- Creating lines and curves
- Creating text objects
- Creating symbols

- Supported shapes not discussed in this section include polygons (or **Polygon** objects) and customized shapes (or **CustomShape** objects).
- Customized shapes that are supported include tables (or **TableShape** objects).

- Shapes are measured in document units. You can specify the unit of measurement for a document by using the **Document.Unit** property (see Setting document properties).

If you want, you can use event handlers to respond to events that are triggered by creating a shape:

- **AddinHook.ShapeCreated**
- **Document.ShapeCreate**

Creating rectangles

You can add rectangles (or **Rectangle** objects) to your documents by using one of the following methods:

- **Layer.CreateRectangle**
- **Layer.CreateRectangle2**
- **Layer.CreateRectangleRect**

These methods return a reference to the new **Shape** object. They differ only in the parameters that they take, so you can choose the method that best suits your macro solution.

- You can also use these rectangle-creation methods to create squares.

The **CreateRectangle** method creates a rectangle by using four parameters that specify the following:

- the distance between the left, top, right, and bottom sides of the rectangle (in that order)
- the corresponding edges of the page frame

The following VBA example uses the **CreateRectangle** method to create a 2" × 1" rectangle that is positioned 6" up from the bottom of the page frame and 3" in from the left side of the page frame:

```
Dim sh As Shape
ActiveDocument.Unit = cdrInch
Set sh = ActiveLayer.CreateRectangle(3, 7, 6, 5)
```

The **CreateRectangle2** method creates a rectangle based on the coordinates of its lower-left corner, its width, and its height.

The following VBA example uses the **CreateRectangle2** method to create the same rectangle as the previous example:

```
Dim sh As Shape
ActiveDocument.Unit = cdrInch
Set sh = ActiveLayer.CreateRectangle2(3, 6, 2, 1)
```

Finally, the **CreateRectangleRect** method creates a rectangle based on its bounding box (or **Rect** object).

These three rectangle-creation methods provide optional parameters for specifying corner roundness.

The **CreateRectangle** method specifies corner roundness by using parameters for the upper-left, upper-right, lower-left, and lower-right corners (in that order). These parameters take integer values (which range from the default **0** to **100**) that define the radius of the four corners as a whole-number percentage of half of the length of the shortest side.

The following VBA example re-creates the 2" × 1" rectangle from the previous examples. However, this time, the four corner radii are set to 100%, 75%, 50%, and 0% of half of the length of the shortest side (that is, to 0.5", 0.375", 0.25", and a cusp):

```
Dim sh As Shape
ActiveDocument.Unit = cdrInch
Set sh = ActiveLayer.CreateRectangle(3, 7, 6, 5, 100, 75, 50, 0)
```

The **CreateRectangle2** method and the **CreateRectangleRect** method define the corner radii in the same order as the **CreateRectangle** method (that is, upper-left, upper-right, lower-left, and lower-right). However, **CreateRectangle2** and **CreateRectangleRect** take double (floating-point) values that

measure the corner radii in document units.

- When using **CreateRectangle2** or **CreateRectangleRect**, you must limit the size of the corner radii to less than half of the length of the shortest side of the rectangle.

The following VBA example uses the **CreateRectangle2** method to create the same round-cornered rectangle as the previous example:

```
Dim sh As Shape
ActiveDocument.Unit = cdrInch
ActiveDocument.ReferencePoint = cdrBottomLeft
Set sh = ActiveLayer.CreateRectangle2(3, 6, 2, 1, 0.5, 0.375, 0.25, _
0)
```

Creating ellipses

You can add ellipses (or **Ellipse** objects) to your documents by using one of the following methods:

- **Layer.CreateEllipse**
- **Layer.CreateEllipse2**
- **Layer.CreateEllipseRect**

These methods return a reference to the new **Shape** object. They differ only in the parameters that they take, so you can choose the method that best suits your macro solution.

- You can also use the ellipse-creation methods to create circles, arcs, and pie shapes.

The **CreateEllipse** method creates an ellipse by using four parameters that specify the following:

- the distance between the left, top, right, and bottom sides of the ellipse (in that order)
- the corresponding edges of the page frame

The following VBA example creates a 50-millimeter circle:

```
Dim sh As Shape
ActiveDocument.Unit = cdrMillimeter
Set sh = ActiveLayer.CreateEllipse(75, 150, 125, 100)
```

The **CreateEllipse2** method creates an ellipse based on its center point, its horizontal radius, and its vertical radius. (If only one radius is given, a circle is created.)

The following VBA example uses the **CreateEllipse2** method to create an ellipse:

```
Dim sh As Shape
ActiveDocument.Unit = cdrMillimeter
Set sh = ActiveLayer.CreateEllipse2(100, 125, 50, 25)
```

The following VBA example uses the **CreateEllipse2** method to create the same 50-millimeter circle as the **CreateEllipse** example:

```
Dim sh As Shape
ActiveDocument.Unit = cdrMillimeter
Set sh = ActiveLayer.CreateEllipse2(100, 125, 25)
```

Finally, the **CreateEllipseRect** method creates an ellipse based on its bounding box (or **Rect** object).

These three ellipse-creation methods provide three optional parameters for creating an arc or a pie shape. The **StartAngle** and **EndAngle** parameters — which are double values that are measured with zero being horizontally right on the page and with positive values being degrees from zero and moving counterclockwise — are used to define the start angle and end angle of the shape (respectively). In addition, the **Pie** parameter — which is a Boolean value — defines whether the shape is an arc (**False**) or a pie shape (**True**).

The following VBA code uses the **CreateEllipse** method to create a “C” shape:

```
Dim sh As Shape
ActiveDocument.Unit = cdrMillimeter
Set sh = ActiveLayer.CreateEllipse(75, 150, 125, 100, 60, 290, False)
```

Creating lines and curves

You can add lines and curves (or **Curve** objects) to your documents. To create a line or a curve, you must first create a **Curve** object “in memory” by using the **Application.CreateCurve** method.

Each **Curve** object has at least one subpath (or **SubPath** object). You can add a subpath to a line or a curve by using the **Curve.CreateSubPath** method.

Each **SubPath** object has at least one segment (or **Segment** object), which can be line-type or curve-type. You can add a line-type segment to the end of a subpath by using the **SubPath.AppendLineSegment** method; you can add a curve-type segment by using the **SubPath.AppendCurveSegment** method or the **SubPath.AppendCurveSegment2** method.

- The **SubPath.AppendLineSegment** method requires one set of Cartesian coordinates, which defines the end of the new segment.
- The **SubPath.AppendCurveSegment** method requires one set of Cartesian coordinates, which defines the end of the new segment. Optionally, you can specify two sets of polar coordinates if you want to define the lengths and angles of the starting and ending control handles for the segment.
- The **SubPath.AppendCurveSegment2** method requires three sets of Cartesian coordinates: one to define the end of the new segment, and two to define the positions of the starting and ending control handles for the segment.

- You can add a segment to the beginning of a subpath by setting the **AppendAtBeginning** parameter for the segment-creation method to **True**.

Finally, each **Segment** object has at least one node (or **Node** object). You can add a node to a segment by using the **Segment.AddNodeAt** method.

- You can close a **Curve** object by setting its **Closed** property to **True**.

After creating a curve “in memory,” you can apply it to a layer by using the **Layer.CreateCurve** method. A reference to the new **Shape** object is returned.

The following VBA code creates a D-shaped curve that is closed:

```
Dim sh As Shape, spath As SubPath, crv As Curve
ActiveDocument.Unit = cdrCentimeter
Set crv = Application.CreateCurve(ActiveDocument)
'Create Curve object
Set spath = crv.CreateSubPath(6, 6) ' Create a SubPath
spath.AppendLineSegment 6, 3 ' Add the short vertical segment
spath.AppendCurveSegment 3, 0, 2, 270, 2, 0 ' Lower curve
spath.AppendLineSegment 0, 0 ' Bottom straight edge
spath.AppendLineSegment 0, 9 ' Left straight edge
spath.AppendLineSegment 3, 9 ' Top straight edge
spath.AppendCurveSegment 6, 6, 2, 0, 2, 90 ' Upper curve
spath.Closed = True ' Close the curve
Set sh = ActiveLayer.CreateCurve(crv) ' Create curve shape
```

The **Layer** class provides three additional methods that act as shortcuts for creating a basic line or basic curve that has a single segment on a single subpath:

- **Layer.CreateLineSegment** — creates a basic line based on the given starting point and ending point
- **Layer.CreateCurveSegment** — creates a basic curve based on the given starting point and ending point and, optionally, on the lengths and angles of the starting and ending control handles for the curve
- **Layer.CreateCurveSegment2** — creates a basic curve based on the given starting point and ending point and on the given positions of the starting and ending control handles for the curve

- These three methods return a reference to the new **Shape** object.

Creating text objects

You can add text (or **Text** objects) to your documents. Two types of text are supported: artistic text and paragraph text. An artistic-text object is a short line of text to which you can apply graphical effects. In contrast, a paragraph-text object is a large block of text — stored in a rectangular container called a “frame” — to which you can apply more complex formatting.

To create an artistic-text object, you can use one of the following methods:

- **Layer.CreateArtisticText** — creates basic artistic text
- **Layer.CreateArtisticTextWide** — creates artistic text that is in Unicode format

- Both of these methods require you to specify the position and content of the artistic-text object. Optionally, both of these methods let you set such text attributes as font style, font size, formatting, and alignment. In addition, both of these methods return a reference to the new **Shape** object.

The following VBA code uses the **CreateArtisticText** method to create a basic artistic-text object that places the words “Hello World” at the specified position:

```
Dim sh As Shape
ActiveDocument.Unit = cdrInch
Set sh = ActiveLayer.CreateArtisticText(1, 4, "Hello World")
```

You can fit artistic text to a path by using the **Text.FitTextToPath** method, which simply attaches the text to the outline of a shape such that the text flows along the path of that shape.

The following VBA code creates a new artistic-text object and attaches it to the selected shape:

```
Dim sh As Shape, sPath As Shape
ActiveDocument.Unit = cdrInch
Set sPath = ActiveShape
Set sh = ActiveLayer.CreateArtisticText(1, 4, "Hello World")
sh.Text.FitToPath sPath
```

To create a paragraph-text object, you can use one of the following methods:

- **Layer.CreateParagraphText** — creates basic paragraph text
- **Layer.CreateParagraphTextWide** — creates paragraph text that is in Unicode format

- Both of these methods require you to specify the size of the paragraph-text frame by setting its position from the left, top, right, and bottom sides of the page frame (in that order). Optionally, both of these methods let you specify the desired text and set such text attributes as font style, font size, formatting, and alignment. In addition, both of these methods return a reference to the new **Shape** object.

The following VBA code uses the **CreateParagraphText** method to create a basic paragraph-text object that centers the words “Hi There” in a frame of the specified size:

```
Dim sh As Shape
ActiveDocument.Unit = cdrInch
Set sh = ActiveLayer.CreateParagraphText(1, 4, 5, 2, "Hi There", _
Alignment := cdrCenterAlignment)
```

You can format existing paragraph text by using text ranges (or **TextRange** objects). Text ranges are handled in two ways, both of which involve frames (or **TextFrame** objects):

- “frames” method — The **Text.Frames** property represents a series of text frames, each of which has its own text range.
- “story” method — The **Text.Story** property represents a text range that includes all text in a series of text frames.

A text range can be treated as a single block of text, such that any changes to text properties (such as font style and font size) are applied to all text in that text range. Alternatively, a text range can be broken down into the following smaller text ranges:

- columns (or **TextColumns** objects)
- paragraphs (or **TextParagraphs** objects)
- lines (or **TextLines** objects)
- words (or **TextWords** objects)
- characters (or **TextCharacters** objects)

- The object model supports all paragraph-formatting options and character-formatting options that are offered by the application.

The following VBA code formats the specified text range by using the **Text.Story** property. The first paragraph of the story is changed to a heading style and the second and third paragraphs into a body-text style:

```
Dim txt As TextRange
' Format the first paragraph
Set txt = ActiveShape.Text.Story.Paragraphs(1)
txt.ChangeCase cdrTextUpperCase
txt.Font = "Verdana"
txt.Size = 18
txt.Bold = True
' Format the second and third paragraphs
Set txt = ActiveShape.Text.Story.Paragraphs(2, 2)
txt.Font = "Times New Roman"
txt.Size = 12
txt.Style = cdrNormalFontStyle
```

You can place selected text inside closed shapes by using the **Shape.PlaceTextInside** method.

The following VBA code creates a 5" × 2" ellipse and places the selected text inside it:

```
Dim txt As Shape, sh As Shape
ActiveDocument.Unit = cdrInch
Set txt = ActiveShape
Set sh = ActiveLayer.CreateEllipse(0, 2, 5, 0)
sh.PlaceTextInside txt
```

Creating symbols

A symbol (or **Symbol** object) is a reusable graphic element that is defined in a symbol library. Using symbols in your documents provides the following benefits:

- lower file-size — Each symbol is defined only once, regardless of how many actual instances of that symbol appear in the document.
- increased productivity — Any changes made to a symbol definition are automatically propagated to all instances of that symbol in the document.
- improved workflow — Symbol libraries are a convenient way to store and reuse common graphic elements.

Symbol libraries come in two varieties: external and internal.

External symbol libraries use the filename extension CSL and contain symbol definitions that must be manually added to the workspace at the application level. You cannot modify a symbol that is defined in an external library unless you open the associated external library (CSL) file; simply importing the file as a library does not let you modify its contents.

- External symbol libraries must be published to a location that all users can access. A common mapped drive is a good solution, but a corporate intranet is a better one. However, if the security of the symbols is not important, the best solution is a corporate Internet site.

Internal symbol libraries exist at the document level. Defining a new symbol in a document — or adding an instance of an external-library symbol to a document — automatically adds that symbol to the internal library for that document. For this reason, each document has its own unique internal symbol library.

- Inserting an instance of a symbol from an external symbol library creates a link to the definition for that symbol in that external symbol library. If, at any point, the document cannot access the external symbol library, the symbol definition in the internal symbol library for that document is used instead.

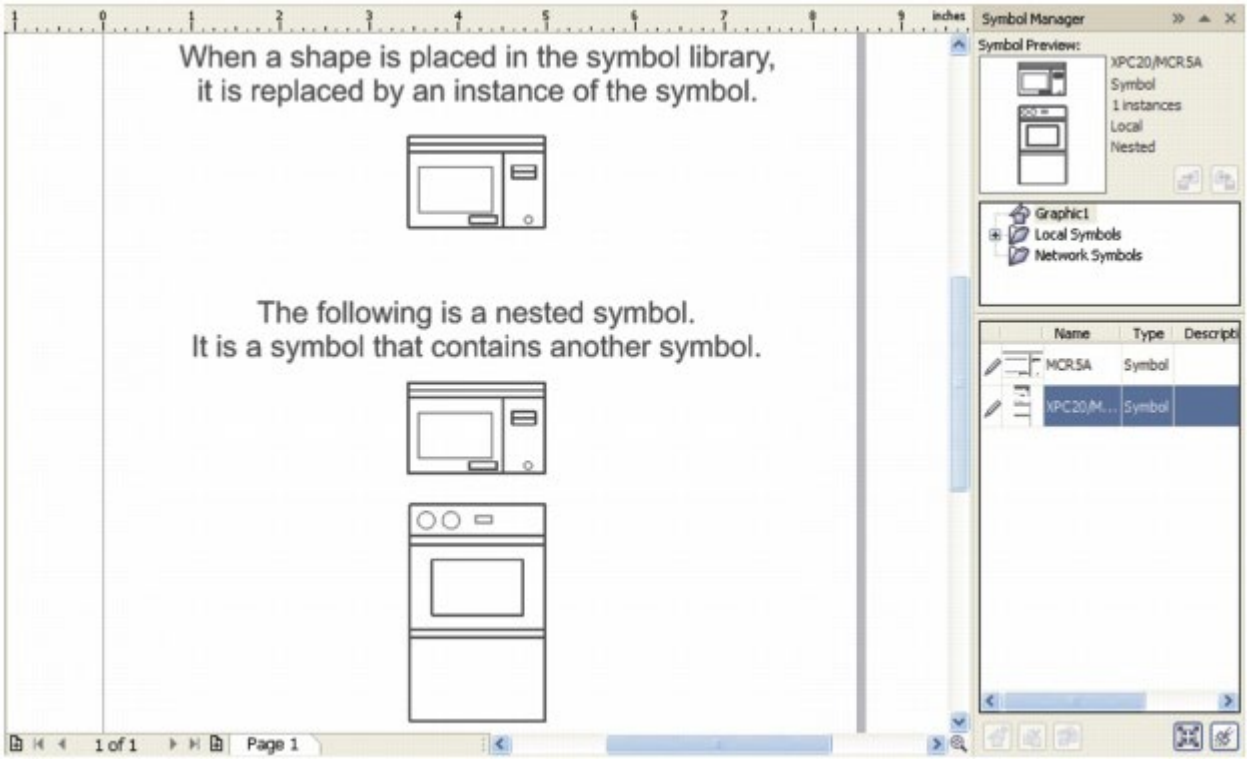
The **Application.SymbolLibraries** property contains the collection of all external symbol libraries (or **SymbolLibrary** objects) that are available to the application; the **Document.SymbolLibrary** property returns just the internal symbol library for that document. The **SymbolLibrary.Symbols** property contains the collection of all symbol definitions (or **SymbolDefinition** objects) in that symbol library. A **SymbolDefinition** object is also returned by the **Symbol.Defintion** property; therefore, you can modify the defintion of a symbol by using the various properties and methods of the **SymbolDefinition** class.

- To remove a symbol definition from the internal symbol library for a document, you must delete all instances of the symbol from the document and then run the **SymbolLibrary.PurgeUnusedSymbols** method. Simply removing all instances of a symbol from a document does not automatically remove its symbol definition from the internal symbol library for that document.

The following VBA code demonstrates the basics of using symbols:

```
Sub AddRemoveSymbols()
Dim objSymLibSwitchA As SymbolLibrary
Dim shpSymBreaker2 As Shape, shpSymBreaker2A As Shape
ActiveDocument.Unit = cdrMillimeter
'Add the switchesA external symbol library to the global
'workspace.
Set objSymLibSwitchA = SymbolLibraries.Add _
("C:\libs\switches\switchesA.csl")
'Add the breaker2 symbol to the active layer.
'NOTE: This automatically adds the symbol definition to the
'internal symbol library for the document.
Set shpSymBreaker2 = ActiveLayer.CreateSymbol(15, 20, _
"breaker2", SymbolLibraries("switchesA"))
'Add another instance of the breaker2 symbol, this time from the
'internal symbol library. NOTE: We did not specify a library, so
'the library for the local document is used by default.
Set shpSymBreaker2A = ActiveLayer.CreateSymbol(30, 20, _
"breaker2")
'Remove the switchesA library from the global workspace.
SymbolLibraries("switchesA").Delete
'Delete the two breaker2 symbols.
shpSymBreaker2.Delete
shpSymBreaker2A.Delete
'At this point, the internal symbol library for the document
'still has the definition of breaker2 stored. To remove this
'definition, we must purge the unused symbols from the library.
'The definition is unused because there are no instances that
'reference it.
ActiveDocument.SymbolLibrary.PurgeUnusedSymbols
End Sub
```

A symbol can contain (or “nest”) other symbols. A top-level symbol can contain symbols, and each of those symbols can contain a symbol, and so forth. In the object model, the **SymbolDefinition.NestedSymbols** property returns (as a **SymbolDefinitions** object) the collection of nested symbols for a symbol definition. While there is no restriction on how many nesting levels can be created, the symbol cannot be rendered without access to its symbol definition (whether external or internal). In addition, even if the first and second nesting layers of a symbol are rendered correctly, a symbol on the third nesting layer may not be rendered correctly without access to its required symbol definition.



Symbols and nested symbols

The following VBA code demonstrates the basics of using nested symbols:

```
Sub MakeNestedSymbol()
Dim shp1 As Shape, shp2 As Shape, shp3 As Shape, shpSym As Shape
Dim shpRng As New ShapeRange
'Create a pair of rectangles and a circle.
Set shp1 = ActiveLayer.CreateRectangle2(0, 0, 10, 20)
Set shp2 = ActiveLayer.CreateRectangle2(50, 50, 20 ,10)
Set shp3 = ActiveLayer.CreateEllipse(10, 10, 20)
'Make a symbol out of the circle. NOTE: This circle is
'automatically added to the internal symbol library for the
'document.
Set shpSym = shp3.ConvertToSymbol("circle")
'Add the rectangles and the circle symbol to a shape range.
shpRng.Add shp1
shpRng.Add shp2
```

```
shpRng.Add shpSym
'Convert the shape range into a symbol. NOTE: This symbol is
'added to the internal symbol library for the document. It is
'also is a nested symbol because it contains the symbol circle.
shpRng.ConvertToSymbol "shapes"
End Sub
```

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Debugging macros

To ensure that your macros run as expected, it's important to debug them.

The Macro Editor provides four windows for debugging VBA code. The Macro Editor also provides two strong debugging facilities that are common to language editors: setting breakpoints, and stepping through code.

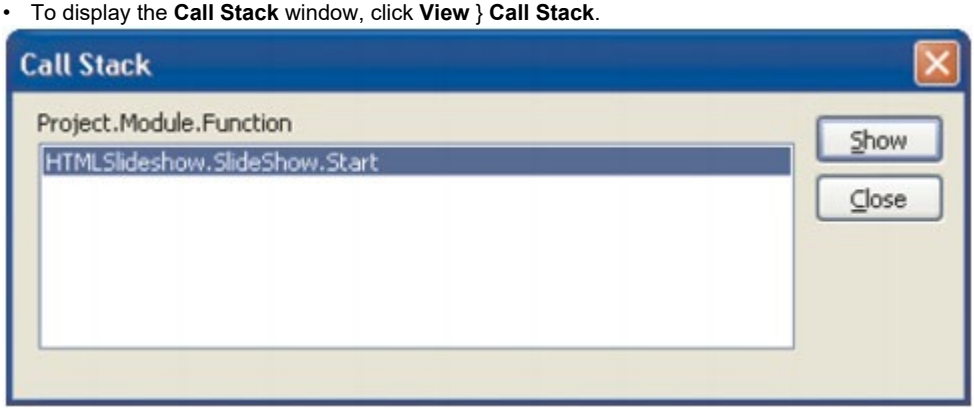
- The Macro Editor also supports two advanced debugging techniques that are not discussed in this documentation: Making changes to the code while it is running, and watching and changing variables.

Using the debugging windows

The Macro Editor provides four windows for debugging VBA code: the **Call Stack** window, the **Immediate** window, the **Locals** window, and the **Watches** window. All of these windows provide important information about the state of functions and variables while an application is running.

The **Call Stack** window is a modal dialog box that lists which function calls which function. In long, complicated applications, this information is useful for tracing the steps to a particular function being called. To visit a function listed in the window, select the function name and then click **Show**, or else close the window.

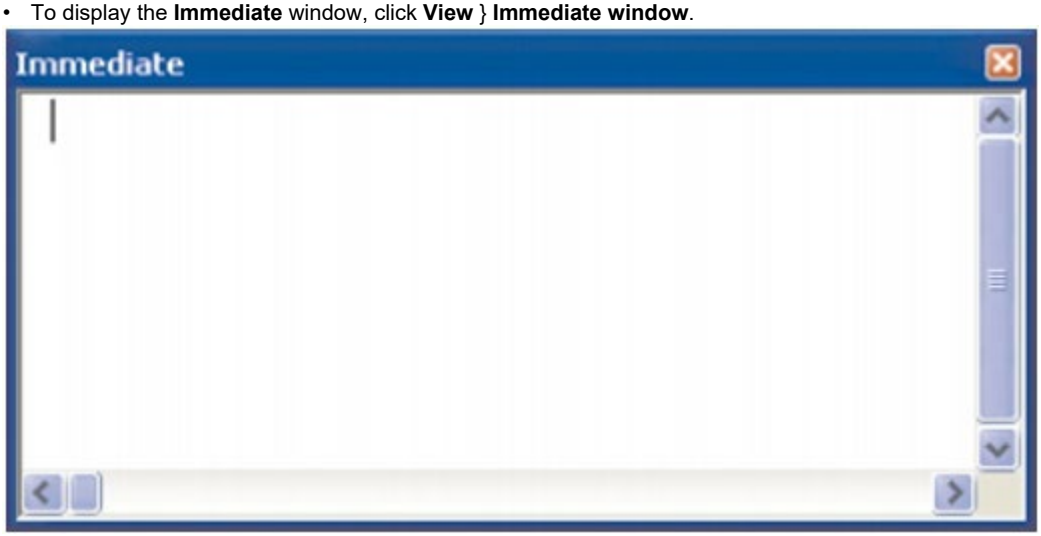
- To display the **Call Stack** window, click **View** } **Call Stack**.



*The **Call Stack** window*

The **Immediate** window lets you type in and run arbitrary lines of code while a macro is paused. This functionality is useful for getting or setting the property of an object in the document, or for setting the value of a variable in the code. To run a piece of code, type it in the **Immediate** window, and then press **Enter**; the code is executed immediately.

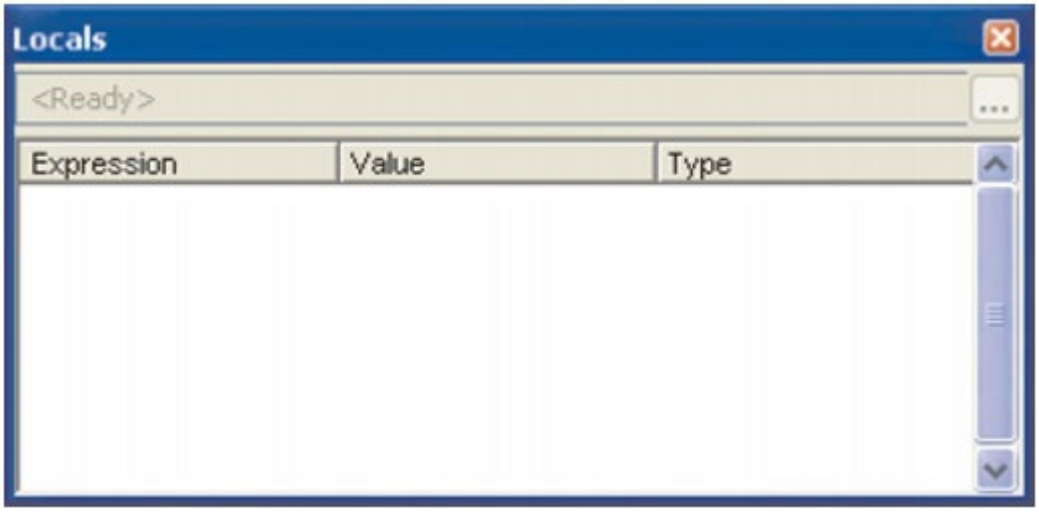
- To display the **Immediate** window, click **View** } **Immediate window**.



*The **Immediate** window*

The **Locals** window displays all variables and objects that exist within the current scope. The type and value for each variable are listed in the columns next to the name of the variable. Some variables and objects have several children, which can be displayed by clicking the button next to the parent. Many variables let you edit their value by clicking it.

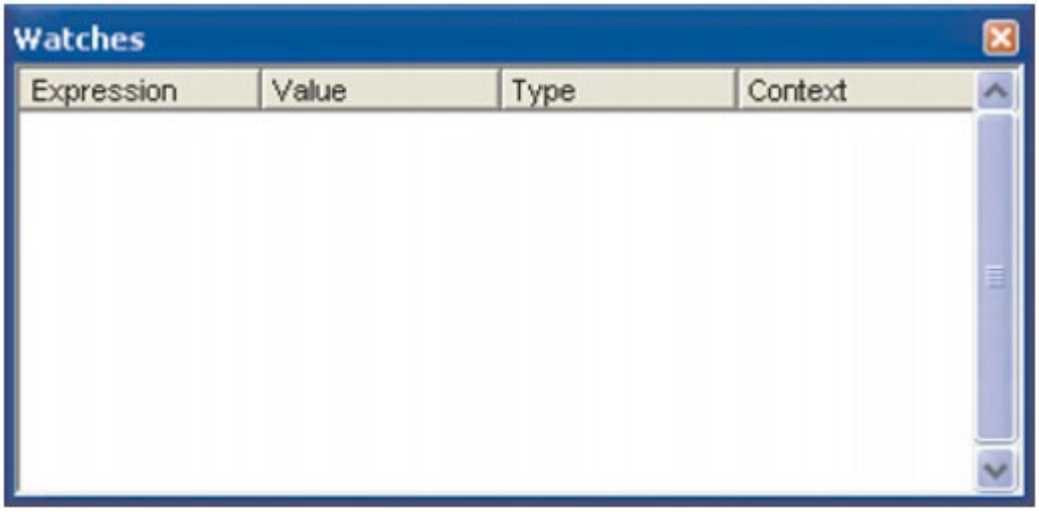
- To display the **Locals** window, click **View** } **Locals window**.



*The **Locals** window*

The **Watches** window is used to watch specific variables or object properties. This functionality is useful for watching just one or two values rather than searching for them among all the values in the **Locals** window.

- To display the **Watches** window, click **View** } **Watch window**.

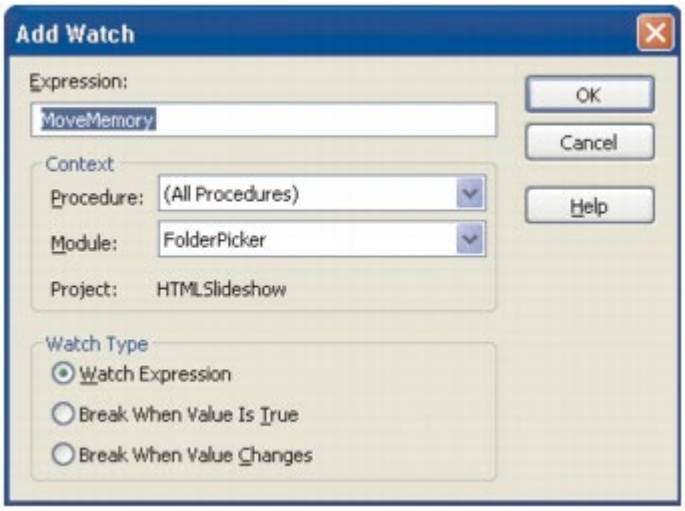


*The **Watches** window*

To add a value to the **Watches** window, do one of the following:

- Select the variable or object and its property, and then drag the selection onto the **Watches** window.
- Click the item, and then click **Debug** } **Quick watch**.

The **Add Watch** dialog appears.



*The **Add Watch** dialog box*

Select the item that you want to watch, select any conditions for this watch, and then click **OK**. If the condition becomes true, the application pauses to let you examine the code.

Setting breakpoints

A breakpoint is a marker in a line of code that causes execution to pause. To continue, you must either restart the execution or step through the subsequent lines of code.

To set or clear a breakpoint, click the line, and then click **Debug** } **Toggle breakpoint**. By default, the line is highlighted in dark red, and a red dot is placed in the margin.

To restart the code after it pauses at a breakpoint, click **Run** } **Continue**. To pause the execution of the code (immediately exiting from all functions and discarding all return values), click **Run** } **Reset**.

You can also “run to cursor” — that is, execute the code until it reaches the line that the cursor is on, and then pause at that line. To do this, click the line where you want execution to pause, and then click **Debug** } **Run to cursor**.

To clear all breakpoints, click **Debug** } **Clear all breakpoints**.

- If the line with the breakpoint (or the cursor, when “running to cursor”) is not executed because it is in a conditional (if-then-else) block, the code does not stop at that line.
- Breakpoints are not saved. They are lost when you close the Macro Editor.

Stepping through the code

When execution pauses at a breakpoint, you can continue through the code one line at a time. This functionality, called “stepping through the code,” lets you do the following:

- examine the values of individual variables after each line
- determine how the code affects the values
- determine how the values affect the code

To step through the code, click **Debug** } **Step into**. The execution advances to every line in all called functions and subroutines.

To step through each line of the current function or subroutine but not through the lines of each called function or subroutine, click **Debug** } **Step over**. The called functions and subroutines are executed, but not line-by-line.

To execute the rest of the current function or subroutine but pause when the function or subroutine returns to the point where it was called, click **Debug** } **Step out**. This technique is a quick way of returning to the point of entry of a function, to continue stepping through the code of the calling function.

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Deleting layers

As previously discussed, each layer is a member of the **Page.Layers** collection for the page on which it appears.

You can delete a layer by calling its **Delete** method. Deleting a layer removes that layer from the document, taking with it all shapes on that layer on all pages in the document.

The following VBA code deletes the layer called “Layer 1”:

```
ActivePage.Layers("Layer 1").Delete
```

If you want, you can use event handlers to respond to events that are triggered by deleting a layer:

- **Document.LayerDelete**

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Deleting pages

You can delete a page by using its **Delete** method, as in the following VBA example:

```
ActivePage.Delete
```

The **Page.Delete** method deletes all shapes on that page, deletes the page from the **Pages** collection for that document, and then updates that collection to reflect the change.

If you want to delete more than one page, you must use the **Delete** method for each unwanted page. However, you cannot delete all pages in a document. You can avoid trying to delete the last remaining page in a document by using the following VBA code:

```
If ActiveDocument.Pages.Count > 1 Then ActivePage.Delete
```

If you want, you can use event handlers to respond to events that are triggered by deleting a page:

- **Document.PageDelete**

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Deleting shapes

If you want, you can use event handlers to respond to events that are triggered by deleting a shape:

- **Document.ShapeDelete**

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Deploying macros

You can deploy macro solutions to users for installation. You can deploy GMS files or workspaces, or both.

Deploying GMS files

Every document has an intrinsic GMS file. For this reason, you can explicitly distribute a macro as part of a document because when that document is opened, the user has immediate access to its macros. This deployment technique lets you, for example, set up a macro to track how much time the user has spent editing a document.

Alternatively, you can distribute the code module that contains the macro. However, this deployment method requires users to manually integrate the code module into an existing project file.

The simplest and most reliable way to deploy a macro project is to use its GMS file. To begin, you must export the GMS file from your computer. Then, each user must import the GMS file by using the **Macro Manager** docker. For more information, see the following procedures:

- To export a GMS file
- To import a GMS file

Deploying workspaces

Some macro solutions include a customized workspace that contains relevant toolbars, menus, and shortcut keys. You can deploy the features of a customized workspace to users by creating a Corel workspace (XSLT) file. You can export a subset of workspace features — such as individual menus, individual toolbars, or complete sets of shortcut keys — if you want users to install only those features, or you can export the entire workspace if you prefer. For more information, see To export workspace features.

Users can install customized workspace features by importing the XSLT files that you provide. For more information, see To import workspace features.

- In CorelDRAW, users can also import workspace features by using the **Application.ImportWorkspace** method.

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Determining shape type

Each **Shape** object has a read-only **Type** property, which returns the shape type (for example, rectangle, ellipse, curve, text, or group). The properties and methods that are available to a shape vary with shape type; therefore, it's a good idea to determine the shape type before applying any properties or methods to that shape.

The following sample VBA code determines whether a shape is text. If the shape is text, the code determines whether it is artistic text or paragraph text. If the shape is artistic text, it is rotated by 10 degrees.

```
Dim sh As Shape
Set sh = ActiveShape
If sh.Type = cdrTextShape Then
If sh.Text.IsArtisticText = True Then
sh.Rotate 10
End If
End If
```

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Displaying documents

You can simultaneously display multiple windows for a single document. For example, a large document can be displayed with one window zoomed in to the upper-right corner of the document and another zoomed in to the lower-right corner. Although the individual windows can be zoomed and panned independently, turning the page in one window affects all windows.

By using the View Manager, you can create views that have individual display settings. Choosing a saved view displays the page according to the settings for that view.

- In VBA, the **Window** object provides access to the windows that contain each **View** object for (or view of) a given document. The **Window** object represents a frame, while the **View** object displays the document inside that frame.

Besides letting you work with windows and views, the application lets you display documents by zooming and panning.

Working with windows

Each **Document** object has a **Windows** collection for displaying that document. To switch between windows, use the **Window.Activate** method:

```
ActiveDocument.Windows(2).Activate
```

The **Document.ActiveWindow** property provides direct access to the active window — that is, the document window that is in front of all other document windows.

The next window and previous window for the active document are referenced in the **Window.Next** and **Window.Previous** properties:

```
ActiveWindow.Next.Activate
```

To create a new window, use the **Window.NewWindow** method:

```
ActiveWindow.NewWindow
```

To close a window (and the document, if it has only one open window), use the **Window.Close** method:

```
ActiveWindow.Close
```

If you want, you can use event handlers to respond to events that are triggered by activating a window:

- **Application.WindowActivate**
- **GlobalMacroStorage.WindowActivate**

You can also use event handlers to respond to events that are triggered by deactivating a window:

- **Application.WindowDeactivate**
- **GlobalMacroStorage.WindowDeactivate**

Working with views

The **Window.ActiveView** property and the **Document.Views** property both represent document views. Each **Window** object has one **ActiveView** object, which represents the current view of the document; saving the display settings for an **ActiveView** object creates a view. In contrast, each **Document** object has a collection of **View** objects in its **Views** property; choosing a **View** object activates the corresponding saved view, which contains the display settings for the corresponding **ActiveView** object.

- The only way to access an **ActiveView** object is from the **Window.ActiveView** property.

You can create a **View** object and add it to a **Document.Views** collection. The following VBA code adds the current **ActiveView** settings to the **Views** collection:

```
ActiveDocument.Views.AddActiveView "New View"
```

You can also create a view with specific settings by using the **Document.CreateView** method. The following VBA code creates a new **View** object that accesses the position (3, 4) in inches, uses a zoom factor of 95%, and displays page 6:

```
ActiveDocument.Unit = cdrInch
ActiveDocument.CreateView "New View 2", 3, 4, 95, 6
```

To apply a saved view to the active window, call the **View.Activate** method:

```
ActiveDocument.Views("New View").Activate
```

Zooming

To zoom an **ActiveView** object by a set amount, set the **ActiveView.Zoom** property by specifying a double value in percent. For example, the following VBA code sets the zoom factor to 200%:

```
ActiveWindow.ActiveView.Zoom = 200.0
```

- You can also zoom by using the following methods of the **ActiveView** class:
  - **SetActualSize**

- **ToFitAllObjects**
- **ToFitArea**
- **ToFitPage**
- **ToFitPageHeight**
- **ToFitPageWidth**
- **ToFitSelection**
- **ToFitShape**
- **ToFitShapeRange**

Panning

To pan an **ActiveView** object, you can move its origin by modifying the **ActiveView.OriginX** and **ActiveView.OriginY** properties. The following VBA code pans the document 5 inches to the left and 3 inches up:

```
Dim av As ActiveView
ActiveDocument.Unit = cdrInch
Set av = ActiveWindow.ActiveView
av.OriginX = av.OriginX - 5
av.OriginY = av.OriginY + 3
```

Alternatively, you can use the **ActiveView**.**SetViewPoint** method:

```
Dim av As ActiveView
ActiveDocument.Unit = cdrInch
Set av = ActiveWindow.ActiveView
av.SetViewPoint av.OriginX - 5, av.OriginY + 3
```

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Duplicating shapes

You can use the **Shape.Duplicate** method to duplicate a shape, and you can use the **ShapeRange.Duplicate** method to duplicate a range of shapes.

```
ActiveSelection.Duplicate
```

The **Duplicate** method provides two optional parameters, **OffsetX** and **OffsetY**, which offset the duplicate from the original (horizontally and vertically, respectively). The following VBA code positions the duplicate two inches to the right and one inch above the original:

```
ActiveDocument.Unit = cdrInch
ActiveSelection.Duplicate 2, 1
```

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cdr

The enumerations in this section begin with the letters *cdr*.

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clr

The enumerations in this section begin with the letters *clr*.

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cui

The enumerations in this section begin with the letters *cui*.

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Enumerations

The CorelDRAW object model contains several enumerations, which are arranged alphabetically in this section.

cdr clr cui pdf Prn

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pdf

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Prn

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A

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B

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C

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D

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E

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L

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Events

The CorelDRAW object model contains several events, which are arranged alphabetically in this section.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

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N

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O

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P

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Q

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S

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W

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Exporting files from documents

Files of all supported formats can be exported.

- Files are exported at the **Document** level because the range of exported objects can extend over multiple layers and multiple pages. However, files are imported at the **Layer** level because each imported object is assigned to a specified layer on a specified page (see Importing files into layers).

The **Document** class has three file-export methods — **Export**, **ExportEx**, and **ExportBitmap** — all of which can export to the bitmap and vector formats.

- The wide selection of supported file formats is due to the vast number of filters that are available to the application. Each filter lets you work with the files from another graphics application. To learn more about working with these filters, see Working with import filters and export filters.

If you want, you can use event handlers to respond to events that are triggered by opening the **Export** dialog box:

- **Application.DocumentBeforeExport**
- **Document.BeforeExport**
- **GlobalMacroStorage.DocumentBeforeExport**

You can also use event handlers to respond to events that are triggered by exporting a document and closing the **Export** dialog box:

- **Application.DocumentAfterExport**
- **Document.AfterExport**
- **GlobalMacroStorage.DocumentAfterExport**

Finally, you can also use event handlers to respond to events that are triggered when the user responds to a request to export a document:

- **Application.QueryDocumentExport**
- **Document.QueryExport**
- **GlobalMacroStorage.QueryDocumentExport**

Understanding the Document.Export method

To export a page, you require only the **Document.Export** method, a filename, and a filter type. The following VBA code exports the current page to a TIFF bitmap file:

```
ActiveDocument.Export "C:\ThisPage.eps", cdrTIFF
```

However, the preceding code gives little control over the output of the image. More control is obtained by including a **StructExportOptions** object, as in the following VBA code:

```
Dim expOpts As New StructExportOptions
expOpts.ImageType = cdrCMYKColorImage
expOpts.AntiAliasingType = cdrNormalAntiAliasing
expOpts.ResolutionX = 72
expOpts.ResolutionY = 72
expOpts.SizeX = 210
expOpts.SizeY = 297
ActiveDocument.Export "C:\ThisPage.eps", cdrTIFF, cdrCurrentPage, expOpts
```

- A **StructPaletteOptions** object can also be included in the function call for palette-based image formats, if you want to provide the settings for automatically generating the palette.

Understanding the Document.ExportEx method

The **Document.ExportEx** method is the same as the **Document.Export** method, except that **ExportEx** can retreive the dialog-box settings for a filter and apply those settings to the export:

```
Dim eFilt As ExportFilter
Set eFilt = ActiveDocument.ExportEx("C:\ThisPage.eps", cdrEPS)
If eFilt.HasDialog = True Then
If eFilt.ShowDialog = True Then
eFilt.Finish
End If
Else
eFilt.Finish
End If
```

Understanding the Document.ExportBitmap method

The **Document.ExportBitmap** method is similar to the **Document.ExportEx** method in that it returns an **ExportFilter** object that can be used to display the **Export** dialog box. However, the **ExportBitmap** method simplifies the coding by taking the individual members of the **StructExportOptions** object as parameters:

```
Dim eFilt As ExportFilter
Set eFilt = ActiveDocument.ExportBitmap("C:\Selection.eps",
```

```
cdrTIFF, cdrSelection, cdrCMYKColorImage, _
210, 297, 72, 72, cdrNormalAntiAliasing, _
False, True, False, cdrCompressionLZW)
eFilt.Finish
```

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Getting started with macros

Now that you understand a bit about automation, you're ready to get started with macros. This section provides an overview of the macro-related tools and features of CorelDRAW

In this section

This section contains the following topics:

- Setting up the automation feature
- Using the Macros toolbar
- Using the Macro Manager docker
- Using the Macro Editor
- Using the VSTA Editor

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Glossary

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

A

accelerator table

A file that contains a list of shortcut keys

The active accelerator table depends on the task that you are performing. (For example, when you highlight text, the Text Editing accelerator table becomes active; if no text is selected, the Main accelerator table is active.)

anti-aliasing

A method of smoothing the curved and diagonal edges in bitmap images

Anti-aliasing partially fills intermediate pixels along curved and diagonal edges to smooth the transition between the edge and the surrounding image. Anti-aliasing reduces or eliminates jagged edges.

argument

See “parameter.”

array

A set of sequentially indexed objects of the same data type (or “array elements”)

Each array element has the same data type (although elements can have different values), and the entire array is stored contiguously in memory (with no gaps between elements). For example, you could have an array of integers or an array of characters or an array of anything that has a defined data type.

By default, array indexes are zero-based.

Arrays can have more than one dimension. A one-dimensional array is called a “vector,” while a two-dimensional array is called a “matrix.”

artistic text

A text type that is used to add single lines of text (such as titles) or to apply graphic text effects (such as fitting text to a path or creating extrusions or blends)

Compare with “paragraph text.”

automation

The process of recording or scripting a macro

B

bevel

An effect that creates the illusion that the edges of an object are cut at an angle other than 90 degrees

You specify the appropriate angle value and depth value for the size of the object being cut. The illusion itself is created through the addition of objects on top of the control object (the object you extruded). These objects work together to give the object a three-dimensional look.

bitmap

An image composed of grids of pixels or dots

bitmap fill

A fill created from any bitmap

black-and-white

A 1-bit color mode that stores images as two solid colors — black and white — with no gradations

The black-and-white color mode is useful for line art and simple graphics.

bleed

The part of the printed image that extends beyond the edge of the page

The bleed ensures that the final image goes right to the edge of the paper after binding and trimming.

blend

A special effect that is created by blending one object with another through a progression of shapes and colors

You can choose from a variety of features for creating interesting blend effects, such as straight-line blends, blends along a path, compound blends, and copied and cloned blends.

brightness

The amount of light that is transmitted or reflected from a given pixel

In the HSB color mode, brightness is a measure of how much white a color contains. For example, a brightness value of 0 produces black, and a brightness value of 255 produces white.

C

CERN

Short for Conseil Europeén pour la Recherche Nucléaire, the scientific laboratory in which the World Wide Web was developed

There are two World Wide Web server systems: CERN and NCSA. Contact your server administrator to find out which system is used by your server.

channel

An 8-bit grayscale version of an image that functions like a plate used in the commercial-printing process

Each channel represents one level of color in an image such that the entire range of colors in the image is produced when all the channels are printed together. For example, an RGB image has three channels (red, green, and blue) that create the entire range of colors in the image when printed or displayed together.

choke

In commercial printing, a form of trapping created by extending the background object into the foreground object

class

The definition of each property, method, and event that applies to a type of object in the application

class module

A type of module that contains the definition of an object-oriented Visual Basic class, including the definitions of the properties and methods for that class

client-side image map

An image map that does not depend on the server to process the map information but instead on the user's browser

Some users may not have a suitable browser to view the map.

clipart

A ready-made image that can be imported into an application (and edited, if required)

Clipboard

A Windows repository for temporarily storing cut or copied information

The Clipboard stores information until it is replaced by information that is subsequently cut or copied.

clone

A copy of an object or a image area that is linked to the original item (or "parent") and consequently inherits most changes made to the parent

CMY

A color mode made up of cyan (C), magenta (M), and yellow (Y)

The CMY color mode is used in the three-color printing process. In Corel applications, the CMY mode is the inverse of the RGB mode, with values ranging from 0 to 255. The CMY color mode is based on the CMY color model.

CMYK

A color mode made up of cyan (C), magenta (M), yellow (Y), and black (K) that produces true blacks and a wide tonal range

In the CMYK color mode, color values are expressed as percentages; therefore, a value of 100 for an ink means that it is applied at full saturation.

Used in most full-color commercial printing, CMYK is similar to CMY, but the addition of black (K) allows for true blacks and a wider tonal range. The CMYK color mode is based on the CMYK color model.

collection

A group of objects that have similar characteristics and similar actions but that are uniquely identified by index names or index numbers

Collections are always plural. For example, **Documents** is a collection of **Document** objects.

color

An effect that is applied to an object to alter the appearance of that object by the way that it reflects light

color cast

See "tint."

color gamut

The range of colors that can be reproduced or perceived by any device

For example, a monitor displays a different color gamut than a printer, so it is necessary to manage colors from original images to final output.

color mode

A system that defines the number and type of colors that make up an image

Examples of color modes include black-and-white, grayscale, RGB, CMYK, and paletted.

color model

A simple color chart that defines the range of colors that is displayed in a color mode

RGB, CMY, CMYK, HSB, HLS, and LAB are examples of color models.

color palette

A collection of solid colors from which you can choose colors for fills and outlines

You can use the on-screen color palette — or the **Select Color** dialog box or the **Color** docker — to choose colors for fills, outlines, and paper. The on-screen color palette is a toolbar that displays a series of color swatches. You can display multiple on-screen color palettes, and you can dock them or leave them floating in the application window.

color profile

A description of the color-handling capabilities and characteristics of a device

color separation

In commercial printing, the process of splitting colors in a composite image to produce a number of separate grayscale images, one for each primary color

in the original image

In the case of a CMYK image, four separations (one for cyan, magenta, yellow, and black) must be made.

color space

In electronic color management, a virtual representation of device or the color gamut of a color model

The boundaries and contours of the color space for a device are mapped by color-management software.

connector line

A straight line that connects two objects in a drawing

Connector lines can be set to remain attached to specified nodes or to shift to the closest nodes on the two connected objects. You can reposition a connector line only by moving the objects it is attached to.

When you use connector lines, special points on each object (called "snap points") are activated. When a mouse passes over a snap point, it becomes visible. Objects can be linked by connector lines only at snap points, which are generally located at each node and at the center of the object.

Unlike normal connector lines that are attached to the nodes of objects, a free connector is an unbound connector line that can be unattached on a layer. A free connector line is not attached to any object.

constant

A value in an automation-programming structure that remains fixed while the macro is being executed

Unlike a variable, which temporarily stores a changing data value in a code procedure or code function, constant values do not change.

A constant is an instance of an enumeration.

contrast

The difference in tone between the dark areas and light areas of an image

Higher contrast values indicate greater differences and fewer gradations between dark and light.

control point

A point that extends from a node along a curve that is being edited with the **Shape** tool

Control points determine the angle at which the curve passes through the node.

coordinate

A parameter used to specify a location on a page

Coordinate values use the set document unit as their base unit of measurement. In addition, coordinate values are expressed as being relative to the center of the current page, which has the coordinates (0,0).

Most commands that use coordinates (such as the **CreateSubPath** command) are affected by the current reference point for the application — that is, the point on the bounding box for the selected object on which the coordinates operate. You can set the current reference point by using the **ReferencePoint** property.

Corel SCRIPT

A type of application command that comprises any script that you create by saving a recording of your operations

Corel SCRIPT application commands instruct the application to perform specific actions. For example, a command might instruct the application to open or to close a document. The application commands are easy to understand, since most are one-word equivalents of the user interface for the application.

Corel SCRIPT programming statements and functions send instructions or perform actions that are not part of another Corel application. For example, Corel SCRIPT programming statements can be used to display a customized dialog box; to include statements and constructs for flow control, such as loops; to create and manipulate variables; or to retrieve information about your computer setup. On their own, Corel SCRIPT programming statements form a powerful programming language. A script that contains only Corel SCRIPT programming statements can be executed even if another Corel application is not running.

cropping

The act of cutting unwanted areas of an image without affecting the resolution of the information that remains

cubist

An abstract style of art that stresses several aspects of the same object simultaneously, generally in the form of squares or cubes

curve

An object that can be any shape

Curve objects have nodes and control points that you manipulate to change the shape of the object. Curve objects can be drawn by using the curve tools; however, you can use the **Arrange** } **Convert To Curves** command to create curve objects from text or from objects that are drawn by using the **Rectangle**, **Ellipse**, or **Polygon** tools.

cusp node

A node with control points that move independently from each other

A curve that passes through a cusp node can bend at a sharp angle.

custom color map lens

A lens that changes all the colors of objects behind it to a color range between any two colors you select

Areas of the lens that do not cover other objects are filled with the color at the end of the color map

custom palette

A color palette in which the user can save a collection of colors

When you create a custom palette, the palette is empty and ready for you to choose the colors you want to include in it. You can create a custom palette that includes all colors in a selected object or in the current document; you can also copy a color swatch to a custom palette by dragging the color swatch from any color palette.

A color palette is saved as a CPL file. There is no limit to the number of custom palettes that you can create.

D

dash

A broken line segment in an outline style

destination application

In Object Linking and Embedding (OLE), the application in which you insert a linked or embedded object

distortion

A lack of proportionality in an image that results from defects in the optical system, or an effect (Push and pull, Twister, or Zipper) that can be applied to simulate such a distortion

distortion handle

One of the outward-facing, double-headed arrows located at each corner of the highlighting box

dithering

The process of simulating color by putting dots of another color very close together

The Windows operating system uses dithering to display colors that the graphics adapter cannot display.

document

A single-page drawing or a collection of single-page drawings that make up a multi-page document

A document page is sometimes called a “page.” You can access any page in a multi-page document by clicking a page tab in the Navigator of the drawing window. Each page in a multi-page document is accessible through the Navigator, located in the bottom-left corner of the drawing window. The Navigator allows you to move through your document quickly and shows the total number of pages in your drawing and the number of the page that is currently displayed. You can move to a page in your document by clicking that page number in the Navigator.

dpi

Short for “dots per inch,” a measure of printer resolution

Typical desktop laser printers print at 300 dpi. Image setters print at 1270 or 2540 dpi. Printers with higher dpi capabilities produce smoother and cleaner output. The term dpi is also used to measure scanning resolution and to indicate bitmap resolution.

Draft view

A view-quality setting that shows uniform fills and low-resolution bitmaps, displaying lenses and fountain fills as colors

Draft view also displays a unique pattern to represent each fill. The checkerboard pattern represents two-color fills. The two-way arrow pattern represents full-color fills. The hatched line pattern represents the bitmap fill. The PS pattern represents the PostScript fill.

View-quality settings affect how a drawing is displayed on the monitor, not the actual size of the drawing.

drawing page

The portion of the drawing window that appears on the printed page

The drawing page is enclosed by a rectangle with a shadow effect. Although you can draw anywhere in the drawing window, only objects on the drawing page appear in your print jobs.

drawing window

The area in the application window that contains a drawing

You can draw anywhere in the drawing window, but only the objects on the drawing page appear in your print jobs.

drop shadow

An effect that applies an offset dark area, in the shape of the object, around that object

When you apply a drop shadow to an object, you create a linked group that has an illusion of depth between the objects. The object to which you applied the drop shadow is the control object. Changes you make to the control object are reflected in the drop shadow. For example, if you change the size of the of the control object, the size drop shadow changes to reflect the size of the control object. However, you can separate the drop shadow (which is a bitmap) from the control object.

duotone

An 8-bit color mode that displays images using 256 shades of up to four tones

An image in the duotone color mode is simply a grayscale image that has been enhanced with one to four additional colors.

E

effect

A feature for altering or adjusting the appearance of an object

Special effects let you alter the appearance of objects. You can distort objects, add new elements to objects, and change the relationship between objects by applying blends, distortions, envelopes, extrusions, perspectives, drop shadows, transparencies, lenses, contours, and PowerClip objects.

You can apply effects to both artistic text and paragraph text. However, some effects are exclusive to artistic text and others to paragraph text because the application treats the two text types differently. Effects you can apply to artistic text include blends, contours, distortions, drop shadows, envelopes, extrusions, lenses, perspectives, and PowerClip objects. Effects you can apply to paragraph-text frames include envelopes, drop shadows, and PowerClip objects.

ellipse

An object that is created by using the **Ellipse** tool

In its simplest form, an ellipse is a variation of a circular shape object. An ellipse is a closed plane curve that is generated by a point moving in such a way that the sums of its distances from two fixed points are constant.

An ellipse is type of curve object.

Enhanced view

A view-quality setting that uses 2× over-sampling to ensure the best possible display quality

View-quality settings affect how a drawing is displayed on the monitor, not the actual size of the drawing.

enumeration

Also called an “enumerated type,” a data type that lists all possible values for the variables that use it

Unlike a variable, which temporarily stores a changing data value, an enumeration stores fixed values.

A constant is an instance of an enumeration.

enumerated type

See “enumeration.”

envelope

A feature that is applied to an object to let you distort its shape

Distortion is created by dragging nodes on an imaginary box (the envelope), which in turn distorts the object.

event

An action that takes place in an object and that is recognized by a form or control

Each object within an object model is defined by a property, method, event, or a combination of each. An event is triggered by an action — such as a click, key press, or system timer — and you can write code that causes an object to respond to that event.

event-driven programming

A style of programming, unlike traditional procedural programming (in which the program starts at line 1 and executes line by line), that executes code in response to events

Visual Basic for Applications is an event-driven programming language. Most of the code you create is written to respond to an event.

Compare with “object-oriented programming.”

event handler

A subroutine that is programmed to cause the application to respond to a specific event

expression

In programming, any legal combination of symbols that represents a value

extrusion

An effect applied to two-dimensional objects that gives them a three-dimensional look by creating the illusion of depth

There are two types of extrusions: vector and bitmap.

Eyedropper tool

A tool lets you use the mouse to select a fill from an object on the drawing window

F

feathering

The level of sharpness along the edge of a drop shadow

filter

An application that translates digital information from one form to another

fixed palette

A color palette that is provided by a third-party manufacturer

Fixed color palettes are most useful when accompanied by a color swatch book.

form

A type of module that is used for customized dialog boxes and user interfaces, and that includes the code to control them

fountain fill

Also called a “gradient” or “graduated fill,” a complex fill that displays a progression between two colors that follow a linear, radial, conical, or square path

You can create a direct blend from one color to another or a cascade of different colors. You can also use preset fountain fills to create neon tubes, metal cylinders, and other effects.

fractal

A mathematical way of generating an image

function

A procedure that performs a given task in a macro and that can be used to return a value

A function procedure begins with a `Function` statement and ends with an `End Function` statement. In VBA and VSTA, functions do not need to be declared before being used, nor before being defined.

G

gap

A space between dashes in an outline style

Gaussian

A frequency distribution that uses bell-shaped distribution curves rather than straight lines

global value

A value that applies to a given project in its entirety

GMS file

Also called a “project file” (and short for “Global Macro Storage file”), the location to which the Macro Editor stores all modules for a project

gradient fill

See “fountain fill.”

gradient node

A square point that represents each color on the gradient arrow of a fountain fill

A gradient node is used to change the start point and end point, colors, and transparency values of the gradient.

graduated fill

See “fountain fill.”

grayscale

A color mode that displays images using 256 shades of gray

Each color is defined as a value between 0 and 255, where 0 is darkest (black) and 255 is lightest (white).

grayscale image

An image that uses the grayscale color model, which can display up to 256 shades of gray ranging from white to black

Grayscale images, especially photographs, are commonly referred to as “black and white.”

group

A set of objects that behave as one unit, such that any operations that you perform on the group apply equally to each of its objects

guideline

A line that is placed anywhere in the drawing window and that is used to help align and position objects

Guidelines are used with the rulers and the grid to draw objects with precision. Guidelines can be horizontal, vertical, or slanted, and any number of them can be created and saved with your drawing.

H

halftone

An image that has been converted from a continuous tone image to a series of dots of various sizes to represent different tones

handle

One of eight black squares that appear at the corners and sides of an object when the object is selected

By dragging individual handles, you can scale, resize, or mirror the object. If you click a selected object, the shape of the handles changes to arrows so that you can rotate and skew the object.

highlighting box

A rectangle with eight handles that encloses a selection in an image

HLS

A color model that defines three components: hue, lightness, and saturation

HSB

A color model that defines three components: hue, saturation, and brightness

Hue determines color (yellow, orange, red, etc.), while brightness determines perceived intensity (lighter or darker color), and saturation determines color depth (from dull to intense).

HTML

Short for HyperText Markup Language, the World Wide Web authoring standard comprised of markup tags that define the structure and components of a document

The HTML markup tags are used to tag text and integrate resources (such as images, sound, video, and animation) when you create a Web page.

hue

The property of a color that allows it to be classified it by name

Hue is the name we give a color in everyday language. Hues form the color wheel. The hue of a lemon is yellow, and that of a strawberry is red.

hyperlink

An electronic link that provides access directly from one place in a document to another place in that document or in another document

I

image map

A hypergraphic in an HTML document that contains clickable areas that link to URLs on the World Wide Web

imagesetter

A machine that makes film or film-based paper used to make printing plates

index number

A reference to an object in a collection that contains more than one object

An index number is used to identify each object in a collection. The index number can range from 1 to the number of available objects within the collection.

intensity

A measure of the brightness of the light pixels in a bitmap image compared with the darker mid-tones and dark pixels

An increase in intensity increases the vividness of whites while maintaining true darks.

interlaced video image

A video image that takes two passes to fill a screen, painting every other line in each pass, such that it may produce a flicker

interlacing

A method that lets you display an image on screen at a low, blocky resolution

As the image data loads, the image quality improves.

J

JPEG

Short for Join Photographic Experts Group, a format for compressed photographic images that offers compression with minimal loss of image quality

Because of their compression (20 to 1) and small file size, JPEG images are widely used in Internet publishing.

K

L

LAB (CIE L*a*b)

A color model that contains a luminance (or lightness) component (“L”) and two chromatic components: “a” (green to red) and “b” (blue to yellow)

In 1931, the Commission Internationale de l’Eclairage (CIE) defined the device-independent LAB color model, based on how the human eye perceives color. The CIE LAB model incorporates the theory that a color cannot be both green and red, nor both blue and yellow, at the same time. As such, single values are used to describe the green/red and blue/yellow components of any color. LAB provides a system for defining color that bases color values on widely accepted standards rather than on individual color-producing devices.

The luminance value (L) ranges from 0 to 100 , while the two chromaticity values (a and b) range from +120 to -120.

layer

One of a series of transparent planes on which you can place objects in a document

You can control how objects in your document overlay one another by moving the layer and the objects they contain. You can also lock layers and make them invisible and nonprintable. Use layers to help you organize different components of a complex document.

When activated, a layer is ready to receive any new objects you draw, import, or paste onto it.

lens

An effect that changes the way that you perceive objects behind the lens, not the actual properties and actual attributes of those objects

You can apply a lens to almost any closed shape that has been created by using the drawing tools. You can choose amongst several types of lenses, each producing distinctive results. These results range from color alteration (for example, as produced by heat-map, inverting, and brightening lenses) to distortion (for example, as produced by magnifying and fish-eye lenses).

lightness

The perceived intesity (lightness or darkness) of a color

looping

When referring to blends, the act of rotating a blend halfway between the centers of rotation for the start object and end object

M

macro

A recorded or scripted set of tasks that can be repeatedly invoked within an application

A macro is a symbol, name, or key that represents a list of commands.

master layer

A layer that contains information that you want to appear on every page of a multi-page document

For example, you can use a master layer to place a header or footer on every page.

master object

An object that has been copied by using the **Clone** command

Most changes you make to the master object are automatically applied to the clone.

master page

A page that contains information that you want to appear on every page of a multi-page document

method

An operation that an object can have performed on itself

mixed node

A node with control points that are a combination of cusp nodes, smooth nodes, and symmetrical nodes

modal dialog box

A dialog box thatlocks the application and must be acted upon (that is, either submitted or cancelled) before the macro can be resumed

Most built-in dialog boxes that can be controlled by automation coding are modal.

modeless dialog box

A dialog box that does not lock the application and can be left open while the user continues working in the application

Modeless dialog boxes behave like dockers.

module

A container that is used by a GMS file for storing project components

Generic modules are used for general code and for macros. Other types of modules include forms and class modules.

mosaic

Decorative artwork that is made by arranging small pieces of variously colored material to form pictures or patterns

N

NCSA

Short for National Center for Supercomputing Applications, one of two World Wide Web servers.

If you are creating an image map, you need to know whether the server you are using runs CERN or NCSA because different codes are required for each. Contact your server administrator to learn whether you are using CERN or NCSA.

node

A square point at the end of a line segment or curve segment

You can change the shape of a line or curve by dragging one or more of its nodes.

There are four types of nodes: cusp, smooth, symmetrical, and mixed.

Normal view

A view-quality setting that shows all fills, all objects, and high-resolution bitmaps

View-quality settings affect how a drawing is displayed on the monitor, not the actual size of the drawing.

NTSC

Short for National Television Standards Committee, a video color filter that is commonly used to define the gamut of colors that is supported by television monitors in North America

nudging

The act of moving an object in increments

See also “super nudge.”

O

object

When referring to an object model, an instance of a class

When referring to a drawing, a generic term for any item — such as a line, a shape, a graphic, or text — that you create or place in that drawing

object model

A high-level structure of the relationship between the parent objects and child objects in an application

For example, the **Application** object represents the beginning of the object hierarchy. From the **Application** object, you can “drill down” and navigate through the object model until you find the desired object. To reference an object with Visual Basic code, you separate each level of the object hierarchy with the dot operator (**.**).

object-oriented programming

A style of programming that places emphasis on creating and using objects

Compare with “event-driven programming.”

offset

A movement out of line with another object

An offset object is distanced from another specified object, and this distance is the offset value.

opacity

The inability to see through an object

If an object is 100% opaque, you cannot see through it. Opacity levels under 100% increase the ability to see through objects.

Compare with“transparency.”

P

page

Also called a “document page”

See “document.”

PAL

A video color filter that is commonly used to define the gamut of colors that is supported by television monitors in Europe and Asia

palette

See “color palette.”

paletted

An 8-bit color mode that displays images using up to 256 colors

You can convert a complex image to the paletted color mode to reduce file size and to control the colors used throughout the conversion process more precisely.

PANTONE process colors

The colors that are available through the PANTONE process color system, which is based on the CMYK color model

paragraph text

A text type that is used to add large blocks of content for text-intensive projects (such as ads or brochures)

Paragraph text is contained in a paragraph-text frame.

Compare with “artistic text.”

paragraph-text frame

A rectangle that contains a block of paragraph text

A text frame can exist alone or be linked to other text frames. When you link paragraph-text frames, the text flow is directed from one frame to the other if the amount of text exceeds the original frame. When you shrink a frame, enlarge a frame, or change text size, the amount of text in the next frame adjusts automatically.

parameter

Synonymous with “argument,” a value that is passed to a routine and that defines a characteristic of an object in the Visual Basic programming environment

Parameters are attributes that appear after a recorded command in the **Recorder** docker. For example, dialog-box options are not recorded as separate commands in the **Recorder** docker; they are recorded as attributes of the command that initially invoked the dialog box.

passing by reference

The act of passing an argument to a function or subroutine by using a reference to the original

By default, function parameters and subroutine parameters are passed by reference. To explicity indicate that you want to pass an argument by reference, prefix the argument with `ByRef`.

passing by value

The act of passing an argument to a function or subroutine by using a copy of the original

To explicity indicate that you want to pass an argument by value, prefix the argument with `ByVal`.

path

The basic component from which objects are constructed

Paths can be open (as in a line or curve) or closed (as in a circle or polygon). Paths can also constitute a single line segment or curve segment or many joined segments.

perspective

An effect that creates the illusion of distance and depth, achieved by lengthening or shortening sides of an object

Although objects are two-dimensional, applying one and two-point perspective adds another dimension to your drawing. By creating one-point perspective, you can make an object look like it is receding from view in one direction. By creating two-point perspective, on the other hand, you can make the object look like it is receding from view in two directions.

perspective handle

One of the hollow circles in the corners of the highlighting box

Pick tool

A tool that lets you select and transform objects

pixel

A colored dot that is the smallest part of a bitmap

See also “resolution.”

point

A unit of measurement used primarily in typesetting to design type sizes

There are approximately 72 points (pts) to an inch and 12 points to a pica.

PowerClip

A feature that lets you place objects (called “contents objects”) inside other objects (called “container objects”)

If the contents object is larger than the container object, the contents object is automatically cropped. Only the contents that fit inside the container object are visible.

You can create a container from any object you create using the application, including shapes, lines, curves, artistic text, and groups. A contents object can be any object you create in the application or import from another program.

pressure-sensitive pen

A pen that you can use to access commands and draw shapes

You must install the pressure-sensitive pen, along with a pressure-sensitive tablet and its corresponding Windows drivers, to use it.

process color

In commercial printing, a color that is produced from a blend of cyan, magenta, yellow, and black

A process color is different from a spot color.

project file

See "GMS file."

property

A characteristic of a class

Properties can be returned or set. In addition, properties can be designated as read-only (to indicate that they are fixed by the design of the class).

Q

R

range

A series of similar objects

range sensitivity

A paletted color mode option that lets you specify a focus color for the paletted conversion

You can adjust the color and specify its importance to guide conversion.

raster graphic

An image that has been rendered into pixels

When you convert vector graphic files to bitmap files, you create rasterized images.

recording

A series of commands that you record in the **Recorder** docker

Recordings let you automate a series of actions to repeat on the same image or on several different images. Recordings are not saved when you end your session.

rendering

The act of capturing a two-dimensional image from a three-dimensional model

resampling

The act of changing the resolution or size of an image to change the number of pixels that it contains

Upsampling increases the resolution and increases the number of pixels, while downsampling reduces the resolution and decreases the number of pixels.

resolution

The amount of detail and information that an image file contains, as well as the level of detail that an input, output, or display device is capable of producing

When you work with bitmaps, resolution affects both the quality of the final output and the file size.

RGB

A color mode in which the three colors of light (red, green, and blue) are combined in varying intensities to produce all other colors

A value between 0 and 255 is assigned to each channel of red, green and blue. Monitors, scanners, and the human eye use RGB to produce or detect color.

Higher RGB values correspond to the presence of greater quantities of white light. Consequently, higher RGB values result in lighter colors. When all 3 components are at the maximum value, the resulting color is white light. Because the RGB model creates colors by adding light, it is considered an additive color model. Monitors and scanners can employ the additive color model because they emit particles of red light, green light, and blue light to create the illusion of millions of different colors.

One of the limitations of the RGB model is that it is device-dependent. This means that not only are there color variations between monitors and scanners by different manufacturers, but there are color variations between identical devices from the same manufacturer. Over time, all monitors "drift" and display colors differently, so it is imperative to regularly calibrate monitors and the other electronic devices. The RGB model cannot be a color standard because its color results are not 100% repeatable.

rotation handle

One of the curved, double arrows in the corners of the highlighting box

S

saturation

The purity or vividness of a color, expressed as the absence of white

A color that has 100% saturation contains no white. A color with 0% saturation is a shade of gray.

scanner

A device that converts images on paper, transparency, or film to digital form

Scanners produce bitmaps or raster graphics.

scope

The visibility of a data type, procedure, or object

script

A recording that has been saved to disk and that can be retrieved at any time

Scripts let you automate a series of actions to repeat on the same image or on several different images. Both a recording and a script are created, edited, and played back by using the controls and commands in the **Recorder** docker.

segment

A line or curve between nodes in a curve object

There are two types of segments: straight and curved. If you click on a node with the **Shape** tool, the status bar displays the type of segment entering the node and the node type itself.

shape

An object that can be displayed as several variations of a rectangle or ellipse

shortcut object

A syntactic replacement for the longhand version of an object

Simple Wireframe view

A view-quality setting that shows objects as outlines; hides fills, extrusions, contours, and intermediate blend shapes; and shows monochrome bitmaps

Editing a drawing in Simple Wireframe view is faster because only the object outlines must be refreshed.

View-quality settings affect how a drawing is displayed on the monitor, not the actual size of the drawing.

skewing handle

One of the straight, double-headed arrows located in the center of each side of the highlighting box

smooth node

A node with control points that are always opposite each other (such that when one point moves, the other point also moves) and that can have different lengths

The control points for a smooth node produce a smooth transition between segments.

snap point

A coordinate point generally located at each node in an object and at the center of that object

Two snap points can be connected by using a connector line.

Snap points can also be free — that is, they can stand alone as a single point without being part of another object.

A snap point is defined by its x and y positions, which are established by the rulers and the document unit of measurement.

source application

In Object Linking and Embedding (OLE), the application used to create the source file

source file

In Object Linking and Embedding (OLE), the file that contains the object that you want to link or embed in a Corel application

special effect

See “effect.”

spot color

In commercial printing, a solid ink color that prints individually, one plate per spot color

spread

A type of trap that is created by extending the foreground object into the background object

stacking order

The sequence in which objects are created in the document window

The stacking order determines the relationship between objects and, therefore, the appearance of your document. The first object you create appears on the bottom; the last object appears on the top.

String

A data type consisting of a sequence of contiguous characters that represent the characters themselves rather than their numeric values

A string can include letters, numbers, spaces, and punctuation. The `String` data type can store fixed-length strings ranging in length from 0 to approximately 63K characters and dynamic strings ranging in length from 0 to approximately 2 billion characters. The dollar sign (**$**) type-declaration character represents a string in Visual Basic.

stylus

A pressure-sensitive pen device that is used in conjunction with a tablet to let you to draw paint strokes by using hand pressure

sub

See “subroutine.”

subpath

A curve or shape within a single curve object

subroutine

Sometimes called a “sub,” a procedure that performs a given task in a macro but cannot be used to return a value

A subroutine procedure begins with a `Sub` statement and ends with an `End Sub` statement. In VBA and VSTA, subroutines do not need to be declared before being used, nor before being defined.

subtractive color model

A color model, such as CMYK, that creates color by subtracting wavelengths of light reflected from an object

For example, a colored ink appears blue if it absorbs all colors except blue.

super nudging

The act of moving an object in large increments

The super nudge value is multiplied by the nudge value to obtain the distance by which the object is moved.

swap disk

Hard-drive space that is used by applications to store temporary files not in use, thereby artificially increasing the amount of memory available to the computer

swatch

One of a series of solid-colored patches used as a sample when selecting a color, such as from the **Color** palette

A printed booklet of swatches is called a “swatch book”.

swatch book

A collection of color samples that shows what each color looks like when printed

symmetrical node

A node with control points that are always opposite each other but (unlike for smooth nodes) are always the same length

The control points for a symmetrical node produce the same curvature on both sides of the node.

T

text style

A variation in a typeface or font

Some common text styles include Roman (regular or normal), bold, italic, and bold italic.

threshold

When referring to bitmaps, a level of tolerance for tonal variation

For example, when you convert an image to the black-and-white color mode, the bitmap threshold determines how many tonal values are converted to black and how many to white.

thumbnail

A miniature, low-resolution version of an image or illustration

tint

Also called a “color cast,” a semitransparent color applied over an image

transparency

The ability to see through an item

The opposite of transparency is opacity. Setting lower levels of transparency causes higher levels of opacity and less visibility of the underlying items or image.

trimming

The act of creating new, irregularly shaped objects by removing the areas that overlap other selected objects

Before you trim objects, you must decide which object you want to trim (the target object) and which object(s) you want to use to trim it. The objects you use to trim must overlap — or be overlapped by — the target object.

true color

Digital RGB color that is composed of 24-bits or 16.7 million colors

TWAIN

An interface that allows Corel graphics applications to acquire an image from imaging hardware such as a scanner or a digital camera, without using any additional applications

By using the TWAIN driver supplied by the manufacturer of the imaging hardware, Corel graphics applications can acquire images directly from a scanner or a digital camera.

U

URL

Short for Uniform Resource Locator, a unique address that defines where a Web page is located on the Internet

V

vanishing point

A marker that appears when you select an extrusion or an object to which perspective has been added

With an extrusion, the vanishing point marker indicates the depth (parallel extrusion) or the point at which the extruded surfaces would meet if extended (perspective extrusion). With the **Perspective** effect, the marker indicates the point (or points) at which the nonparallel lines would meet.

variable

An item that can be created (or "declared") for the purposes of storing data

The built-in data types are `Boolean`, `Double`, `Integer`, `Long`, `Single`, String, Variant, and several other less-used types including `Date`, `Decimal`, and `Object`. If a variable is not declared before being used, the compiler interprets it as a `Variant`.

Variant

The data type for all variables that are not declared as another type, such as `Dim`, `Private`, `Public`, or `Static`

The `Variant` data type has no type-declaration character.

VBA

A built-in programming language that can automate repetitive functions and create intelligent solutions in a software application

VBA is a subset of the Microsoft Visual Basic (VB) object-driven programming environment, but it is considered "for applications" because it is most often integrated into another application to customize the functionality of that application.

vector graphic

An image that is generated from mathematical descriptions that determine the position, length, and direction in which its lines are drawn

Vector graphics are created as collections of lines rather than as patterns of individual dots (or "pixels").

view

A customized way of looking at a drawing in the application

You can customize your views to suit your viewing needs. These views can be saved, deleted, edited, and renamed according to your viewing preferences.

Several preset view-quality settings are provided: Simple Wireframe, Wireframe, Draft, Normal, and Enhanced.

viewpoint

A mark representing the center point of what is being viewed through a lens

A viewpoint is marked by an "X" in the drawing window.

Visual Basic for Applications

See "VBA."

Visual Studio Tools for Applications

See "VSTA."

VSTA

The successor to VBA

VSTA is based on Microsoft Visual Studio 2012. The integrated development environment (IDE) for VSTA can be used to support two additional programming languages (Visual Basic .NET and C#) and to take advantage of the .NET framework natively.

W

welding

The act of creating a single curve object out of two or more objects

If the objects overlap, the result is a single object with one outline. If the objects do not overlap, they form a weld group in which the objects appear separate but are actually treated as one object.

window

See "drawing window."

Wireframe view

A view-quality setting that display object in skeleton form, without fills or outlines

Because the screen redraws faster in this view, you may want to use it when you edit complex drawings.

View-quality settings affect how a drawing is displayed on the monitor, not the actual size of the drawing.

workspace

An enclosed work environment that is ideally suited to meet specific needs when creating a drawing in the application

You can create multiple workspaces for specific users or specific tasks. You can also delete workspaces when they are no longer needed.

X

Y

YIQ

A color model used in television broadcast systems (NTSC video standard)

Colors are split into a luminance value (Y) and two chromaticity values (I and Q). In monitor calibration, chromaticity refers to the chroma (hue adjustment) of your monitor. On a color monitor, all three components are visible. On a monochrome monitor, only the Y component is visible. All values are scaled from 0 to 255.

Z

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Referencing collections

Many objects are members of collections. A collection is similar to an array, except that it contains objects rather than values. However, members of collections can be accessed in the same way as arrays. For example, a collection that is used frequently is the collection of shapes on a layer: The object `ActiveLayer` references either the current layer or the layer that is selected in the **Object Manager** docker.

CorelDRAW contains many collections: A document contains pages, a page contains layers, a layer contains shapes, a curve contains subpaths, a subpath contains segments and nodes, a text range contains lines and words, a group contains shapes, and the application contains windows. All these collections are handled by VBA in the same way.

Referencing collection items

To reference the shapes on a layer, the collection of shapes for that layer is used: `ActiveLayer.Shapes`. To reference the individual shapes in the collection, the `Item()` property is used. Here is a VBA example:

```
Dim sh As Shape
Set sh = ActiveLayer.Shapes.Item(1)
```

Most elements of a collection start at `1` and increase. For the collection `ActiveLayer.Shapes`, `Item(1)` is the item at the “top” or “front” of the layer — in other words, it is the item that is in front of all other shapes. Furthermore, because each item in the `ActiveLayer` collection is an object of type `Shape`, you can reference any item in VBA merely by appending the appropriate dot-notated member:

```
ActiveLayer.Shapes.Item(1).Outline.ConvertToObject
```

Sometimes, individual items have names. If the item you are looking for has an associated name (and you know what the name is and which collection the item is in), you can use that name to reference the item directly, as in the following VBA example:

```
Dim sh1 As Shape, sh2 As Shape
Set sh1 = ActiveLayer.CreateRectangle(0, 5, 7, 0)
sh1.Name = "myShape"
Set sh2 = ActiveLayer.Shapes.Item("myShape")
```

Also, because an item is usually the implied or default member of a collection, it is not strictly required. For this reason, the last line of the preceding VBA code can be rewritten as follows:

```
Set sh2 = ActiveLayer.Shapes("myShape")
```

Counting collection items

All collections have a property called `Count`. This read-only property gives the number of members in the collection, as in the following VBA example:

```
Dim count As Long
count = ActiveLayer.Shapes.Count
```

The returned value is not only the number of items in the collection: Because the collection starts from `1`, it is also the index of the last item.

Parsing collection items

It is often necessary to parse through the members of a collection to check or change the properties of each item.

By using the `Item()` and `Count` members, it is straightforward to step through a collection of items. With each iteration, it is possible to test the properties of the current item, or to call its methods. The following VBA code restricts all shapes on the layer to no wider than ten units:

```
Dim I As Long, count As Long
count = ActiveLayer.Shapes.Count
For I = 1 to count
If ActiveLayer.Shapes.Item(i).SizeWidth > 10 Then
ActiveLayer.Shapes.Item(i).SizeWidth = 10
End If
Next I
```

There is, however, a more convenient way of parsing a collection in VBA. Instead of using the `Count` property and a **For-Next** loop, this technique uses a **For-Each-In** loop:

```
Dim sh As Shape
For Each sh In ActiveLayer.Shapes
If sh.SizeWidth > 10 Then
sh.SizeWidth = 10
End If
Next sh
```

If you want to copy the selection and then parse it later when it is no longer selected, copy the selection into a `ShapeRange` object:

```
Dim sr As ShapeRange
Dim sh As Shape
Set sr = ActiveSelectionRange
For Each sh In sr
' Do something with each shape
Next sh
```

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Including comments

Comments in VBA — similarly to in ANSI, C++, and Java — can be created only at the end of a line. Comments begin with an apostrophe (') and terminate at the end of the line.

Each line of a multi-line comment must begin with its own apostrophe in VBA:

```
a = b ' This is a really interesting piece of code that
' requires so much explanation that I needed to break
' the comment over multiple lines.
```

To comment out large sections of VBA code, use the following syntax (similarly to in C or C++):

```
#If 0 Then ' That's a zero, not the letter 'oh'.
' All this code will be ignored by
' the compiler at run time!
#End If
```

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Providing event handlers

While running, CorelDRAW raises various events to which macros can respond through the use of event handlers — subroutines with specific, defined names. Each macro project defines its event handlers in one of the following code modules:

- **ThisMacroStorage** — for macro projects that are stored in Global Macro Storage (GMS) files
- **ThisDocument** — for macro projects that are stored in documents

The **GlobalMacroStorage** object is a virtual object that represents each and all open documents. The **GlobalMacroStorage** object has several events that are raised at the time of any event, such as opening, printing, saving, or closing a document (although the range of events is actually greater than this because each one has a “before” and “after” event).

To respond to an event, you must provide an event handler — a subroutine in any **ThisMacroStorage** module with a specific name for which the application is pre-programmed to search. However, the application does check all **ThisMacroStorage** modules in all installed projects; for this reason, you can create an event-driven solution and distribute it as a single project file just as you would provide any other solution. Each project can have only one **ThisMacroStorage** module, and it is automatically created when the project is first created.

In VBA, you can add event handlers to a **ThisMacroStorage** module by using the Macro Editor. For example, a CorelDRAW macro solution may need to respond to the closing of a document by logging the closure in a file as part of a workflow-management system. To respond to the opening of a document, the solution must respond to the `OpenDocument` event for the `GlobalMacroStorage` class. To create this event handler in VBA, do the following:

- Open a **ThisMacroStorage** module for editing in the Macro Editor.
- Next, choose **GlobalMacroStorage** from the **Object** list box at the top of the **Code** window, and then choose **DocumentOpen** from the **Procedure** list box.

The Macro Editor creates a new, empty subroutine called `GlobalMacroStorage_DocumentOpen()` — or, if that subroutine already exists, the Macro Editor places the cursor into it. To then add the name of the opened file to the log, you need only write some code. To reduce the size of the **ThisMacroStorage** module, you can assign this event-logging task to a public subroutine in another module. This technique lets the run-time interpreter more easily parse all the **ThisMacroStorage** modules each time an event is raised. The following VBA code illustrates this example for CorelDRAW:

```
Private Sub GlobalMacroStorage_OpenDocument(ByVal Doc As Document, _
ByVal FileName As String)
Call LogFileOpen(FileName)
End Sub
```

Here is a small sample of the events available in CorelDRAW:

Event	Description
`Start`	Raised when the user starts the application
`DocumentNew`	Raised when a document is created; passes a reference to the document
`DocumentOpen`	Raised when a document is opened; passes a reference to the document
`DocumentBeforeSave`	Raised before a document is saved; passes the file name of the document as a parameter
`DocumentAfterSave`	Raised after a document is saved; passes the file name of the document as a parameter
`DocumentBeforePrint`	Raised before the **Print** dialog box is displayed
`DocumentAfterPrint`	Raised after a document is printed
`SelectionChange`	Raised when a selection changes
`DocumentClose`	Raised before a document is closed
`Quit`	Raised when the user quits the application

- Event handlers for frequent events — such as events related to the **Shape** class — should be as efficient as possible, to keep the application running as quickly as possible.

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Using functions and subroutines

VBA uses both functions and subroutines (or “subs”). Functions can be used to return a value, but subs cannot.

In VBA, functions and subs do not need to be declared before they are used, nor before they are defined. In fact, functions and subs need to be declared only if they actually exist in external system DLLs.

Typical functions in a language such as Java or C++ can be structured as follows:

```
void foo( string stringItem ) {
// The body of the function goes here
}
double bar( int numItem ) { return 23.2; }
```

In VBA, however, functions are structured as in the following example:

```
Public Sub foo (stringItem As String)
' The body of the subroutine goes here
End Sub
Public Function bar (numItem As Integer) As Double bar = 23.2
End Function
```

To force a function or sub to exit immediately, you can use `Exit Function` or `Exit Sub` (respectively).

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Ending lines

In VBA, each statement must exist on its own line, but no special character is required to denote the end of each line. (In contrast, many other programming languages use a semicolon to separate individual statements.)

To break a long VBA statement over two or more lines, each of the lines (other than the last line) must end with an underscore (_) preceded by at least one space:

```
newString = fooFunction ("This is a string", _
5, 10, 2)
```

You can combine several statements in a single VBA line by separating them with colons:

```
a = 1 : b = 2 : c = a + b
```

- A VBA line cannot end with a colon. VBA lines that end with a colon are labels that are used by the `Goto` statement.

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Allocating memory

VBA does not support C-style memory pointers. Memory allocation and garbage collection are automatic and transparent, just as in Java and JavaScript (and some C++ code).

Passing arguments

Most languages, including C++ and Java, pass an argument to a procedure as a copy of the original. If the original must be passed, then one of two things can happen:

- a memory pointer is passed that directs the procedure to the original argument in memory
- a reference to the original argument is passed

Microsoft Visual Basic (VB) has the same requirements for passing arguments. In VB, passing a copy of the original argument is called “passing by value” and passing a reference to the original is called “passing by reference.”

By default, function and subroutine parameters are passed by reference. A reference to the original variable is passed in the argument of the procedure; changing the value of that argument, in effect, changes the value of the original variable value as well. In this way, more than one value can be returned from a function or subroutine. To explicitly annotate the code to indicate that an argument is being passed by reference, you can prefix the argument with `ByRef`.

If you want to prevent a procedure from changing the value of the original variable, you can force the copying of an argument. To do this in VBA, prefix the argument with `ByVal`, as shown in the example that follows. The functionality of `ByRef` and `ByVal` is similar to the ability of C and C++ to pass a copy of a variable, or to pass a pointer to the original variable.

```
Private Sub fooFunc (ByVal int1 As Integer, _
ByRef long1 As Long, _
long2 As Long) ' Passed ByRef by default
```

In the preceding VBA example, arguments `long1` and `long2` are both, by default, passed by reference. Modifying either argument within the body of the function modifies the original variable; however, modifying `int1` does not modify the original because it is a copy of the original.

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Providing message boxes and input boxes

In VBA, you can present simple messages to the user by using the `MsgBox` function:

```
Dim retval As Long
retval = MsgBox("Click OK if you agree.", _
vbOKCancel, "Easy Message")
If retval = vbOK Then
MsgBox "You clicked OK.", vbOK, "Affirmative"
End If
```

You can also get strings from the user by using `InputBox` function:

```
Dim inText As String
inText = InputBox("Input some text:", "type here")
If Len(inText) > 0 Then
MsgBox "You typed the following: " & inText & "."
End If
```

If the user clicks **Cancel**, the length of the string returned in `inText` is zero.

- For information on creating more complex user interfaces, see Making macros user-friendly.

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Using object shortcuts

Shortcuts are provided for some frequently accessed objects. Using shortcuts requires less typing, so shortcuts are easier to use than their longhand versions. (Also, using shortcuts can improve run-time performance because the compiler does not need to determine every object in a long dot-separated reference.)

For CorelDRAW, a shortcut can be used on its own as a property of the `Application` object. The following table provides these shortcuts and their long forms. (For a description of any item, see the Object Model ReferenceCorelDRAW .)

Shortcut	Long form
`ActiveLayer`	`ActivePage.ActiveLayer`
`ActivePage`	`ActiveDocument.ActivePage`
`ActiveSelection`	`ActiveDocument.Selection`
`ActiveSelectionRange`	`ActiveDocument.SelectionRange`
`ActiveShape`	`ActiveDocument.Selection.Shapes(1)`
`ActiveView`	`ActiveWindow.ActiveView`
`ActiveWindow`	`ActiveDocument.ActiveWindow`

For CorelDRAW, the following shortcuts can also be used as members of a given `Document` object:

- `ActiveLayer`
- `ActivePage`
- `ActiveShape`
- `ActiveWindow`

For CorelDRAW, the `Document` object also has the properties `Selection` and `SelectionRange`, which let you get the selection or selection range (respectively) from a specified document regardless of whether that document is active.

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Referencing objects

If you want to create a reference to an object so that you can treat that reference like a variable (`sh`, in the following VBA example), you can use the `Set` keyword.

```
Dim sh As Shape
Set sh = ActiveSelection.Shapes.Item(1)
```

After you create this reference, you can treat it as though it were the object itself.

```
sh.Outline.Color.GrayAssign 35
```

If the selection is changed while `sh` is still in scope, `sh` references the original shape from the old selection and is unaffected by the new selection. You cannot simply assign the object to the variable as in the following example:

```
Dim sh As Shape
sh = ActiveSelection.Shapes.Item(1)
```

To release an object, you must set its reference value to `Nothing`.

```
Set sh = Nothing
```

You can also test whether a variable references a valid object by using the `Nothing` keyword.

```
If sh Is Nothing Then MsgBox "sh is de-referenced."
```

Objects do not need to be explicitly released. In most cases, VB releases the object upon disposal of the variable when you exit the function or subroutine.

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Using logical and bitwise operators

In VBA, logical operations are performed by using the keywords `And`, `Not`, `Or`, `Xor`, `Imp`, and `Eqv`, which perform the logical operations AND, NOT, OR, Exclusive-OR, logical implication, and logical equivalence (respectively). These operators also perform Boolean comparisons.

The following code shows a comparison written in C or a similar language:

```
if( ( a && b ) || ( c && d ) )
```

This example would be written as follows in VBA:

```
If ( a And b ) Or ( c And d ) Then
```

Alternatively, the preceding VBA code could be written in the following full long-hand form:

```
If ( a And b = True ) Or ( c And d = True ) = True Then
```

The following table provides a comparison of the four common VBA logical and bitwise operators, and the C-style logical and bitwise operators that are used by C, C++, Java, and JavaScript.

VBA operator	C-style bitwise operator	C-style Boolean operator
`And`	`&`	`&&`
`Not`	`~`	`!`
`Or`	`|`	`||`
`Xor`	`^`	

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Declaring variables

In VBA, the construction for declaring variables is as follows:

```
Dim foobar As Integer
```

The built-in data types are `Byte`, `Boolean`, `Integer`, `Long`, `Single`, `Double`, `String`, `Variant`, and several other less-used types including `Date`, `Decimal`, and `Object`.

Variables can be declared anywhere within the body of a function, or at the top of the current module. However, it is generally a good idea to declare a variable before it is used; otherwise, the compiler interprets it as a `Variant`, and inefficiencies can be incurred at run time.

- Booleans take `False` to be `0` and `True` to be any other value, although converting from a `Boolean` to a `Long` results in `True` being converted to a value of `-1`.

- To get more information about one of the built-in data types, type it in the **Code** window of the Macro Editor, select it, and then press **F1**.

Data structures can be built by using the following VBA syntax:

```
Public Type fooType
item1 As Integer
item2 As String
End Type
Dim myTypedItem As fooType
```

The items within a variable declared as type `fooType` are accessed by using dot notation:

```
myTypedItem.item1 = 5
```

Declaring strings

Using strings is much simpler in VBA than in C. In VBA, strings can be added together, truncated, searched forwards and backwards, and passed as simple arguments to functions.

To add two strings together in VBA, simply use the concatenation operator (`&`) or the addition operator (`+`):

```
Dim string1 As String, string2 As String
string2 = string1 & " more text" + " even more text"
```

In VBA, there are many functions for manipulating strings, including `InStr()`, `Left()`, `Mid()`, `Right()`, `Len()`, and `Trim()`.

Declaring enumerations

To declare an enumeration in VBA, use the following construction:

```
Public Enum fooEnum
ItemOne
ItemTwo
ItemThree
End Enum
```

- By default, the first item in an enumerated type is assigned a value of `0`.

Declaring arrays

To declare an array in VBA, use parentheses — that is, the `(` and `)` symbols:

```
Dim barArray (4) As Integer
```

The value defines the index of the last item in the array. Because array indexes are zero-based by default, there are five elements in the preceding sample array (that is, elements `0` thru `4`, inclusive).

Arrays can be resized by using `ReDim`. For example, the following VBA code adds an extra element to `barArray` but preserves the existing contents of the original five elements:

```
ReDim Preserve barArray (6)
```

Upper and lower bounds for an array can be determined at run time by using the functions `UBound()` and `LBound()`.

Multidimensional arrays can be declared by separating the dimension indexes with commas, as in the following VBA example:

```
Dim barArray (4, 3)
```

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Using Boolean comparison and assignment

In Microsoft Visual Basic (VB), Boolean comparison and Boolean assignment are both performed by using a single equals sign (=):

```
If a = b Then c = d
```

By contrast, many other languages use a double equals sign for Boolean comparison and a single equals sign for Boolean assignment:

```
if( a == b ) c = d;
```

The following code, which is valid in C, C++, Java, and JavaScript, is invalid in VBA:

```
if( ( result = fooBar( ) ) == true )
```

The preceding example would be written in VBA as the following:

```
result = fooBar( )
If result = True Then
```

For other Boolean comparisons, VBA uses the same operators as other languages (except for the operators for “is equal to” and “is not equal to”). All the Boolean-comparison operators are provided in the following table.

Comparison	VBA operator	C-style operator
Is equal to	=	==
Is not equal to	<>	!=
Is greater than	>	>
Is less than	<	<
Is greater than or equal to	>=	>=
Is less than or equal to	<=	<=

The result of using a Boolean operator is always either `True` or `False`.

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How is automation coding structured?

Your programming knowledge should help you learn to automate the software, regardless of your level of experience with Microsoft Visual Basic for Applications (VBA) or Microsoft Visual Studio Tools for Applications (VSTA).

In this topic

This topic contains the following subtopics:

- Declaring variables
- Using functions and subroutines
- Ending lines
- Including comments
- Allocating memory
- Defining scope
- Using Boolean comparison and assignment
- Using logical and bitwise operators
- Providing message boxes and input boxes
- Referencing objects
- Referencing collections
- Using object shortcuts
- Providing event handlers

- The Macro Editor formats all VBA coding for you (see Formatting code automatically). The only way to customize the formatting is to change the size of the indentations.
- VBA can be used to create object-oriented classes. However, this function is a feature of the programming language and is therefore not discussed in detail in this documentation.

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Defining scope

You can define the scope of a data type or procedure (or even an object). Data types, functions, and subroutines (and members of classes) that are declared as private are visible only within that module (or file). By contrast, functions that are declared as public are visible throughout all the modules; however, you may need to use fully qualified referencing if the modules are almost out of scope — for example, if you are referencing a function in a different project.

Unlike C, VBA does not use braces — that is, the `{` and `}` symbols — to define local scope. Local scope in VBA is defined by an opening function or subroutine definition statement (that is, `Function` or `Sub`) and a matching `End` statement (that is, `End Function` or `End Sub`). Any variables declared within the function are available only within the scope of the function itself.

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Events

The Application class contains the following events:

- DocumentAfterExport
- DocumentAfterPrint
- DocumentAfterSave
- DocumentBeforeExport
- DocumentBeforePrint
- DocumentBeforeSave
- DocumentClose
- DocumentNew
- DocumentOpen
- OnPluginCommand
- OnUpdatePluginCommand
- QueryDocumentClose
- QueryDocumentExport
- QueryDocumentPrint
- QueryDocumentSave
- QueryQuit
- Quit
- SelectionChange
- Start
- WindowActivate
- WindowDeactivate

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Events

The Document class contains the following events:

- AfterExport
- AfterPrint
- AfterSave
- BeforeExport
- BeforePrint
- BeforeSave
- Close
- LayerActivate
- LayerChange
- LayerCreate
- LayerDelete
- Open
- PageActivate
- PageChange
- PageCreate
- PageDelete
- QueryClose
- QueryExport
- QueryPrint
- QuerySave
- SelectionChange
- ShapeChange
- ShapeCreate
- ShapeDelete
- ShapeDistort
- ShapeMove
- ShapeTransform

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Events

The GlobalMacroStorage class contains the following events:

- DocumentAfterExport
- DocumentAfterPrint
- DocumentAfterSave
- DocumentBeforeExport
- DocumentBeforePrint
- DocumentBeforeSave
- DocumentClose
- DocumentNew
- DocumentOpen
- OnPluginCommand
- OnUpdatePluginCommand
- QueryDocumentClose
- QueryDocumentExport
- QueryDocumentPrint
- QueryDocumentSave
- QueryQuit
- Quit
- SelectionChange
- Start
- WindowActivate
- WindowDeactivate

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ActiveView

Properties Methods

Description

The **ActiveView** class represents the view properties (zoom, view quality, coordinates of the center of the window) of a window.

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ActiveView.Application

Property **Application** As Application

Description

Member of ActiveView

The Application object defines properties of the main CorelDRAW application. It is the top-level object in the CorelDRAW hierarchy of objects, and you can access all other features of the application through its functions, methods, and properties.

It is not necessary to explicitly specify an **Application** object; it is always available in VBA, and its properties and methods are global.

The **Application** property returns a read-only value that is associated with the main CorelDRAW application.

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ActiveView.GetViewArea

Sub **GetViewArea(***x As Double*, *y As Double*, *Width As Double*, *Height As Double***)**

Description

Member of ActiveView

The **GetViewArea** method returns the current view window (a rectangular area) of the document.

Parameter	Description
x	Specifies the x-coordinate of the rectangular area
y	Specifies the y-coordinate of the rectangular area
Width	Specifies the width of the rectangular area
Height	Specifies the height of the rectangular area

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ActiveView.OriginX

Property **OriginX** As Double

Description

Member of ActiveView

The **OriginX** property returns or specifies the horizontal (x-) coordinate of the drawing window's center in a CorelDRAW document. If you change the x- or y-coordinate of the active view, you are changing the area of the current document that is presented on the screen.

VBA example

The following VBA example sets the coordinates of the drawing window's center to (0,0) in the document's active view. This places the center of the drawing window in the lower- left corner of the page.

```
Sub OriginXOriginY()
 With ActiveWindow.ActiveView
  .OriginX = 0
  .OriginY = 0
 End With
End Sub
```

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ActiveView.OriginY

Property **OriginY** As Double

Description

Member of ActiveView

The **OriginY** property returns or specifies the vertical (y-) coordinate of the drawing window's center in a CorelDRAW document. If you change the x- or y-coordinate of the active view, you are changing the area of the current document which is presented on the screen.

VBA example

The following VBA example sets the coordinates of the drawing window's center to (0,0) in the document's active view. This places the center of the drawing window in the lower-left corner of the page.

```
Sub OriginXOriginY()
 With ActiveWindow.ActiveView
  .OriginX = 0
  .OriginY = 0
 End With
End Sub
```

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ActiveView.Parent

Property **Parent** As Window

Description

Member of ActiveView

The **Parent** property returns a read-only value that is associated with the properties, methods, and controls of an object's parent in the CorelDRAW hierarchy of objects. The application object refers to the CorelDRAW application.

A parent object acts as a container for child objects. For example, a form (as a container) is the parent object of a check box on that form.

Most objects have either a **Parent** property or a **Collection** property, pointing to the object's parent object in the object model. The Collection property is used if the parent object is a collection.

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ActiveView.ProofColorSettings

Property **ProofColorSettings** As ProofColorSettings

Description

Member of ActiveView

The **ProofColorSettings** property specifies the color-proofing settings for the current view window.

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ActiveView.SetActualSize

Sub **SetActualSize()**

Description

Member of ActiveView

The **SetActualSize** method adjusts the current view so that all objects in the view are visible in their actual size. For example, if an ellipse in the active view has a width of two inches, the **SetActualSize** method adjusts the current view so the ellipse has a width of two inches on the screen.

VBA example

The following VBA example sets the current view to reflect the actual size of the object in the view.

```
Sub ViewActual()
 ActiveWindow.ActiveView.SetActualSize
End Sub
```

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ActiveView.SetViewArea

Sub **SetViewArea**(*x As Double*, *y As Double*, *Width As Double*, *Height As Double*)

Description

Member of ActiveView

The **SetViewArea** method sets the current view to display a window of the rectangular area of the document.

Parameter	Description
x	Specifies the x-coordinate of the rectangular area
y	Specifies the y-coordinate of the rectangular area
Width	Specifies the width of the rectangular area
Height	Specifies the height of the rectangular area

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ActiveView.SetViewPoint

Sub **SetViewPoint(***x As Double***,** *y As Double***, [***Zoom As Double***])**

Description

Member of ActiveView

The **SetViewPoint** method sets the viewpoint (which is the center window position) and, optionally, a zoom level.

Parameter	Description
x	Specifies the x-coordinate of the viewpoint, measured in document units
y	Specifies the y-coordinate of the viewpoint, measured in document units
Zoom	Specifies a preset, or sets a user-defined magnification level, measure as a percentage. This parameter is optional, and its default value is **0**.

VBA example

The following VBA example sets the viewpoint of the active view at the point (**2**, **2**) and sets the zoom at **200**%.

```
Sub Test()
 ActiveWindow.ActiveView.SetViewPoint 2, 2, 200
End Sub
```

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ActiveView.ShowProofColors

Property **ShowProofColors** As Boolean

Description

Member of ActiveView

The **ShowProofColors** property specifies whether to display color proofing in the current view window.

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ActiveView.SimulateOverprints

Property **SimulateOverprints** As Boolean

Description

Member of ActiveView

The **SimulateOverprints** property sets the document view to **Simulate Overprints** mode.

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ActiveView.ToFitAllObjects

Sub **ToFitAllObjects()**

Description

Member of ActiveView

The **ToFitAllObjects** method changes the active view to fit all objects on the current page into the application window. This method adjusts the zoom level so that the area surrounding all objects on the current page fills up the active view.

VBA example

The following VBA example uses the **ToFitAllObjects** method to adjust the zoom level so that all objects on the page fill up the active view of CorelDRAW.

```
Sub ToFitAllObjects()
 ActiveWindow.ActiveView.ToFitAllObjects
End Sub
```

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ActiveView.ToFitArea

Sub **ToFitArea(***Left As Double***,** *Top As Double***,** *Right As Double***,** *Bottom As Double***)**

Description

Member of ActiveView

The **ToFitArea** method changes the active view to fit a specified rectangular area, identified by its coordinates, in the current document into the application window. The **ToFitArea** method defines an area, and adjusts the zoom level of the page so that the defined region fills the current view in the application window.

Parameter	Description
Left	A coordinate that specifies the left side of the rectangle of the active view relative to the center of the coordinates set for the document. The left coordinate is measured in document units.
Top	A coordinate that specifies the top side of the rectangle of the active view relative to the center of the coordinates set for the document. The top coordinate is measured in document units.
Right	A coordinate that specifies the right side of the rectangle of the active view relative to the center of the coordinates set for the document. The right coordinate is measured in document units.
Bottom	A coordinate that specifies the bottom side of the rectangle of the active view relative to the center of the coordinates set for the document. The bottom coordinate is measured in document units.

VBA example

The following VBA example creates a rectangular area in the active document. This defined area, using the coordinates **0**, **0**, **5**, and **5**, uses the **ToFitArea** method to adjust the zoom level to display the newly defined rectangular page area as the active view.

```
Sub ToFitArea()
 ActiveWindow.ActiveView.ToFitArea 0, 0, 5, 5
End Sub
```

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ActiveView.ToFitPage

Sub **ToFitPage()**

Description

Member of ActiveView

The **ToFitPage** method changes the active view to fit the entire page of the current document. It ensures that the entire page in a document is visible in CorelDRAW.

The **ToFitPage** method is equivalent to the **Shift+F4** command in CorelDRAW.

VBA example

The following VBA example sets the active view of the current document with the **ToFitPage** method, ensuring that the entire page in the document is visible in the application window.

```
Sub ToFitPage()
 ActiveWindow.ActiveView.ToFitPage
End Sub
```

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ActiveView.ToFitPageHeight

Sub **ToFitPageHeight()**

Description

Member of ActiveView

The **ToFitPageHeight** method changes the active view to fit the entire page height of the current document into the application window. This method places the top and bottom borders of the active page to the top and bottom borders of the application window, adjusting the zoom level so that these page boundaries fit the entire view space.

VBA example

The following VBA example adjusts the zoom level so that the active view fits the height of the page in the active document. This changes the appearance of the active view in that the entire view space is now occupied by the height of the active page.

```
Sub ToFitPageHeight()
 ActiveWindow.ActiveView.ToFitPageHeight
End Sub
```

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ActiveView.ToFitPageWidth

Sub **ToFitPageWidth()**

Description

Member of ActiveView

The **ToFitPageWidth** method changes the active view to fit the entire page width of the current document into the application window. This method places the left and right borders of the active page to the left and right borders of the application window, adjusting the zoom level so that these page boundaries fit the entire window space.

VBA example

The following VBA example adjusts the zoom level in CorelDRAW so that the active view fits the width of the page in the active document. This changes the appearance of the active view in that the entire view space is now occupied by the width of the active page.

```
Sub ToFitPageWidth()
 ActiveWindow.ActiveView.ToFitPageWidth
End Sub
```

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ActiveView.ToFitSelection

Sub **ToFitSelection()**

Description

Member of ActiveView

The **ToFitSelection** method changes the active view to fit a selection of objects in the current document into the application window. This method adjusts the zoom level of the page so that the entire current selection of objects fills the current view in the application window.

ToFitSelection method is equivalent to the **Shift+F2** command in CorelDRAW.

VBA example

The following VBA example adjusts the zoom level in CorelDRAW so that all selected objects on the page appear in the active view of the current document.

```
Sub ToFitSelection()
 ActiveWindow.ActiveView.ToFitSelection
End Sub
```

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ActiveView.ToFitShape

Sub **ToFitShape(**Shape As Shape**)**

Description

Member of ActiveView

The **ToFitShape** method changes the active view to fit an entire shape in the current document into the application window. This method adjusts the zoom level of the page so that the specified shape fills the current view in the application window.

Parameter	Description
Shape	Specifies the shape object that you want to zoom out

VBA example

The following VBA example creates a circle in CorelDRAW and adjusts the active view to zoom in on the circle in the active document.

```
Sub ToFitShape()
 Dim s As Shape
 Set s = ActiveLayer.CreateEllipse2(5, 5, 2)
 ActiveWindow.ActiveView.ToFitShape s
End Sub
```

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ActiveView.ToFitShapeRange

Sub **ToFitShapeRange(**ShapeRange As ShapeRange**)**

Description

Member of ActiveView

The **ToFitShapeRange** method changes the active view to fit an entire collection of shape objects on the current page into the application window. The **ToFitShapeRange** method adjusts the zoom level of the page so that the specified shapes fill the current view in the application window.

Parameter	Description
ShapeRange	Specifies the area to include in the defined active view

VBA example

The following VBA example creates a rectangle and circle on the current page, adds the rectangle and the circle to a shape range, and uses the **ToFitShapeRange** method to adjust the active view and zoom in so that the shapes in the shape range make up the entire view in CorelDRAW.

```
Sub ToFitShapeRange()
 Dim s As Shape
 Dim sr As New ShapeRange
 Set s = ActiveLayer.CreateEllipse2(5, 5, 2)
 sr.Add s 'Creates an ellipse and includes it in the defined range
 Set s = ActiveLayer.CreateRectangle(0, 0, 2, 2)
 sr.Add s 'Creates a rectangle and includes it in the defined range
 ActiveWindow.ActiveView.ToFitShapeRange sr
End Sub
```

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ActiveView.Type

Property **Type** As cdrViewType

Description

Member of ActiveView

The **Type** property returns or specifies the active view quality in CorelDRAW. A view quality lets you choose how CorelDRAW displays a drawing on the screen. If you have a fast computer or want to approximate how a drawing looks when printed, use the Normal or Enhanced view. If you have a slower computer or want to speed up the refresh rate of a complex drawing, you may find the Simple Wireframe or Wireframe view to be the most effective.

VBA example

The following VBA example sets the view type of the current window equal to a value that represents the Enhanced view.

```
Sub DocumentType()
 ActiveView.Type = cdrEnhancedView
End Sub
```

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ActiveView.Zoom

Property **Zoom** As Double

Description

Member of ActiveView

The **Zoom** property returns or specifies a magnification level in the active view of CorelDRAW.

VBA example

The following VBA example sets the zoom level of the active document to 200%, if the current zoom level is less than 200%.

```
Sub Zoom()
 With ActiveWindow.ActiveView
  If .Zoom < 200 Then .Zoom = 200
 End With
End Sub
```

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ActiveView.ZoomIn

Sub **ZoomIn()**

Description

Member of ActiveView

The **ZoomIn** property zooms in at the center of the current document window.

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ActiveView.ZoomInAtPoint

Sub **ZoomInAtPoint(***x As Double***,** *y As Double***)**

Description

Member of ActiveView

The **ZoomInAtPoint** property zooms in at the specified point.

Parameter	Description
x	Specifies the x-coordinate of the point
y	Specifies the y-coordinate of the point

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ActiveView.ZoomOut

Sub **ZoomOut()**

Description

Member of ActiveView

The **ZoomOut** property zooms out from the current document window.

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Application

Properties Methods Events

Description

The Application object defines properties of the main CorelDRAW application. It is the top-level object in the CorelDRAW hierarchy of objects, and you can access all other features of the application through its functions, methods, and properties.

It is not necessary to explicitly specify an **Application** object; it is always available in VBA, and its properties and methods are global.

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Application.ActiveDocument

Property **ActiveDocument** As Document

Description

Member of Application

The **ActiveDocument** property returns a read-only reference to the active document.

You do not need to activate a document before working with it in CorelDRAW. The document stays inactive, but you still can work with it. For example, consider the following code example:

```
Documents(1).Activate
ActiveDocument.AddPages 1
```

The preceding code example produces the same result as the following code example:

```
Documents(1).AddPages 1
```

Both of these examples add a new page to the first open document, which is the first document.

VBA example

The following VBA example creates a new document and draws an ellipse in it. Then another document is created, and a rectangle is added to it. The first document is then activated and a new page is added to it.

```
Sub DocumentActive()
 CreateDocument
 ActiveLayer.CreateEllipse2 3, 2, 1
 CreateDocument
 ActiveLayer.CreateRectangle 2, 3, 4, 5
 Documents(1).Activate
 ActiveDocument.AddPages 1
End Sub
```

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Application.ActiveLayer

Property **ActiveLayer** As Layer

Description

Member of Application

The **ActiveLayer** property returns the active Layer on the active page in the active document.

The **ActiveLayer** property returns a read-only value.

VBA example

The following VBA example creates an ellipse on the active layer in the current document.

```
Sub LayerActive()
 ActiveLayer.CreateEllipse 3, 3, 2, 1
 ActiveDocument.Selection.Shapes(1).Fill.UniformColor.CMYKAssign 100, 0, 100, 0
 'Fills the ellipse green
End Sub
```

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Application.ActivePage

Property ActivePage As Page

Description

Member of Application

The **ActivePage** property returns the active page in the active document.

The **ActivePage** property returns a read-only value.

VBA example

The following VBA example changes the background color of the current page in the active document to red.

```
Sub PageActive()
 ActivePage.Color.RGBAssign 255, 0, 0
End Sub
```

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Application.ActivePalette

Property **ActivePalette** As Palette

Description

Member of Application

The **ActivePalette** property returns a read-only reference to the active palette.

VBA example

The following VBA example displays the name of the default palette, and the number of colors in that palette, in a message box.

```
Sub PaletteActive()
 MsgBox "The default color palette is: " & _
 ActivePalette.Name & _
 "It contains " & ActivePalette.ColorCount & " colors."
End Sub
```

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Application.ActiveSelection

Property **ActiveSelection** As Shape

Description

Member of Application

The **ActiveSelection** property returns the active selection in the active document.

The **ActiveSelection** property returns a read-only value.

VBA example

The following VBA example colors the selected shapes green, if the shape selected is an ellipse.

```
Sub SelectionActive()
 Dim s As Shape
 For Each s In ActiveSelection.Shapes
  If s.Type = cdrEllipseShape Then
   s.Fill.UniformColor.CMYKAssign 100, 0, 100, 0
  End If
 Next s
End Sub
```

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Application.ActiveSelectionRange

Property **ActiveSelectionRange** As ShapeRange

Description

Member of Application

The **ActiveSelectionRange** property returns the active ShapeRange object in the active document.

The **ActiveSelectionRange** property returns a read-only value.

VBA example

The following VBA example displays messages about selected objects in the active document, depending on the number of objects selected.

```
Sub SelectionRangeActive()
 If ActiveSelectionRange.Count = 1 Then
  MsgBox "There is only one shape selected in the shape range."
 ElseIf ActiveSelectionRange.Count = 0 Then
  MsgBox "There are no shapes selected in the shape range."
 Else
  MsgBox "There are " & ActiveSelectionRange.Count & " shapes in the selected shape range."
 End If
End Sub
```

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Application.ActiveShape

Property **ActiveShape** As Shape

Description

Member of Application

The **ActiveShape** property returns the first selected shape in the active document.

The **ActiveShape** property returns a read-only value.

VBA example

The following VBA example applies the Brighten lens to the currently selected shape.

```
Sub Test()
 ActiveShape.CreateLens cdrLensBrighten, 50
End Sub
```

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Application.ActiveSpread

Property **ActiveSpread** As Spread

Description

Member of Application

The **ActiveSpread** property returns a read-only value that is associated with the active page spread in the current document.

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Application.ActiveTool

Property **ActiveTool** As cdrTools

Description

Member of Application

The **ActiveTool** property returns or specifies the active tool in CorelDRAW.

VBA example

The following VBA example changes the active tool from the **Pick** tool to the **Eye Dropper** tool.

```
Sub ToolActive()
 If ActiveTool = cdrToolPick Then
  ActiveTool = cdrToolEyeDropper
 End If
End Sub
```

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Application.ActiveTreeManager

Property **ActiveTreeManager** As TreeManager

Description

Member of Application

The **ActiveTreeManager** property returns the TreeManager object of the active document.

The **ActiveTreeManager** property returns a read-only value.

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Application.ActiveVirtualLayer

Property **ActiveVirtualLayer** As Layer

Description

Member of Application

The **ActiveVirtualLayer** property returns the virtual Layer object of the active document.

The **ActiveVirtualLayer** property returns a read-only value.

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Application.ActiveWindow

Property **ActiveWindow** As Window

Description

Member of Application

The **ActiveWindow** property returns the active window of the active document.

The **ActiveWindow** property returns a read-only value.

VBA example

The following VBA example redraws the active document window.

```
Sub WindowActive()
 ActiveWindow.Refresh
End Sub
```

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Application.ActiveWorkspace

Property **ActiveWorkspace** As Workspace

Description

Member of Application

The **ActiveWorkspace** property returns a read-only reference to the active workspace.

VBA example

The following VBA example displays the name of the active workspace in a message box.

```
Sub WorkspaceActive()
 MsgBox ActiveWorkspace.Name
End Sub
```

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Application.AddonPath

Property **AddonPath** As String

Description

Member of Application

The **AddonPath** property returns a read-only full path to the application's Addons folder.

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Application.AddPluginCommand

Function **AddPluginCommand(***CommandID As String***, [***Caption As String***], [***Tooltip As String***])** As Boolean

Description

Member of Application

The **AddPluginCommand** method registers a new command to be used with plugins.

Parameter	Description
CommandID	Specifies the ID of the command
Caption	Specifies a caption for the command. This parameter is optional.
Tooltip	Specifies a ToolTip for the command. This parameter is optional.

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Application.AdviseEvents

Function **AdviseEvents(**EventSink As IDrawApplicationEvents**)** As Long

Description

Member of Application

The **AdviseEvents** method advises application events.

Parameter	Description
EventSink	

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Application.Application

Property Application As Application

Description

Member of Application

The Application object defines properties of the main CorelDRAW application. It is the top-level object in the CorelDRAW hierarchy of objects, and you can access all other features of the application through its functions, methods, and properties.

It is not necessary to explicitly specify an **Application** object; it is always available in VBA, and its properties and methods are global.

The **Application** property returns a read-only value that is associated with the main CorelDRAW application.

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Application.AppWindow

Property **AppWindow** As AppWindow

Description

Member of Application

The **AppWindow** property returns the main application window.

The **AppWindow** property returns a read-only value.

VBA example

The following VBA example displays the caption of the main application window in a message box.

```
Sub CaptionAppWindow()
 MsgBox AppWindow.Caption
End Sub
```

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Application.ArrowHeads

Property **ArrowHeads** As ArrowHeads

Description

Member of Application

The **Arrowheads** property returns the ArrowHeads collection.

The **Arrowheads** property returns a read-only value.

VBA example

The following VBA example applies an arrowhead with an index of 5 to the beginning of the outlines of each selected object.

```
Sub ItemArrowHead()
 Dim s As Shape
 For Each s In ActiveDocument.Selection.Shapes
  If s.Outline.Type = cdrOutline Then
   s.Outline.StartArrow = ArrowHeads.Item(5)
  'can also be referenced as s.Outline.StartArrow = ArrowHeads (5)
  End If
 Next s
End Sub
```

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Application.cdrMixedDouble

Function **cdrMixedDouble()** As Double

Description

Member of Application

The **cdrMixedDouble** function returns a double value corresponding to a mixed (uneven) double. An undefined value is returned by some functions when ambiguity (different values) exists. For example, text that contains font characters of different sizes has mixed settings.

VBA example

The following VBA example checks whether the specified selection is a text object. If this text object contains characters with varying font sizes, it sets a uniform font size for the entire text object.

```
Sub MixedDouble()
 Dim s As Shape
 Set s = ActiveDocument.Selection.Shapes(1)
 If s.Type = cdrTextShape Then
  If s.Text.FontProperties.Size = cdrMixedDouble Then
   s.Text.FontProperties.Size = 12
  End If
 End If
End Sub
```

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Application.cdrMixedLong

Function **cdrMixedLong()** As Long

Description

Member of Application

The **cdrMixedLong** property returns a long value corresponding to a mixed (uneven) long. An undefined value is returned by some functions when ambiguity (different values) exists. For example, text that contains font characters of different sizes has mixed settings.

VBA example

The following VBA example checks whether the specified selection is a text object. If this text object contains characters with varying font sizes, it sets a uniform font size for the entire text object.

```
Sub MixedLong()
 Dim s As Shape
 Set s = ActiveDocument.Selection.Shapes(1)
 If s.Type = cdrTextShape Then
  If s.Text.FontProperties.Size = cdrMixedLong Then
   s.Text.FontProperties.Size = 12
  End If
 End If
End Sub
```

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Application.cdrMixedSingle

Function **cdrMixedSingle()** As Single

Description

Member of Application

The **cdrMixedSingle** property returns a value corresponding to a mixed (uneven) single. An undefined value is returned by some functions when ambiguity (different values) exists. For example, text that contains font characters of different sizes has mixed settings.

VBA example

The following VBA example checks whether the specified selection is a text object. If this text object contains characters with varying font sizes, it sets a uniform font size for the entire text object.

```
Sub MixedSingle()
 Dim s As Shape
 Set s = ActiveDocument.Selection.Shapes(1)
 If s.Type = cdrTextShape Then
  If s.Text.FontProperties.Size = cdrMixedSingle Then
   s.Text.FontProperties.Size = 12
  End If
 End If
End Sub
```

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Application.CheckPlatformVersion

Function **CheckPlatformVersion(VersionMajor As Long, [VersionMinor As Long])** As Long

Description

Member of Application

The **CheckPlatformVersion** method compares the specified version to that of the automation engine.

Parameter	Description
VersionMajor	Specifies the major version (see Application.VersionMajor)
VersionMinor	Specifies the minor version (see Application.VersionMinor). This parameter is optional, and its default value is **0**.

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Application.Class

Property **Class** As cdrApplicationClass

Description

Member of Application

The **Class** property returns the application class.

The **Class** property returns a read-only value.

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Application.Clipboard

Property **Clipboard** As Clipboard

Description

Member of Application

The **Clipboard** property returns the system Clipboard.

The **Clipboard** property returns a read-only value.

VBA example

The following VBA example checks whether the Clipboard is empty. If data is present, it is pasted into the current document. If no data is present, a message displays in a message box.

```
Sub ClipboardData()
 If Not Clipboard.Empty Then
  ActiveLayer.Paste
 Else
  MsgBox "There is no data in the clipboard."
 End If
End Sub
```

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Application.ColorManager

Property **ColorManager** As ColorManager

Description

Member of Application

The **ColorManager** property returns the application's ColorManager object.

The **ColorManager** property returns a read-only value.

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Application.CommandBars

Property **CommandBars** As CommandBars

Description

Member of Application

The **CommandBars** property returns a collection of all command bars.

The **CommandBars** property returns a read-only value.

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Application.Components

Property **Components** As Components

Description

Member of Application

The **Components** property returns a collection of application-specific components.

The **Components** property returns a read-only value.

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Application.ConfigPath

Property **ConfigPath** As String

Description

Member of Application

The **ConfigPath** property returns the path to the configuration folder.

The **ConfigPath** property returns a read-only value.

VBA example

The following VBA example displays the path of the CorelDRAW configuration folder in a message box.

```
Sub PathConfig()
 MsgBox ConfigPath
End Sub
```

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Application.ConvertFromUnicode

Function **ConvertFromUnicode(***String As String***,** *CodePage As Long***)** As String

Description

Member of Application

The **ConvertFromUnicode** method converts a string from Unicode into the specified code page.

Parameter	Description
String	Specifies the string to convert
CodePage	Specifies the code page into which to convert the string

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Application.ConvertToUnicode

Function **ConvertToUnicode(***String As String***,** *CodePage As Long***)** As String

Description

Member of Application

The **ConvertToUnicode** method converts a string from a given code page into a Unicode string.

Parameter	Description
String	Specifies the string to convert
CodePage	Specifies the code page from which to convert the string

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Application.ConvertUnits

Function **ConvertUnits(***Value As Double***,** FromUnit As cdrUnit**,** ToUnit As cdrUnit**)** As Double

Description

Member of Application

The **ConvertUnits** method converts a measurement from one unit to another.

Parameter	Description
Value	Specifies the value to convert
FromUnit	Specifies the unit from which to convert
ToUnit	Specifies the unit to which to convert

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Application.CorelScriptTools

Function **CorelScriptTools()** As CorelScriptTools

Description

Member of Application

The **CorelScriptTools** method returns a CorelScriptTools object as a generic object type.

VBA example

The following VBA example displays the standard **File Open** dialog box and asks the user to choose a file. Then the FileAttr function determines whether the file exists at the specified location.

```
Sub ScriptTools()
 Dim cst As Object
 Dim FilePath As String
 Set cst = CorelScriptTools
 FilePath = cst.GetFileBox()
 If FilePath <> "" Then
  If cst.FileAttr(FilePath) = 0 Then
   MsgBox "The file you selected doesn't exist."
  End If
 End If
End Sub
```

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Application.CreateBWColor

Function **CreateBWColor(***White As Boolean***)** As Color

Description

Member of Application

The **CreateBWColor** method creates a new color object based on the black-and-white color model.

Parameter	Description
White	Specifies the appearance of white in the black-and-white color model

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateCMYColor

Function **CreateCMYColor(***Cyan As Long*, *Magenta As Long*, *Yellow As Long***)** As Color

Description

Member of Application

The **CreateCMYColor** method creates a new color object based on the CMY color model.

Parameter	Description
Cyan	Specifies the level of cyan in the CMY color model. Values range from **0** to **255**.
Magenta	Specifies the level of magenta in the CMY color model. Values range from **0** to **255**.
Yellow	Specifies the level of magenta in the CMY color model. Values range from **0** to **255**.

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateCMYKColor

Function **CreateCMYKColor(***Cyan As Long*, *Magenta As Long*, *Yellow As Long, Black As Long***)** As Color

Description

Member of Application

The **CreateCMYKColor** method creates a new color object based on the CMYK color model.

Parameter	Description
Cyan	Specifies the level of cyan in the CMYK color model. Values range from **0** to **100**.
Magenta	Specifies the level of magenta in the CMYK color model. Values range from **0** to **100**.
Yellow	Specifies the level of magenta in the CMYK color model. Values range from **0** to **100**.
Black	Specifies the level of black in the CMYK color model. Values are percentages and range from **0** to **100**.

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateColor

Function **CreateColor([***ColorString As String***])** As Color

Description

Member of Application

The **CreateColor** method creates a color object with default color properties.

This function is now obsolete because the **Color** object can be used with the **New** keyword.

Parameter	Description
ColorString	This parameter is optional.

VBA example

The following VBA examples show several ways to apply a cyan fill to the first selected object in the active document.

```
Sub CreateColor1()
 Dim c As Color
 Set c = CreateColor
 c.CMYKAssign 100, 0, 0, 0
 ActiveDocument.Selection.Shapes(1).Fill.ApplyUniformFill c
End Sub
Sub CreateColor2()
 Dim c As New Color
 c.CMYKAssign 100, 0, 0, 0
 ActiveSelection.Shapes(1).Fill.ApplyUniformFill c
End Sub
Sub CreateColor3()
 ActiveShape.Fill.UniformColor.CMYKAssign 100, 0, 0 ,0
End Sub
```

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Application.CreateColorContext

Function **CreateColorContext([**RGBProfile As ColorProfile**], [**CMYKProfile As ColorProfile**], [**GrayscaleProfile As ColorProfile**], [**RenderingIntent As clrRenderingIntent**], [**BlendingColorModel As clrColorModel**])** As ColorContext

Description

Member of Application

The **CreateColorContext** method creates a color context by using the specified color profiles.

Parameter	Description
RGBProfile	Specifies an RGB color profile. This parameter is optional, and its default value is **Nothing**.
CMYKProfile	Specifies a CMYK color profile. This parameter is optional, and its default value is **Nothing**.
Grayscale Profile	Specifies a grayscale color profile. This parameter is optional, and its default value is **Nothing**.
RenderingIntent	Specifies a rendering intent. This parameter is optional, and its default value is clrRenderRelative.
BlendingColorModel	Specifies a color model for blending. This parameter is optional, and its default value is clrColorModelRGB.

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Application.CreateColorContext2

Function **CreateColorContext2(***ColorProfileList As String***, [**RenderingIntent As clrRenderingIntent**], [**BlendingColorModel As clrColorModel**])** As ColorContext

Description

Member of Application

The **CreateColorContext2** method creates a color context by using a comma-separated list of color-profile names.

Parameter	Description
ColorProfileList	Specifies a comma-separated list of color-profile names
RenderingIntent	Specifies a rendering intent. This parameter is optional, and its default value is clrRenderRelative.
BlendingColorModel	Specifies a color model for blending. This parameter is optional, and its default value is clrColorModelRGB.

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Application.CreateColorEx

Function **CreateColorEx([***ColorModel As Long***], [***V1 As Long***], [***V2 As Long***], [***V3 As Long***], [***V4 As Long***], [***V5 As Long***], [***V6 As Long***], [***V7 As Long***])** As Color

Description

Member of Application

The **CreateColorEx** method creates a color object and initializes it, with specified color information using a Corel SCRIPT color specification.

Parameter	Description
ColorModel	Specifies the color model used to create the new color. This parameter specifies the numeric variable that is assigned to the color model. This paramater is optional, and its default value is **0**.
V1	Specifies the numeric variable that is assigned to the first color component of the selected color model. For example, cyan is the first color component of the CMYK color model. The value is a percentage and can range from **1** to **100**. A color that is selected at 100 is applied with full saturation. This paramater is optional, and its default value is **0**.
V2	Specifies the numeric variable that is assigned to the second color component of the selected color model. For example, magenta is the second color component of the CMYK color model. The value is a percentage and can range from **1** to **100**. A color that is selected at 100 is applied with full saturation. This paramater is optional, and its default value is **0**.
V3	Specifies the numeric variable that is assigned to the third color component of the selected color model. For example, yellow is the third color component of the CMYK color model. The value is a percentage and can range from **1** to **100**. A color that is selected at 100 is applied with full saturation. This paramater is optional, and its default value is **0**.
V4	Specifies the numeric variable that is assigned to the fourth color component of the selected color model. For example, black is the fourth color component of the CMYK color model. The value is a percentage and can range from **1** to **100**. A color that is selected at 100 is applied with full saturation. This paramater is optional, and its default value is **0**.
V5	Specifies the numeric variable that is assigned to the fifth color component of the selected color model. Even though the most popular color models use only four levels of color saturations, CorelDRAW can create colors using seven color channels. This paramater is optional, and its default value is **0**.
V6	Specifies the numeric variable that is assigned to the sixth color component of the selected color model. Even though the most popular color models use only four levels of color saturations, CorelDRAW can create colors using seven color channels. This paramater is optional, and its default value is **0**.
V7	Specifies the numeric variable that is assigned to the seventh color component of the selected color model. Even though the most popular color models use only four levels of color saturations, CorelDRAW can create colors using seven color channels. This paramater is optional, and its default value is **0**.

VBA example

The following VBA example adds a new color to the active color palette.

```
Sub CreatePaletteColor()
 ActivePalette.AddColor CreateColorEx(5002, 90, 90, 0, 0)
End Sub
```

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Application.CreateCurve

Function **CreateCurve([**Document As Document**])** As Curve

Description

Member of Application

The **CreateCurve** method creates a curve in memory and optionally binds it to a document.

Any changes made to the curve in memory do not affect the document, and are, therefore, performed very fast.

When you work with coordinates (for example, setting the position of a node, getting the length of a curve, and so on), it is important to know which document to reference. You can attach a curve in memory to a document to use that document's system of measurement. This can be done by specifying the optional **Document** parameter in the **CreateCurve** method, or by using the Curve.BindToDocument method.

Parameter	Description
Document	Specifies the settings to apply This paramater is optional, and its default value is **Nothing**.

VBA example

The following VBA example creates a curve on the active layer using the settings of the active document.

```
Sub Test()
 Dim crv As Curve
 Dim s As Shape
 Dim sp As SubPath
 Set crv = Application.CreateCurve(ActiveDocument)
 Set sp = crv.CreateSubPath(1, 5)
 sp.AppendLineSegment 2, 0
 sp.AppendCurveSegment2 5, 7, 6, 6, 3, 5
 sp.AppendCurveSegment2 7, 2, 3, 6, 7, 1
 sp.AppendLineSegment 9, 6
 sp.Nodes(3).Type = cdrSmoothNode
 Set s = ActiveLayer.CreateCurve(crv)
End Sub
You can also use the New keyword to create a curve in memory.
Dim crv As Curve
Set crv = New Curve
```

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Application.CreateDocument

Function **CreateDocument()** As Document

Description

Member of Application

The **CreateDocument** method creates a new CorelDRAW document.

VBA example

The following VBA example creates a new CorelDRAW document, adds a colored ellipse to the active layer of the document, and saves the new document with a new name.

```
Sub DocumentCreate()
 Dim doc As Document
 Set doc = CreateDocument()
 doc.ActiveLayer.CreateEllipse (0, 3, 5, 1).Fill.UniformColor.CMYKAssign 0, 100, 100, 0
 doc.SaveAs "C:\My New Document"
End Sub
```

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Application.CreateDocumentEx

Function **CreateDocumentEx(**Options As StructCreateOptions**)** As Document

Description

Member of Application

The **CreateDocumentEx** method creates a new document by using the specified settings.

Parameter	Description
Options	Specifies the settings to apply to the new document

Example

Here is a code example:

```
Sub Test()
Dim d As Document
Dim so As StructCreateOptions
Set so = CreateStructCreateOptions
With so
.Name = "CreateDocumentEx Sample"
.PageHeight = 10
.PageWidth = 10
.Resolution = 300
.Units = cdrPixel
End With
Set d = CreateDocumentEx(so)
End Sub
```

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Application.CreateDocumentFromTemplate

Function **CreateDocumentFromTemplate(***Template As String***, [**IncludeGraphics As Boolean = True**])** As Document

Description

Member of Application

The **CreateDocumentFromTemplate** method creates a new document based on a template in CorelDRAW.

Parameter	Description
Template	Specifies the filename of a template
IncludeGraphics	Specifies whether to include graphics from the template in the new document. This parameter is optional, and its default value is **True**.

VBA example

The following VBA example creates a new document based on the template **webtemplate_001.cdt** and adds text content to each text block.

```
Sub Test()
 Dim d As Document
 Dim s As Shape
 Set d = CreateDocumentFromTemplate(Path & "Template\Web\webtemplate_001.cdt", True)
 For Each s In d.ActivePage.Shapes
  If s.Type = cdrTextShape Then
   s.Text.Contents = s.Name
  End If
 Next s
End Sub
```

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Application.CreateDuotone

Function **CreateDuotone()** As Duotone

Description

Member of Application

The **CreateDuotone** method creates a new duotone information object.

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Application.CreateFillMetadata

Function **CreateFillMetadata()** As FillMetadata

Description

Member of Application

The **CreateFillMetadata** method creates a new FillMetadata object instance.

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Application.CreateGrayColor

Function **CreateGrayColor(***GrayValue As Long***)** As Color

Description

Member of Application

The **CreateGrayColor** method creates a color object based on the grayscale color model.

Parameter	Description
GrayValue	Specifies the level of gray in the grayscale color model. Each color is defined as a value between **0** and **255**, where **0** is darkest (black) and **255** is lightest (white).

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateHLSColor

Function **CreateHLSColor(***Hue As Long*, *Lightness As Long*, *Saturation As Long***)** As Color

Description

Member of Application

The **CreateHLSColor** method creates a new color from the HLS color model.

Parameter	Description
Hue	Specifies the hue for the HLS color model. Hue is the property of a color that allows us to classify it by its name. For example, blue, green, and red are all hues. Values range from **0** to **360**.
Lightness	Specifies the lightness for the HLS color model. Lightness determines the intensity of the color. Values range from **0** to **255**.
Saturation	Specifies the saturation for the HLS color model. Saturation is the purity, vividness, or color depth (from dull to intense) of a color, expressed as the absence of white. A color that has 100% saturation contains no white. A color with 0% saturation is a shade of gray. Values range from **0** (0%) to **255** (100%).

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateHSBColor

Function **CreateHSBColor(***Hue As Long*, *Saturation As Long*, *Brightness As Long***)** As Color

Description

Member of Application

The **CreateHSBColor** method creates a color object from the HSB color model.

Parameter	Description
Hue	Specifies the hue for the HSB color model. Hue is the property of a color that allows us to classify it by its name. For example, blue, green, and red are all hues. Values range from **0** to **360**.
Saturation	Specifies the saturation for the HSB color model. Saturation is the purity, vividness, or color depth (from dull to intense) of a color, expressed as the absence of white. A color that has 100% saturation contains no white. A color with 0% saturation is a shade of gray. Values range from **0** to **255**.
Brightness	Specifies the brightness for the HSB color model. Brightness is the intensity (lighter or darker color) or the amount of light that is transmitted or reflected from a given pixel. In the HSB color model, brightness is a measure of how much white a color contains. In this case, a brightness value of 0 produces black and a brightness value of 255 produces white. Values range from **0** to **255**.

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateLabColor

Function **CreateLabColor(***L As Long*, *A As Long*, *B As Long***)** As Color

Description

Member of Application

The **CreateLabColor** method creates a color object based on the LAB color model.

Parameter	Description
L	Specifies the luminance, or brightness, value in the LAB color model. Values range from **0** to **255**.
A	Specifies one of two chromatic components in the LAB color model. “A” refers to a range of colors between green and red. Values range from **-128** to **127**.
B	Specifies one of two chromatic components in the LAB color model. “B” refers to a range of colors between blue and yellow. Values range from **-128** to **127**.

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateNodeRange

Function **CreateNodeRange()** As NodeRange

Description

Member of Application

The **CreateNodeRange** method creates an empty NodeRange object.

VBA example

The following VBA example creates an ellipse and converts it to curves. Every second node is added to the node range, and the nodes are moved along the y-axis.

```
Sub Test()
 Dim s As Shape
 Dim crv As Curve
 Dim nr As NodeRange
 Dim n As Node
 Dim i As Integer
 Set s = ActiveLayer.CreateEllipse(0, 0, 5, 5)
 s.ConvertToCurves
 Set nr = CreateNodeRange
 For i = 1 To s.Curve.Nodes.Count Step 2
  Set n = s.Curve.Nodes(i)
  nr.Add n
 Next i
 nr.Move 0, 2
End Sub
```

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Application.CreateOutlineStyle

Function **CreateOutlineStyle(***DashDotCount As Long*, *DashDotLengths() As Variant***)** As OutlineStyle

Description

Member of Application

The **CreateOutlineStyle** method creates a custom dash-dot outline pattern by specifying dash or gap lengths directly.

Parameter	Description
DashDotCount	Specifies the number of dashes or dots
DashDotLengths	Specifies the length of the dashes or dots

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Application.CreatePaletteColor

Function **CreatePaletteColor(***PaletteIdentifier As String***,** *ColorIndex As Long***)** As Color

Description

Member of Application

The **CreatePaletteColor** method creates a Color object that is obtained from an existing color palette.

Parameter	Description
PaletteIdentifier	Specifies the color palette to use
ColorIndex	Specifies the index number of the color in the specified color palette

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Application.CreatePatternCanvas

Function **CreatePatternCanvas()** As PatternCanvas

Description

Member of Application

The **CreatePatternCanvas** method creates an empty PatternCanvas object.

VBA example

The following VBA example creates an empty pattern canvas and copies another pattern to this canvas. A rectangle is filled with this pattern, and then the data of the pattern is changed.

```
Sub Test()
 Dim c As PatternCanvas
 Set c = CreatePatternCanvas
 c.PutCopy PatternCanvases(29)
 With ActiveLayer.CreateRectangle(0, 0, 2, 2)
  .Fill.ApplyPatternFill cdrTwoColorPattern
  .Fill.Pattern.Canvas = c
  c.Data = "%%['$aH,&A5&0C"
 End With
End Sub
```

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Application.CreateProofColorSettings

Function **CreateProofColorSettings(***ProfileName As String***, [**RenderingIntent As clrRendingIntent**], [**PreserveColorValues As Boolean = False**])** As ProofColorSettings

Description

Member of Application

The **CreateProofColorSettings** method creates an instance of a ProofColorSettings object.

Parameter	Description
ProfileName	Specifies the color profile to use
RenderingIntent	Specifies the rendering intent to use. This parameter is optional, and its default value is clrRenderRelative.
PreserveColorValues	Specifies whether to preserve color values. This parameter is optional, and its default value is **False**.

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Application.CreateRect

Function **CreateRect([***x As Double***], [***y As Double***], [***Width As Double***], [***Height As Double***])** As Rect

Description

Member of Application

The **CreateRect** method creates an object represeneting a rectangular area in a document.

Parameter	Description
x	Specifies the horizontal position of the rectangular area. This parameter is optional, and its default value is **0**.
y	Specifies the vertical position of the rectangular area. This parameter is optional, and its default value is **0**.
Width	Specifies the width of the rectangular area. This parameter is optional, and its default value is **0**.
Height	Specifies the height of the rectangular area. This parameter is optional, and its default value is **0**.

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Application.CreateRegistrationColor

Function **CreateRegistrationColor()** As Color

Description

Member of Application

The **CreateRegistrationColor** method creates a registration Color object. Registration color is printed on all separation plates.

VBA example

The following VBA example creates cropmarks around one or more selected objects and assigns registration color to each of them.

```
Sub Test()
 Const Offset As Double = 1 ' Cropmark offset
 Const Length As Double = 5 ' Cropmark length
 Dim x As Double, y As Double, w As Double, h As Double
 Dim c As Color
 ActiveDocument.Unit = cdrMillimeter
 Set c = CreateRegistrationColor()
 ActiveSelection.GetBoundingBox x, y, w, h, True
 ActiveLayer.CreateLineSegment(x - Length - Offset, y, x - Offset, y).Outline.Color = c
 ActiveLayer.CreateLineSegment(x, y - Length - Offset, x, y - Offset).Outline.Color = c
 ActiveLayer.CreateLineSegment(x + w + Offset, y, x + w + Length + Offset, y).Outline.Color = c
 ActiveLayer.CreateLineSegment(x + w, y - Length - Offset, x + w, y - Offset).Outline.Color = c
 ActiveLayer.CreateLineSegment(x - Length - Offset, y + h, x - Offset, y + h).Outline.Color = c
 ActiveLayer.CreateLineSegment(x, y + h + Length + Offset, x, y + h + Offset).Outline.Color = c
 ActiveLayer.CreateLineSegment(x + w + Offset, y + h, x + w + Length + Offset, y + h).Outline.Color = c
 ActiveLayer.CreateLineSegment(x + w, y + h + Length + Offset, x + w, y + h + Offset).Outline.Color = c
End Sub
```

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Application.CreateRGBColor

Function **CreateRGBColor(***Red As Long*, *Green As Long*, *Blue As Long***)** As Color

Description

Member of Application

The **CreateRGBColor** method creates a color object based on the RGB color model.

Parameter	Description
Red	Specifies the level of red in the RGB color model. Values range from **0** to **255**.
Green	Specifies the level of green in the RGB color model. Values range from **0** to **255**.
Blue	Specifies the level of blue in the RGB color model. Values range from **0** to **255**.

VBA example 1

The following VBA example creates a rectangle that covers the current page, and applies a red-to-black fountain fill to the rectangle.

```
Sub Test()
 Dim s As Shape
 Set s = ActiveLayer.CreateRectangle(0, 0, ActivePage.SizeWidth, ActivePage.SizeHeight)
 s.Fill.ApplyFountainFill CreateRGBColor(255, 0, 0), CreateRGBColor(0, 0, 0)
End Sub
```

VBA Example 2

Here is another code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.CreateSegmentRange

Function **CreateSegmentRange()** As SegmentRange

Description

Member of Application

The **CreateSegmentRange** method creates an empty SegmentRange object.

VBA example

The following VBA example creates a curve, adds its segments to a segment range, and changes the segments in the range to line segments.

```
Sub Test()
 Dim s As Shape, sp As SubPath, crv As Curve
 Dim sr As SegmentRange
 Dim i As Integer
 Set crv = Application.CreateCurve(ActiveDocument)
 Set sp = crv.CreateSubPath(1.351, 8.545)
 sp.AppendCurveSegment 1.351, 8.926, 0.127, 89.901, 0.127, -64.56
 sp.AppendCurveSegment 1.156, 8.952, 0.066, 115.44, 0.066, -48.906
 sp.AppendCurveSegment 1.156, 9.15, 0.065, 131.09, 0.065, -133.149
 sp.AppendCurveSegment 1.351, 9.163, 0.065, 46.846, 0.065, -116.315
 sp.AppendCurveSegment 1.351, 9.545, 0.127, 63.683, 0.127, -89.902
 sp.AppendCurveSegment 0.976, 9.545, 0.125, 179.951, 0.125, 25.612
 sp.AppendCurveSegment 0.96, 9.342, 0.063, -154.391, 0.063, 40.943
 sp.AppendCurveSegment 0.767, 9.339, 0.067, -139.06, 0.067, -41.987
 sp.AppendCurveSegment 0.752, 9.547, 0.063, 138.014, 0.065, -33.906
 sp.AppendCurveSegment 0.351, 9.545, 0.134, 146.087, 0.134, 0.045
 sp.AppendCurveSegment 0.351, 9.163, 0.127, -90#, 0.127, 63.681
 sp.AppendCurveSegment 0.156, 9.15, 0.065, -116.317, 0.065, 46.846
 sp.AppendCurveSegment 0.156, 8.952, 0.065, -133.152, 0.065, 131.093
 sp.AppendCurveSegment 0.351, 8.926, 0.066, -48.906, 0.066, 115.439
 sp.AppendCurveSegment 0.351, 8.545, 0.127, -64.561, 0.127, 90#
 sp.AppendCurveSegment 0.752, 8.547, 0.134, 0.002, 0.134, 146.087
 sp.AppendCurveSegment 0.767, 8.339, 0.065, -33.908, 0.063, 138.012
 sp.AppendCurveSegment 0.96, 8.342, 0.067, -41.987, 0.067, -139.058
 sp.AppendCurveSegment 0.976, 8.545, 0.063, 40.943, 0.063, -154.388
 sp.AppendCurveSegment 1.351, 8.545, 0.125, 25.613, 0.125, 179.998
 sp.Closed = True
 Set s = ActiveLayer.CreateCurve(crv)
 Set sr = CreateSegmentRange
 For i = 1 To s.Curve.Segments.Count
  sr.Add s.Curve.Segments(i)
 Next i
 sr.SetType cdrLineSegment
End Sub
```

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Application.CreateShapeRange

Function **CreateShapeRange()** As ShapeRange

Description

Member of Application

The **CreateShapeRange** creates an empty ShapeRange object.

VBA example

The following VBA example creates six red ellipses and then rotates them.

```
Sub Test()
 Dim n As Long
 Dim sr As ShapeRange
 Dim s As Shape
 Set sr = CreateShapeRange
 Set s = ActiveLayer.CreateEllipse2(0, 0, 0.5)
 s.RotationCenterX = 1
 s.RotationCenterY = 1
 For n = 1 To 5
  sr.Add s.Duplicate
  s.Rotate 60
 Next n
 sr.Add s
 sr.ApplyUniformFill CreateRGBColor(255, 0, 0)
End Sub
```

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Application.CreateSpotColor

Function **CreateSpotColor(***PaletteIdentifier As String***,** *SpotColorID As Long***,** **[***Tint As Long=100***])** As Color

Description

Member of Application

The **CreateSpotColor** method creates a spot-color object by using its ID.

Parameter	Description
PaletteIdentifier	Specifies the unique identifier of the color palette
SpotColorID	Specifies the ID of the spot color
Tint	Specifies the tint of the spot color. This parameter is optional, and its default value is **100**.

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Application.CreateSpotColorByName

Function **CreateSpotColorByName(***PaletteIdentifier As String***,** *SpotColorName As String***, [***Tint As Long=100***])** As Color

Description

Member of Application

The **CreateSpotColorByName** method creates a spot-color object by using its name.

Parameter	Description
PaletteIdentifier	Specifies the unique identifier of the color palette
SpotColorName	Specifies the name of the spot color
Tint	Specifies the tint of the spot color. This parameter is optional, and its default value is **100**.

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Application.CreateStructCreateOptions

Function **CreateStructCreateOptions()** As StructCreateOptions

Description

Member of Application

The **CreateStructCreateOptions** method creates a StructCreateOptions object.

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Application.CreateStructExportOptions

Function **CreateStructExportOptions()** As StructExportOptions

Description

Member of Application

The **CreateStructExportOptions** method creates a StructExportOptions object.

VBA example

The following VBA example creates a few shapes and exports the document to the JPEG format.

```
Sub Test()
 Dim exp As ExportFilter
 Dim se As StructExportOptions
 ActiveLayer.CreateRectangle 1, 2, 3, 4
 ActiveLayer.CreateRectangle2 4, 4, 5, 5
 ActiveLayer.CreatePolygon 1, 3, 4, 5, 6
 Set se = CreateStructExportOptions
 With se
  .AntiAliasingType = cdrNormalAntiAliasing
  .ImageType = cdrRGBColorImage
  .ResolutionX = 97
  .ResolutionY = 97
 End With
 Set exp = ActiveDocument.ExportEx("c:\primitives.jpg", cdrJPEG, cdrCurrentPage, se)
 exp.Finish
End Sub
```

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Application.CreateStructImportOptions

Function **CreateStructImportOptions()** As StructImportOptions

Description

Member of Application

The **CreateStructImportOptions** method creates a StructImportOptions object.

VBA example

The following VBA example imports a JPEG file and links it externally.

```
Sub TestImport()
 Dim im As ImportFilter
 Dim si As StructImportOptions
 Set si = CreateStructImportOptions
 si.LinkBitmapExternally = True
 Set im = ActiveLayer.ImportEx("c:\primitives.jpg", , si)
 im.Finish
End Sub
```

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Application.CreateStructOpenOptions

Function **CreateStructOpenOptions()** As StructOpenOptions

Description

Member of Application

The **CreateStructOpenOptions** method creates a StructOpenOptions object.

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Application.CreateStructPaletteOptions

Function **CreateStructPaletteOptions()** As StructPaletteOptions

Description

Member of Application

The **CreateStructPaletteOptions** method creates a StructPaletteOptions object.

VBA example

The following VBA example exports a document to GIF format and sets white to transparent.

```
Sub Test()
 Dim exp As ExportFilter
 Dim pal As StructPaletteOptions
 Dim se As StructExportOptions
 Dim n As Long
 Dim i As Integer
 Set pal = CreateStructPaletteOptions
 Set se = CreateStructExportOptions
 With pal
  .DitherType = cdrDitherNone
  .Smoothing = 5
  .PaletteType = cdrPaletteOptimized
 End With
 With se
  .AntiAliasingType = cdrNormalAntiAliasing
  .Dithered = False
  .ResolutionX = 96
  .ResolutionY = 96
 End With
 Set exp = ActiveDocument.ExportEx("c:\transparent.gif", cdrGIF, , se, pal)
 With exp
  n = .NumColors
  For i = 1 To n
   If .PaletteColor(i) = RGB(255, 255, 255) Then
     .ColorIndex = i
     .Transparency = 1
    Exit For
   End If
  Next i
   .Finish
 End With
End Sub
```

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Application.CreateStructPasteOptions

Function **CreateStructPasteOptions()** As StructPasteOptions

Description

Member of Application

The **CreateStructPasteOptions** method creates a StructPasteOptions object.

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Application.CreateStructSaveAsOptions

Function **CreateStructSaveAsOptions()** As StructSaveAsOptions

Description

Member of Application

The **CreateStructSaveAsOptions** creates a StructSaveAsOptions object.

VBA example

The following VBA example saves each open document to the drive labelled as “C.”

```
Sub TestSaveAs()
 Dim sa As StructSaveAsOptions
 Dim d As Document
 Set sa = CreateStructSaveAsOptions
 With sa
  .EmbedICCProfile = True
  .EmbedVBAProject = True
  .Filter = cdrCDR
  .IncludeCMXData = True
  .Overwrite = True
  .ThumbnailSize = cdr10KColorThumbnail
  .Version = cdrCurrentVersion
 End With
 For Each d In Documents
  d.SaveAs "c:\" & CStr(d.Index) & ".cdr", sa
 Next d
End Sub
```

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Application.CreateYIQColor

Function **CreateYIQColor(***y As Long*, *I As Long*, *Q As Long***)** As Color

Description

Member of Application

The **CreateYIQColor** method creates a color object based on the YIQ color model.

Parameter	Description
y	Specifies the luminance, or brightness, value in the YIQ color model. Values range from **0** to **255**.
I	Specifies one of two chromaticity values in the YIQ color model. Values range from **0** to **255**.
Q	Specifies one of two chromaticity values in the YIQ color model. Values range from **0** to **255**.

Example

Here is a code example:

```
Sub CreateColorExample()
CreateDocument
Dim sFace As Shape
Set sFace = ActiveLayer.CreateEllipse(1, 10, 8, 1.5)
sFace.Fill.UniformColor = CreateCMYKColor(0, 0, 100, 0) 'Create a yellow fill
Dim sEyeLeft As Shape
Set sEyeLeft = ActiveLayer.CreateEllipse2(3.25, 7.25, 1, -1)
sEyeLeft.Fill.UniformColor = CreateBWColor(True) 'Create a white fill
Dim sEyeRight As Shape
Set sEyeRight = ActiveLayer.CreateEllipse2(6, 7, 1, -1)
sEyeRight.Fill.UniformColor = CreateRGBColor(255, 255, 255) 'Create a white fill
Dim sEyeLeftPupil As Shape
Set sEyeLeftPupil = ActiveLayer.CreateEllipse2(3.5, 7, 0.35, -0.4)
sEyeLeftPupil.Fill.UniformColor = CreateBWColor(False)
Dim sEyeRightPupil As Shape
Set sEyeRightPupil = sEyeLeftPupil.Clone
sEyeRightPupil.Fill.ApplyFountainFill CreateHSBColor(210, 10, 10), CreateGrayColor(72), cdrConicalFountainFill
sEyeRightPupil.Move 2.75, -0.35
Dim sNose As Shape
Set sNose = ActiveLayer.CreateEllipse2(4.3, 5.4, -0.75, -0.5)
sNose.Rotate -10
sNose.Fill.ApplyFountainFill CreateCMYColor(0, 255, 255), CreateHLSColor(0, 75, 75), cdrRadialFountainFill
Dim sMouth As Shape
Set sMouth = ActiveLayer.CreateEllipse2(4.2, 3.75, 1.75, 0.75, 0, 180, True)
sMouth.Rotate -15
sMouth.Fill.ApplyFountainFill CreateYIQColor(97, 97, 100), CreateLabColor(215, -1, -2), cdrSquareFountainFill
End Sub
```

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Application.DefaultColorContext

Property **DefaultColorContext** As ColorContext

Description

Member of Application

The **DefaultColorContext** property returns a read-only value: the default color context for the application, which is used to supply color context to new documents.

- Because the **Application.DefaultColorContext** property is read-only, it cannot be used to modify the application's default color context. Instead, you must use the ColorManager.DefaultColorContext property for this purpose.

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Application.Documents

Property **Documents** As Documents

Description

Member of Application

The **Documents** property returns the collection of all open documents in the application.

The **Documents** property returns a read-only value.

VBA example

The following VBA example checks for open documents. If a document is open, text is added and if there is no open document, a message displays in a message box.

```
Sub ApplicationDocument()
 If Documents.Count > 0 Then
  ActiveLayer.CreateArtisticText 0, 0, "Text"
 Else
  MsgBox " There is no document open."
 End If
End Sub
```

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Application.EnhancedOutlines

Property **EnhancedOutlines** As OutlineStyles

Description

Member of Application

The **EnhancedOutlines** property returns a collection of enhanced outline styles.

The **EnhancedOutlines** property returns a read-only value.

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Application.Evaluate

Function **Evaluate(***Expression As String***)** As Variant

Description

Member of Application

The **Evaluate** method evaluates the given expression and returns the result of the expression.

- You can use the **Evaluate** method to perform queries with the Corel Query Language (CQL). For information on queries, see the section "Including queries in macros."

Parameter	Description
Expression	Specifies the CQL expression

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Application.EventsEnabled

Property **EventsEnabled** As Boolean

Description

Member of Application

The **EventsEnabled** property returns or specifies a value indicating the enabled status of an event in CorelDRAW. An event is enabled if it is ready and able to execute in the application. If an event is enabled, its **EventsEnabled** status is **True**. If the event is disabled, the status is set to **False**. Events can be disabled to temporarily improve performance when performing routine operations

VBA example

The following VBA example disables events before creating a second ellipse.

```
Sub EnabledEvents()
 Dim state As Boolean
 ActiveLayer.CreateEllipse2 0, 0, 5
 'GlobalDocument_ShapeCreated event will be generated here.
 state = EventsEnabled 'Remember the flag state.
 EventsEnabled = False 'Disable events.
 ActiveLayer.CreateEllipse2 3, 3, 5
 ' This event wil not be generated.
 EventsEnabled = state 'Restore event state.
End Sub
```

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Application.FontList

Property **FontList** As FontList

Description

Member of Application

The **FontList** property returns the list of all fonts installed on the system.

The **FontList** property returns a read-only value.

VBA example

The following VBA example displays the name of the fourth installed font.

```
Sub ListFontName()
 MsgBox FontList(4)
End Sub
```

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Application.ForceUpdateFontTable

Function **ForceUpdateFontTable()**

Description

Member of Application

The **ForceUpdateFontTable** method forces the application to pick up all newly installed fonts immediately.

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Application.FrameWork

Property **FrameWork** As FrameWork

Description

Member of Application

The **FrameWork** property returns a FrameWork object.

The **FrameWork** property returns a read-only value.

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Application.GetSupportedOpenTypeFeatures

Function **GetSupportedOpenTypeFeatures()** As String**()**

Description

Member of Application

The **GetSupportedOpenTypeFeatures** method returns a list of OpenType features currently supported by the application.

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Application.GlobalUserData

Property **GlobalUserData** As Properties

Description

Member of Application

The **GlobalUserData** property returns a set of custom user settings, which are preserved between application sessions.

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Application.GMSManager

Property **GMSManager** As GMSManager

Description

Member of Application

The **GMSManager** property returns a Global Macro Storage (GMS) Manager, or GMSManager, object. A GMS manager lets you run VBA macros from other programs, as well as manage GMS modules.

The **GMSManager** property returns a read-only value.

VBA example

The following VBA example runs a macro, **Cropmarks**, in the **Main** module of GMS **CropMarks**.

```
Sub Test()
 GMSManager.RunMacro "CropMarks", "Main.Cropmarks"
End Sub
```

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Application.ID

Property **ID** As cdrApplicationID

Description

Member of Application

The **ID** property returns the application ID.

The **ID** property returns a read-only value.

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Application.ImportWorkspace

Sub **ImportWorkspace**(*FileName As String*)

Description

Member of Application

The **ImportWorkspace** method imports workspace elements into the current workspace.

Parameter	Description
FileName	Specifies the full path name of the workspace file that is imported into CorelDRAW

VBA example

The following VBA example imports the specified workspace and merges it with the current workspace.

```
Sub Test()
 ImportWorkspace "C:\Temp\MyToolbar.cwf"
End Sub
```

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Application.InitializeVBA

Function **InitializeVBA()** As Boolean

Description

Member of Application

The **InitializeVBA** method loads and initializes the Macro Editor.

This method is called from an automation controller such as Visual Basic (VB) or C++.

VBA example

The following VBA example runs the specified VBA macro, “CreatePuzzlePiece,” from a VB project.

The first subroutine should be placed in a code module named “TestModule.” The second subroutine should be placed in the code of a form module of a standard EXE project in VB. There must be a command button on the form.

```
Public Sub CreatePuzzlePiece()
 Dim s As Shape, sp As SubPath, crv As Curve
 Dim i As Integer
 Optimization = True
 Set crv = Application.CreateCurve(ActiveDocument)
 Set sp = crv.CreateSubPath(1.351, 8.545)
 sp.AppendCurveSegment 1.351, 8.926, 0.127, 89.901, 0.127, -64.56
 sp.AppendCurveSegment 1.156, 8.952, 0.066, 115.44, 0.066, -48.906
 sp.AppendCurveSegment 1.156, 9.15, 0.065, 131.09, 0.065, -133.149
 sp.AppendCurveSegment 1.351, 9.163, 0.065, 46.846, 0.065, -116.315
 sp.AppendCurveSegment 1.351, 9.545, 0.127, 63.683, 0.127, -89.902
 sp.AppendCurveSegment 0.976, 9.545, 0.125, 179.951, 0.125, 25.612
 sp.AppendCurveSegment 0.96, 9.342, 0.063, -154.391, 0.063, 40.943
 sp.AppendCurveSegment 0.767, 9.339, 0.067, -139.06, 0.067, -41.987
 sp.AppendCurveSegment 0.752, 9.547, 0.063, 138.014, 0.065, -33.906
 sp.AppendCurveSegment 0.351, 9.545, 0.134, 146.087, 0.134, 0.045
 sp.AppendCurveSegment 0.351, 9.163, 0.127, -90#, 0.127, 63.681
 sp.AppendCurveSegment 0.156, 9.15, 0.065, -116.317, 0.065, 46.846
 sp.AppendCurveSegment 0.156, 8.952, 0.065, -133.152, 0.065, 131.093
 sp.AppendCurveSegment 0.351, 8.926, 0.066, -48.906, 0.066, 115.439
 sp.AppendCurveSegment 0.351, 8.545, 0.127, -64.561, 0.127, 90#
 sp.AppendCurveSegment 0.752, 8.547, 0.134, 0.002, 0.134, 146.087
 sp.AppendCurveSegment 0.767, 8.339, 0.065, -33.908, 0.063, 138.012
 sp.AppendCurveSegment 0.96, 8.342, 0.067, -41.987, 0.067, -139.058
 sp.AppendCurveSegment 0.976, 8.545, 0.063, 40.943, 0.063, -154.388
 sp.AppendCurveSegment 1.351, 8.545, 0.125, 25.613, 0.125, 179.998
 sp.Closed = True
 Set s = ActiveLayer.CreateCurve(crv)
 Optimization = False
 ActiveWindow.Refresh
 Application.Refresh
End Sub
Private Sub Command1_Click()
 Dim cDraw As Object
 Dim gms As CorelDRAW.GMSManager
 Set cDraw = CreateObject("CorelDRAW.Application")
 cDraw.CreateDocument
 cDraw.InitializeVBA
 cDraw.Visible = True
 Set gms = cDraw.GMSManager
 gms.RunMacro "GlobalMacros", "TestModule.CreatePuzzlePiece"
End Sub
```

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Application.IsUILanguageAvailable

Function **IsUILanguageAvailable(**Language As cdrTextLanguage**)** As Boolean

Description

Member of Application

The **IsUILanguageAvailable** method determines whether a specified user-interface language is installed.

Parameter	Description
Language	Specifies the user-interface language for which to check

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Application.LanguagePath

Property **LanguagePath** As String

Description

Member of Application

The **LanguagePath** property returns the path to the language-specific data files.

The **LanguagePath** property returns a read-only value.

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Application.MainMenu

Property **MainMenu** As CommandBar

Description

Member of Application

The **MainMenu** property returns the main menu.

The **MainMenu** property returns a read-only value.

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Application.Name

Property **Name** As String

Description

Member of Application

The **Name** property returns the application name. It is the default property of the Application class.

The **Name** property returns a read-only value.

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Application.OpenCorelScriptFile

Function **OpenCorelScriptFile(***FileName As String***)** As CorelScriptFile

Description

Member of Application

The **OpenCorelScriptFile** method opens a Corel SCRIPT file and returns a reference to the **CorelScriptFile** object which allows you to load and execute a Corel SCRIPT program.

Parameter	Description
FileName	Specifies the full path name of the Corel SCRIPT file to open and execute

VBA example

The following VBA example opens an existing Corel SCRIPT file. An ellipse is created on the active layer of the current document and the script runs to create a fill effect, Alien Sky, on the ellipse.

```
Sub Test()
 Dim csf As CorelScriptFile
 Set csf = OpenCorelScriptFile("C:\Corel\Graphics10\Draw\Fill_Out\Fills\Fountain\aliensky.csc")
 ActiveDocument.ActiveLayer.CreateEllipse2 3, 3, 2
 csf.Play
End Sub
```

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Application.OpenDocument

Function **OpenDocument(***FileName As String***, [***CodePage As Long***])** As Document

Description

Member of Application

The **OpenDocument** method opens an existing document in CorelDRAW.

Parameter	Description
FileName	Specifies the full path name of the existing document to open
CodePage	Specifies the document encoding (code page or character set) to use for the document. This parameter is optional, and its default value is **0**.

VBA example

The following VBA example opens an existing CorelDRAW (CDR) document, adds a page to the document, creates a colored ellipse on the new page, and saves the entire document.

```
Sub DocumentOpen()
 Dim doc As Document
 Set doc = OpenDocument("C:\Flower.cdr")
 With doc
  .AddPages (1)
  .ActiveLayer.CreateEllipse(0, 3, 5, 1).Fill.UniformColor.CMYKAssign 0, 100, 100, 0
  .Save
 End With
End Sub
```

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Application.OpenDocumentAsCopy

Function **OpenDocumentAsCopy(***FileName As String***, [**Options As StructOpenOptions**])** As Document

Description

Member of Application

The **OpenDocumentAsCopy** method opens an existing document without maintaining a link to the filename of the original document.

Parameter	Description
FileName	Specifies the full path name of the existing document to open
Options	Specifies, through a value of StructOpenOptions, the settings to apply to the document. This parameter is optional, and its default value is **Nothing**.

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Application.OpenDocumentEx

Function **OpenDocumentEx(***FileName As String***,** Options As StructOpenOptions**)** As Document

Description

Member of Application

The **OpenDocumentEx** method opens an existing document and applies the specified settings to it.

Parameter	Description
FileName	Specifies the full path name of the existing document to open
StructOpenOptions	Specifies the settings to apply to the document

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Application.Optimization

Property **Optimization** As Boolean

Description

Member of Application

The **Optimization** property returns or specifies the optimization, the ability to suppress screen redraws when working with documents and shapes.

Setting the **Optimization** property to **True** disables some internal updating. It is important to disable optimization when exiting a macro, or CorelDRAW may not update its screens. You may also want to refresh the document window after disabling optimization so the display is updated and reflects all changes.

VBA example

The following VBA example creates 100 random circles in the active document, with optimization enabled for increased performance.

```
Sub Optimize()
 Dim i As Long
 Dim x As Double, y As Double, r As Double
 Dim n As Long
 Dim num As Long
 Dim MaxX As Double, MaxY As Double, MaxR As Double
 MaxX = ActivePage.SizeWidth
 MaxY = ActivePage.SizeHeight
 MaxR = 1
 num = ActivePalette.ColorCount
 Optimization = True
 For i = 1 To 100
  x = Rnd() * MaxX
  y = Rnd() * MaxY
  r = Rnd() * MaxR
  n = CLng(Fix(Rnd() * num)) + 1
  Set s = ActiveLayer.CreateEllipse2(x, y, r)
  s.Fill.ApplyUniformFill ActivePalette.Color(n)
 Next i
 Optimization = False
 ActiveWindow.Refresh
End Sub
```

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Application.OutlineStyles

Property **OutlineStyles** As OutlineStyles

Description

Member of Application

The **OutlineStyles** property returns the collection of available outline style presets.

The **OutlineStyles** property returns a read-only value.

VBA example

The following VBA example applies predefined outline styles to selected objects. The style that is applied depends on the type of shape selected; for example, if the shape is a rectangle, the first preset outline style is applied to the outline of the rectangle. Only shapes that have an outline are included the selection.

```
Sub StylesOutline()
 Dim s As Shape
 For Each s In ActiveSelection.Shapes
  If s.Outline.Type = cdrOutline Then
   Select Case s.Type
    Case cdrRectangleShape
     s.Outline.Style = OutlineStyles(1)
    Case cdrEllipseShape
     s.Outline.Style = OutlineStyles(2)
    Case cdrPolygonShape
     s.Outline.Style = OutlineStyles(3)
   End Select
  End If
 Next s
End Sub
```

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Application.PageSizes

Property **PageSizes** As PageSizes

Description

Member of Application

The **PageSizes** property returns the collection of all page sizes defined for the application.

The **PageSizes** property returns a read-only value.

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Application.PaletteManager

Property **PaletteManager** As PaletteManager

Description

Member of Application

The **PaletteManager** property returns an instance of a PaletteManager object.

The **PaletteManager** property returns a read-only value.

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Application.Palettes

Property **Palettes** As Palettes

Description

Member of Application

The **Palettes** property returns the collection of open color palettes.

The **Palettes** property returns a read-only value.

VBA example

The following VBA example displays the name of all palettes in use, and the number of colors in each palette, in a message box.

```
Sub PalettesList()
 Dim pal As Palette
 Dim s As String
 s = "Active palettes. "
 For Each pal In Palettes
  s = s & pal.Name & " (" & pal.ColorCount & " colors)"
 Next pal
 MsgBox s
End Sub
```

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Application.PanoseMatching

Property **PanoseMatching** As cdrPanoseMatchingType

Description

Member of Application

The **PanoseMatching** property returns or specifies the default behavior when the document being opened contains fonts that are not installed on the system.

VBA example

The following VBA example specifies that non-existing fonts are replaced permanently and automatically with the closest font matches available on the system.

```
Sub MatchingPanose()
 PanoseMatching = cdrPanosePermanent
 OpenDocument "C:\My Documents\Graphic1.cdr"
End Sub
```

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Application.Parent

Property **Parent** As Application

Description

Member of Application

The **Parent** property returns a read-only value that is associated with the properties, methods, and controls of an object's parent in the CorelDRAW hierarchy of objects. The application object refers to the CorelDRAW application.

A parent object acts as a container for child objects. For example, a form (as a container) is the parent object of a check box on that form.

Most objects have either a **Parent** property or a **Collection** property, pointing to the object's parent object in the object model. The **Collection** property is used if the parent object is a collection.

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Application.Path

Property **Path** As String

Description

Member of Application

The **Path** property returns a string value associated with the full path identity of the CorelDRAW folder.

The **Path** property returns a read-only value.

VBA example

The following VBA example displays the path of the CorelDRAW folder in a message box.

```
Sub PathApplication()
 MsgBox Path
End Sub
```

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Application.PatternCanvases

Property **PatternCanvases** As PatternCanvases

Description

Member of Application

The **PatternCanvases** property returns the collection of available PatternCanvas objects (two-color pattern fills).

The **PatternCanvases** property returns a read-only value.

VBA example

The following VBA example applies a predefined pattern fill, overlaid with a checkerboard pattern, to an ellipse.

```
Sub CanvasPattern()
 Dim s As Shape
 Dim pf As PatternFill
 Dim cnv As New PatternCanvas
 cnv.PutCopy PatternCanvases(2)
 Set s = ActiveLayer.CreateEllipse(8, 7, 3, 4)
 s.Fill.ApplyNoFill
 Set pf = s.Fill.ApplyPatternFill(cdrTwoColorPattern, , , CreateColorEx(5005, 255, 0, 0), _
 CreateColorEx(5005, 255, 255, 0))
 pf.Canvas = cnv
End Sub
```

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Application.PlatformVersionMajor

Property **PlatformVersionMajor** As Long

Description

Member of Application

The **PlatformVersionMajor** property returns the major version of the application automation engine.

The **PlatformVersionMajor** property returns a read-only value.

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Application.PlatformVersionMinor

Property **PlatformVersionMinor** As Long

Description

Member of Application

The **PlatformVersionMinor** property returns the minor version of the application automation engine.

The **PlatformVersionMinor** property returns a read-only value.

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Application.Printers

Property **Printers** As SystemPrinters

Description

Member of Application

The **Printers** property returns the SystemPrinters collection object that contains a list of installed printers on the user's system.

The **Printers** property returns a read-only value.

VBA example

The following VBA example displays the name of the default printer and the port to which it is connected.

```
Sub Test()
 If Printers.Count = 0 Then
  MsgBox "There's no printer installed on the system."
  Exit Sub
 End If
 MsgBox "Default printer is: " & Printers.Default.Name & _
  " on port: " & Printers.Default.Port
End Sub
```

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Application.PrintJob

Property **PrintJob** As PrintJob

Description

Member of Application

The **PrintJob** property returns a read-only value that is associated with a print job in CorelDRAW.

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Application.ProgramPath

Property **ProgramPath** As String

Description

Member of Application

The **ProgramPath** property returns a read-only string that shows the full path to the application's **Programs** folder.

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Application.Quit

Sub **Quit()**

Description

Member of Application

The **Quit** method closes the current session of CorelDRAW.

VBA example

The following VBA example terminates the current session of CorelDRAW.

```
Sub QuitApp()
 Quit
End Sub
```

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Application.RecentFiles

Property **RecentFiles** As RecentFiles

Description

Member of Application

The **RecentFiles** property returns the collection of all files in the recent-file list of CorelDRAW.

The **RecentFiles** property returns a read-only value.

VBA example

The following VBA example displays a list of all recent files, with full file names, in a message box.

```
Sub RecentFilesList()
 Dim s As String
 Dim rf As RecentFile
 s = "Recent Files" & vbCrLf
 For Each rf In RecentFiles
  s = s & vbCrLf & rf.Name & " (" & rf.FullName & ")"
 Next rf
 MsgBox s
End Sub
```

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Application.Refresh

Sub **Refresh()**

Description

Member of Application

The **Refresh** method updates all open dockers.

VBA example

The following VBA example updates the **Properties** docker with the new width for the outline. The **Outline** tab of the **Properites** docker must be open.

```
Sub Test()
 Dim rect As Shape
 Optimization = True
 Set rect = ActiveLayer.CreateRectangle(0, 0, 3, 3)
 rect.Outline.Width = 0.4
 Optimization = False
 ActiveWindow.Refresh
 Refresh
End Sub
```

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Application.RemovePluginCommand

Function **RemovePluginCommand(***CommandID As String***)** As Boolean

Description

Member of Application

The **RemovePluginCommand** method unregisters a plugin command.

Parameter	Description
CommandID	Specifies the ID of the plugin command

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Application.SessionUserData

Property **SessionUserData** As Properties

Description

Member of Application

The **SessionUserData** property returns a set of custom user settings that are available only during the current application session.

The **SessionUserData** property returns a read-only value.

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Application.SetupPath

Property **SetupPath** As String

Description

Member of Application

The **SetupPath** property returns the root path of the software installation.

The **SetupPath** property returns a read-only value.

VBA example

The following VBA example displays, in a message box, the path of the setup folder for the software.

```
Sub PathSetup()
 MsgBox SetupPath
End Sub
```

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Application.StartupMode

Property **StartupMode** As cdrAppStartupMode

Description

Member of Application

The **StartupMode** property returns or specifies the default application action at startup.

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Application.Status

Property **Status** As AppStatus

Description

Member of Application

The **Status** property returns the application-status (or AppStatus) object.

The **Status** property returns a read-only value.

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Application.StatusBar

Property **StatusBar** As CommandBar

Description

Member of Application

The **StatusBar** property returns the status bar.

The **StatusBar** property returns a read-only value.

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Application.SymbolLibraries

Property **SymbolLibraries** As SymbolLibraries

Description

Member of Application

The **SymbolLibraries** property lists all symbol libraries visible in the **Symbols** docker.

The **SymbolLibraries** property returns a read-only value.

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Application.UILanguage

Property **UILanguage** As cdrTextLanguage

Description

Member of Application

The **UILanguage** property returns the current user-interface language.

The **UILanguage** property returns a read-only value.

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Application.UnadviseEvents

Sub **UnadviseEvents**(*Cookie As Long*)

Description

Member of Application

The **UnadviseEvents** method unadvises application events.

Parameter	Description
Cookie	

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Application.Unit

Property **Unit** As cdrUnit

Description

Member of Application

The **Unit** property specifies the default unit of measurements for methods and properties when there is no document context.

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Application.UserDataPath

Property **UserDataPath** As String

Description

Member of Application

The **UserDataPath** property returns the path to the folder where the user's personal data resides.

The **UserDataPath** property returns a read-only value.

VBA example

The following VBA example displays all installed GMS files in the GMS folder.

```
Sub Test()
 Dim s As String
 Dim s1 As String
 s = Dir(UserDataPath & "GMS\*.gms")
 Do
  If s = "" Then Exit Do
  s = s & s1 & vbCrLf
  s1 = Dir
 Loop Until s1 = ""
 s = s & Dir(SetupPath & "Draw\GMS\*.gms")
 Do
  If s = "" Then Exit Do
  s = s & s1 & vbCrLf
  s1 = Dir
 Loop Until s1 = ""
 MsgBox s, , "The following GMS files are installed:"
End Sub
```

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Application.UserWorkspacePath

Property **UserWorkspacePath** As String

Description

Member of Application

The **UserWorkspacePath** property returns the path to the user workspace.

The **UserWorkspacePath** property returns a read-only value.

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Application.VBE

Property **VBE** As Object

Description

Member of Application

The **VBE** property returns a read-only reference to the Macro Editor.

VBA example

The following VBA example displays the editor type name in the **Debug** window of the Macro Editor in CorelDRAW.

```
Sub AppVBE()
 Dim obj as Object
 Set obj = Application.VBE
 Debug.Print TypeName (obj)
End Sub
```

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Application.Version

Property **Version** As String

Description

Member of Application

The **Version** property returns version information about CorelDRAW. The **Version** property contains the version major number (before the decimal) and version build number (after the decimal). For example if the current version is 14.123, the **Version** value would be **Version 14.123**.

The **Version** property returns a read-only value.

VBA example

The following VBA example displays the current version of CorelDRAW in a message box.

```
Sub AppVersion()
 MsgBox "You are using: " & Version
End Sub
```

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Application.VersionBuild

Property **VersionBuild** As Long

Description

Member of Application

The **VersionBuild** property returns version information about CorelDRAW. The **VersionBuild** property returns only the build number of the current version. The **VersionBuild** property returns the second part of the **Version** number (to the right of the decimal point). For example, if the current version is 14.123, the **VersionBuild** value is **123**.

The **VersionBuild** property returns a read-only value.

VBA example 1

The following VBA example displays the build number of the current CorelDRAW application in a message box.

```
Sub BuildVersion()
 MsgBox "You are using CorelDRAW build " & VersionBuild
End Sub
```

VBA example 2

The following VBA example detects earlier versions of CorelDRAW. Version properties were introduced in CorelDRAW 9 Office Edition.

```
Sub Main()
 Dim v&, b&
 GetDrawVersion v, b
 MsgBox v & "." & b
End Sub
Sub GetDrawVersion(v As Long, b As Long)
 On Error Resume Next
 Err.Clear
 v = CorelDRAW.Application.VersionMajor
 If Err.Number <> 0 Then
  v = 9
  b = 439
 Else
  b = CorelDRAW.Application.VersionBuild
 End If
 On Error GoTo 0
 MsgBox "You are using CorelDRAW " & v & "." & b
End Sub
```

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Application.VersionMajor

Property **VersionMajor** As Long

Description

Member of Application

The **VersionMajor** property returns version information about CorelDRAW. The **VersionMajor** property returns only the first part of the version number (to the left of the decimal place). For example, if the current version is 14.123, the major version value would be **14**.

The **VersionMajor** property returns a read-only value.

VBA example

The following VBA example checks the current version of CorelDRAW to ensure that the version number is at least 14.123.

```
Sub MajorVersion()
 If VersionMajor = 14 And VersionBuild < 123 Then
  MsgBox "This script required that CorelDRAW version 14.123 or later be installed."
 End If
End Sub
```

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Application.VersionMinor

Property **VersionMinor** As Long

Description

Member of Application

The **VersionMinor** property returns version information about CorelDRAW. The **VersionMinor** property is set to **0** (zero) for a mainstream application.

The **VersionMinor** property returns a read-only value.

VBA example

The following VBA example checks the current version of CorelDRAW to ensure that the version number is at least 14.123.

```
Sub MajorVersion()
 If VersionMajor = 14 And VersionBuild < 123 Then
  MsgBox "This script required that CorelDRAW version 14.123 or later be installed."
 End If
End Sub
```

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Application.Visible

Property **Visible** As Boolean

Description

Member of Application

The **Visible** property returns or specifies a value indicating the visibility of the application window. This property allows you to show or hide the CorelDRAW application window. If the **Visible** property is **True**, the application is running and visible on the monitor display.

VBA example

The following VBA example hides CorelDRAW by setting the **Visible** property to **False**, and it then shows CorelDRAW by setting the **Visible** property to **True**. Message boxes display messages indicating the visibility of the application.

```
Sub ApplicationVisible()
 Visible = False
 MsgBox "CorelDRAW is hidden."
 Visible = True
 MsgBox "CorelDRAW is now visible."
End Sub
```

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Application.Windows

Property **Windows** As Windows

Description

Member of Application

The **Windows** property returns the collection of document windows open in the application.

The **Windows** property returns a read-only value.

VBA example 1

The following VBA example sets the zoom level of all document windows to **200**%.

```
Sub WindowsZoom()
 Dim w As Window
 For Each w In Windows
  w.ActiveView.Zoom = 200
 Next w
End Sub
```

VBA example 2

The following VBA example tiles the document windows horizontally.

```
Sub WindowsTile()
 Windows.Arrange cdrTileHorizontally
End Sub
```

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Application.Workspaces

Property **Workspaces** As Workspaces

Description

Member of Application

The **Workspaces** property returns the collection of all workspaces available in the application.

The **Workspaces** property returns a read-only value.

VBA example

The following VBA example displays the available workspaces in a message box and adds an asterisk next to the name of the active workspace.

```
Sub WorkspaceCollection()
 Dim s As String
 Dim wks As Workspace
 s = "Available Workspaces: "
 For Each wks In Workspaces
  s = s & wks.Name
  If wks.Active Then s = s & " *"
 Next wks
 MsgBox s
End Sub
```

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AppStatus

Properties Methods

Description

The **AppStatus** class represents application status.

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AppStatus.Aborted

Property **Aborted** As Boolean

Description

Member of AppStatus

The **Aborted** property returns **True** if **Esc** was pressed.

The **Aborted** property returns a read-only value.

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AppStatus.BeginProgress

Sub **BeginProgress([***Message As String***], [**CanAbort As Boolean = False**])**

Description

Member of AppStatus

The **BeginProgress** method shows a progress meter in the application status bar area.

Parameter	Description
Message	Specifies a message. This parameter is optional.
CanAbort	Specifies whether the process can be aborted. This parameter is optional, and its default value is **False**.

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AppStatus.EndProgress

Sub **EndProgress()**

Description

Member of AppStatus

The **EndProgress** method hides the progress meter from the status bar.

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AppStatus.Progress

Property **Progress** As Long

Description

Member of AppStatus

The **Progress** property returns or specifies the current progress meter position.

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AppStatus.SetProgressMessage

Sub **SetProgressMessage(***Message As String***)**

Description

Member of AppStatus

The **SetProgressMessage** method specifies a custom message for the progress meter.

Parameter	Description
Message	Specifies the message

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AppStatus.UpdateProgress

Sub **UpdateProgress([**Step As Long = 1**])**

Description

Member of AppStatus

The **UpdateProgress** method updates the progres meter position.

Parameter	Description
Step	This parameter is optional, and its default value is **1**.

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AppWindow

Properties Methods

Description

The **AppWindow** class represents the application-window object — that is, the main CorelDRAW window.

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AppWindow.Activate

Sub **Activate()**

Description

Member of AppWindow

The **Activate** method makes the main application window active and gives it keyboard focus.

Some operating systems do not allow the application to activate itself, to prevent intervention with the current activity. If the application cannot be activated, the application window flashes on the taskbar, but it does not appear in the foreground.

VBA example

The following VBA example activates the CorelDRAW main application window if it is currently inactive. A window is inactive if it is open but not the currently used Windows application.

```
Sub Activate()
 If Not AppWindow.Active Then AppWindow.Activate
End Sub
```

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AppWindow.Active

Property **Active** As Boolean

Description

Member of AppWindow

The **Active** property returns a value indicating the status of the CorelDRAW main application window. If the window is active, a value of **True** is returned, indicating that it is currently in use. If the window is inactive, the return value is **False**, indicating that the main application window is currently open but not in use.

You can use the Activate method to change the inactive status of the main application window.

The **Active** property returns a read-only value.

VBA example

The following VBA example displays the status of the CorelDRAW main application window in a message box.

```
Sub AppWindowActive()
 If AppWindow.Active Then
  MsgBox "CorelDRAW is currently active."
 Else
  MsgBox "CorelDRAW is not active at this time."
 End If
End Sub
```

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AppWindow.Application

Property **Application** As Object

Description

Member of AppWindow

The Application object defines properties of the main CorelDRAW application. It is the top-level object in the CorelDRAW hierarchy of objects, and you can access all other features of the application through its functions, methods, and properties.

It is not necessary to explicitly specify an **Application** object; it is always available in VBA, and its properties and methods are global.

The **Application** property returns a read-only value that is associated with the main CorelDRAW application.

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AppWindow.Caption

Property **Caption** As String

Description

Member of AppWindow

The **Caption** property returns or specifies the display text of the main application window. The caption appears as text in the title bar of the application window.

If the document window is maximized when using the **Caption** property, the document title also displays after the caption in the title bar of the main application window.

VBA example 1

The following VBA example sets the caption of the CorelDRAW main application window.

```
Sub AppWindowCaption()
 AppWindow.Caption = "My Draw Application"
End Sub
```

VBA example 2

The following VBA example resets the caption.

```
Sub ResetCaption()
 Dim NewCaption As String
 NewCaption = "My New Caption"
 If AppWindow.Caption <> NewCaption Then
  AppWindow.Caption = NewCaption
 End If
End Sub
```

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AppWindow.ClientHeight

Property **ClientHeight** As Long

Description

Member of AppWindow

The **ClientHeight** property returns a numerical value that measures the height of a client area, the gray area where document windows display within the main application window of CorelDRAW. The area excludes dockers, docked toolbars, menus, the status bar, and palettes.

The **ClientHeight** property is measured in screen pixels.

The **ClientHeight** property returns a read-only value.

VBA example

The following VBA example displays the client height, in screen pixels.

```
Sub AppWindowClient()
 With AppWindow
  MsgBox .ClientHeight
 End With
End Sub
```

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AppWindow.ClientWidth

Property **ClientWidth** As Long

Description

Member of AppWindow

The **ClientWidth** property returns a numerical value that measures the width of a client area, the gray area where document windows display within the main application window of CorelDRAW. The area excludes dockers, docked toolbars, menus, the status bar, and palettes.

The **ClientWidth** property is measured in screen pixels.

The **ClientWidth** property returns a read-only value.

VBA example

The following VBA example displays the client width, in screen pixels, in a message box.

```
Sub AppWindowClient()
 With AppWindow
  MsgBox .ClientWidth
 End With
End Sub
```

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AppWindow.Handle

Property **Handle** As Long

Description

Member of AppWindow

The **Handle** property returns the window handle value of the main application window. The value can be used with calls to Windows API functions that require handles.

The **Handle** property returns a read-only value.

VBA example

The following VBA example shows a message using the Windows API function, MessageBox, and specifying the CorelDRAW window as the parent to the message box.

```
Private Declare Function MessageBox Lib "user32" Alias "MessageBoxA" _
 (hwnd As Long, lpText As String, _
 lpCaption As String, wType As Long) As Long
Sub Test()
 MessageBox AppWindow.Handle, "Some Message", "Some Caption", 64
End Sub
```

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AppWindow.Height

Property **Height** As Long

Description

Member of AppWindow

The **Height** property sets the height of the CorelDRAW main application window on the desktop.

This property applies only when the main application window is in a normal window state, that is, the window is not minimized or maximized.

The height of the application window is a numerical value measured in screen pixels.

VBA example

The following VBA example sets the current window state to normal, defines the new dimensions of the window, and resizes the CorelDRAW main application window according to the new dimensions.

```
Sub AppWindowHeight()
 With AppWindow
  .WindowState = cdrWindowNormal
  .Top = 10
  .Left = 10
  .Height = 500
  .Width = 600
 End With
End Sub
```

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AppWindow.Left

Property **Left** As Long

Description

Member of AppWindow

The **Left** property sets the distance, in screen pixels, from the CorelDRAW main application window's left border to the left side of the monitor's display area.

This property only applies when the main application window is in a normal window state; that is, the window is not minimized or maximized.

VBA example

The following VBA example sets the current window state to normal, defines the new dimensions of the window, and resizes the CorelDRAW main application window according to the new dimensions.

```
Sub AppWindowLeft()
 With AppWindow
  .WindowState = cdrWindowNormal
  .Top = 10
  .Left = 10
  .Height = 500
  .Width = 600
 End With
End Sub
```

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AppWindow.Parent

Property **Parent** As Object

Description

Member of AppWindow

The **Parent** property returns a read-only value that is associated with the properties, methods, and controls of an object's parent in the CorelDRAW hierarchy of objects. The application object refers to the CorelDRAW application.

A parent object acts as a container for child objects. For example, a form (as a container) is the parent object of a check box on that form.

Most objects have either a **Parent** property or a **Collection** property, pointing to the object's parent object in the object model. The **Collection** property is used if the parent object is a collection.

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AppWindow.Top

Property **Top** As Long

Description

Member of AppWindow

The **Top** property sets the distance, in screen pixels, from the CorelDRAW window's top border to the top of the monitor's display area.

This property only applies when the main application window is in a normal window state; that is, the window is not minimized or maximized.

VBA example

The following VBA example sets the current window state to normal, defines the new dimensions of the window, and resizes the CorelDRAW main application window according to the new dimensions.

```
Sub AppWindowTop()
 With AppWindow
  .WindowState = cdrWindowNormal
  .Top = 10
  .Left = 10
  .Height = 500
  .Width = 600
 End With
End Sub
```

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AppWindow.Width

Property **Width** As Long

Description

Member of AppWindow

The **Width** property sets the height of the CorelDRAW main application window on the desktop.

This property only applies when the main application window is in a normal window state; that is, the window is not minimized or maximized.

The width of the application window is a numerical value measured in screen pixels.

VBA example

The following VBA example sets the current window state to normal, defines the new dimensions of the window, and resizes the CorelDRAW main application window according to the new dimensions.

```
Sub AppWindowWidth()
 With AppWindow
  .WindowState = cdrWindowNormal
  .Top = 10
  .Left = 10
  .Height = 500
  .Width = 600
 End With
End Sub
```

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AppWindow.WindowState

Property **WindowState** As cdrWindowState

Description

Member of AppWindow

The **WindowState** property returns or specifies the current window state of the CorelDRAW main application window. This window can be minimized, maximized or in a normal state.

VBA example

The following VBA example begins with the application window maximized, sets the current window state to minimized, and displays a message that the window has been minimized. The main application window is then restored to a maximized state.

```
Sub AppWindowWindowState()
 Dim state As cdrWindowState
 state = cdrWindowMaximized
 AppWindow.WindowState = cdrWindowMinimized
 MsgBox "CorelDRAW is now minimized."
 AppWindow.WindowState = state
End Sub
```

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ArrowHead

Properties Methods

Description

The **ArrowHead** class represents an arrowhead on a curve or outline.

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ArrowHead.BaseOutlineScale

Property **BaseOutlineScale** As Double

Description

Member of ArrowHead

The **BaseOutlineScale** property returns a read-only value that is associated with the relative width of the outline in relation to an arrowhead curve.

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ArrowHead.BindToDocument

Function **BindToDocument(**Document As Document**)** As ArrowHead

Description

Member of ArrowHead

The **BindToDocument** method returns a copy of an arrowhead, bound to a specified document. This causes the arrowhead to report coordinates and sizes in the coordinate system of the specified document.

Parameter	Description
Document	Specifies the document

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ArrowHead.CenterX

Property **CenterX** As Double

Description

Member of ArrowHead

The **CenterX** property returns a read-only value that is associated with the relative horizontal position of the center of an arrowhead.

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ArrowHead.CenterY

Property **CenterY** As Double

Description

Member of ArrowHead

The **CenterY** property returns a read-only value that is associated with the relative vertical position of the center of an arrowhead.

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ArrowHead.CompareWith

Function **CompareWith(**ArrowHead As ArrowHead**)** As Boolean

Description

Member of ArrowHead

The **CompareWith** method determines whether two arrowheads are the same. If they are the same, **True** is returned.

Parameter	Description
ArrowHead	Specifies the arrowhead with which to compare

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ArrowHead.Curve

Property **Curve** As Curve

Description

Member of ArrowHead

The **Curve** property returns a read-only value that is associated with a curve representing the shape of an arrowhead.

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ArrowHead.DisplayName

Property **DisplayName** As String

Description

Member of ArrowHead

The **DisplayName** property returns a read-only value that is associated with the name for an arrowhead.

- If no custom name is specified, a generated name is returned.

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ArrowHead.Index

Property **Index** As Long

Description

Member of ArrowHead

The **Index** property returns the index number associated with an arrowhead type. For example, **ArrowHeads(5)** refers to the fifth arrowhead in the collection.

The **Index** property returns a read-only value.

VBA example

The following VBA example takes all of the shapes in the current selection, and determines which shapes have outlines. If the shape has an outline, an arrowhead with an index of five replaces all of the arrowheads that have an index of 1. This is done for both the start arrows and end arrows of the shape objects.

```
Sub Index()
 Dim s As Shape
 For Each s In ActiveDocument.Selection.Shapes
  If s.Outline.Type = cdrOutline Then
   With s.Outline
    If .StartArrow.Index = 1 Then
     .StartArrow = ArrowHeads(5)
    End If
    If s.Outline.EndArrow.Index = 1 Then
     s.Outline.EndArrow = ArrowHeads(5)
    End If
   End With
  End If
 Next s
End Sub
```

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ArrowHead.LineOffset

Property **LineOffset** As Double

Description

Member of ArrowHead

The **LineOffset** property returns a read-only value that is associated with the relative length by which the outline should be reduced to fit an arrowhead properly.

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ArrowHead.Name

Property **Name** As String

Description

Member of ArrowHead

The **Name** property pecifies the name for an arrowhead.

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ArrowHeadOptions

Properties Methods

Description

The **ArrowHeadOptions** class represents various options for arrowheads (or ArrowHead objects).

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ArrowHeadOptions.CopyAssign

Sub **CopyAssign(**Source As ArrowHeadOptions**)**

Description

Member of ArrowHeadOptions

The **CopyAssign** method copies the properties of a given arrowhead (or ArrowHead object).

Parameter	Description
Source	Specifies, through a value of ArrowHeadOptions, the source of the copy

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ArrowHeadOptions.Flip

Sub **Flip(**Axes As cdrFlipAxes**)**

Description

Member of ArrowHeadOptions

The **Flip** method flips an arrowhead (or ArrowHead object) horizontally or vertically (or both).

Parameter	Description
Axes	Specifies, through a value of cdrFlipAxes, whether to flip the arrowhead horizontally or vertically (or both)

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ArrowHeadOptions.FlipHorizontal

Property **FlipHorizontal** As Boolean

Description

Member of ArrowHeadOptions

The **FlipHorizontal** property specifies whether to mirror an arrowhead (or ArrowHead object) horizontally.

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ArrowHeadOptions.FlipVertical

Property **FlipVertical** As Boolean

Description

Member of ArrowHeadOptions

The **FlipVertical** property specifies whether to mirror an arrowhead (or ArrowHead object) vertically.

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ArrowHeadOptions.GetCopy

Function **GetCopy()** As ArrowHeadOptions

Description

Member of ArrowHeadOptions

The **GetCopy** method makes a copy of an arrowhead (or ArrowHead object).

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ArrowHeadOptions.Length

Property **Length** As Double

Description

Member of ArrowHeadOptions

The **Length** property specifies the horizontal size (or length) of an arrowhead (or ArrowHead object).

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ArrowHeadOptions.OffsetX

Property **OffsetX** As Double

Description

Member of ArrowHeadOptions

The **OffSetX** property specifies the relative horizontal offset of an arrowhead (or ArrowHead object) from its origin point, measured as a percentage of the length of the arrowhead.

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ArrowHeadOptions.OffsetY

Property **OffsetY** As Double

Description

Member of ArrowHeadOptions

The **OffsetY** property specifies the relative vertical offset of an arrowhead (or ArrowHead object) from its origin point, measured as a percentage of the width of the arrowhead.

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ArrowHeadOptions.RotationAngle

Property **RotationAngle** As Double

Description

Member of ArrowHeadOptions

The **RotationAngle** property specifies the rotation angle of an arrowhead (or ArrowHead object) around its origin point.

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ArrowHeadOptions.Width

Property **Width** As Double

Description

Member of ArrowHeadOptions

The **Width** property specifies the vertical size (or width) of an arrowhead (or ArrowHead object).

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ArrowHeads

Properties Methods

Description

The **ArrowHeads** class represents a collection of arrowheads (or ArrowHead objects).

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ArrowHeads.Add

Function **Add(**ArrowHead As ArrowHead**)** As ArrowHead

Description

Member of ArrowHeads

The **Add** method adds a given customized arrowhead to the global list of arrowheads. Returned is a new arrowhead object that references the global arrowhead added.

Parameter	Description
ArrowHead	Specifies the arrowhead

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ArrowHeads.Application

Property **Application** As Application

Description

Member of ArrowHeads

The Application object defines properties of the main CorelDRAW application. It is the top-level object in the CorelDRAW hierarchy of objects, and you can access all other features of the application through its functions, methods, and properties.

It is not necessary to explicitly specify an **Application** object; it is always available in VBA, and its properties and methods are global.

The **Application** property returns a read-only value that is associated with the main CorelDRAW application.

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ArrowHeads.Count

Property **Count** As Long

Description

Member of ArrowHeads

The **Count** property returns the number of arrowheads in an ArrowHeads collection.

The **Count** property returns a read-only value.

VBA example

The following VBA example identifies all arrowheads of selected objects that have an index of one and replaces them with arrowheads that have an index equal to the total number of arrowhead objects. This replaces the selected arrowheads with the last arrowhead in the **ArrowHeads** collection.

```
Sub Count()
 Dim s As Shape
 Dim Arrow As ArrowHead
 Set Arrow = ArrowHeads(ArrowHeads.Count)
 For Each s In ActiveDocument.Selection.Shapes
  If s.Outline.Type = cdrOutline Then
   With s.Outline
    If .StartArrow.Index = 1 Then
     .StartArrow = Arrow
    End If
    If .EndArrow.Index = 1 Then
     .StartArrow = Arrow
    End If
   End With
  End If
 Next s
End Sub
```

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ArrowHeads.Item

Property **Item(I***ndex As Long***)** As ArrowHead

Description

Member of ArrowHeads

The **Item** property returns the index number of an ArrowHead object in an ArrowHeads collection. Each type of arrowhead is identified by an index number represented in base 1. For example, **ArrowHeads(5)** refers to the fifth arrowhead in the collection.

Item is the default property of the **ArrowHeads** class, and its reference can be omitted when you are accessing items in an **ArrowHeads** collection. For example, **ArrowHeads.Item(5)** is the same as **ArrowHeads(5)** — they both reference the fifth arrowhead in the collection.

The **Item** property returns a read-only value.

Parameter	Description
Index	Specifies the preset placeholder that uniquely identifies each member of the **ArrowHeads** collection

VBA example

The following VBA example takes all of the shapes in the current selection, and determines which shapes have outlines. If the shape has an outline, an arrowhead with an index of five is applied to the start arrow position of each shape.

```
Sub Item()
 Dim s As Shape
 For Each s In ActiveDocument.Selection.Shapes
  If s.Outline.Type = cdrOutline Then
   s.Outline.StartArrow = ArrowHeads.Item(5)
  End If
 Next s
End Sub
```

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ArrowHeads.Parent

Property **Parent** As Application

Description

Member of ArrowHeads

The **Parent** property returns a read-only value that is associated with the properties, methods, and controls of an object's parent in the CorelDRAW hierarchy of objects. The application object refers to the CorelDRAW application.

A parent object acts as a container for child objects. For example, a form (as a container) is the parent object of a check box on that form.

Most objects have either a **Parent** property or a **Collection** property, pointing to the object's parent object in the object model. The **Collection** property is used if the parent object is a collection.

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ArrowHeads.Remove

Sub **Remove(***Index As Long***)**

Description

Member of ArrowHeads

The **Remove** method deletes an arrowhead object from an ArrowHeads collection.

Parameter	Description
Index	Specifies the preset placeholder that uniquely identifies each member of the **ArrowHeads** collection

VBA example

The following VBA example removes the last arrowhead object from the **ArrowHeads** collection.

```
Sub ArrowHeadDelete()
 ArrowHeads.Remove (ArrowHeads.Count)
End Sub
```

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ArrowHeads.Replace

Function **Replace(***Index As Long***,** ArrowHead As ArrowHead**)** As ArrowHead

Description

Member of ArrowHeads

The **Replace** method replaces an existing arrowhead with a new one.

Parameter	Description
Index	Specifies the preset placeholder that uniquely identifies the arrowhead
ArrowHead	Specifies the arrowhead

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BBoxSnapPoint

Properties

Description

The **BBoxSnapPoint** class represents bounding-box snap points.

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BBoxSnapPoint.Type

Property **Type** As cdrReferencePoint

Description

Member of BBoxSnapPoint

The **Type** property returns a read-only value that is associated with the type of a bounding-box snap point.

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Bitmap

Properties Methods

Description

The **Bitmap** class represents the bitmap-specific settings for a shape (or Shape object).

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Bitmap.ApplyBitmapEffect

Sub **ApplyBitmapEffect(***UndoString As String*, *Command As String***)**

Description

Member of Bitmap

The **ApplyBitmapEffect** method applies the specified bitmap effect.

Parameter	Description
UndoString	Specifies the name in the **Undo** list
Command	Specifies the effect to apply

VBA example

The following VBA example applies the Whirlpool effect to all bitmaps on the active page.

```
Sub ApplyBitmapEffect()
Dim s As Shape
For Each s In ActivePage.Shapes
If s.Type = cdrBitmapShape Then
If s.Bitmap.Mode <> cdrRGBColorImage Then
s.Bitmap.ConvertTo cdrRGBColorImage
End If
s.Bitmap.ApplyBitmapEffect "Whirlpool", "WhirlpoolEffect
WhirpoolSpacing=20,WhirlpoolSmear=9,WhirlpoolTwist=70,WhirlpoolStreak=60,WhirpoolWarp=1,WhirlpoolRandomSeed=0"
End If
Next s
End Sub
```

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Bitmap.BoundingBoxPath

Property **BoundingBoxPath** As Curve

Description

Member of Bitmap

The **BoundingBoxPath** property returns the uncropped read-only bitmap bounding box curve.

The **BoundingBoxPath** property returns a read-only value.

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Bitmap.ConvertTo

Sub **ConvertTo(**Mode As cdrImageType**)**

Description

Member of Bitmap

The **ConvertTo** method changes the color mode of a bitmap.

Parameter	Description
Mode	Specifies the color mode to apply

VBA example

The following VBA example converts all bitmaps on the active page to the RGB color mode.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Mode <> cdrRGBColorImage Then
    s.Bitmap.ConvertTo cdrRGBColorImage
   End If
  End If
 Next s
End Sub
```

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Bitmap.ConvertToBW

Sub **ConvertToBW(**RenderType As cdrRenderType**, [**Intensity As Long = 100**], [**Threshold As Long = 128**], [**Halftone As cdrHalftoneType = cdrHalftoneSquare**], [**HalftoneAngle As Long = 45**], [***HalftoneSize As Long***])**

Description

Member of Bitmap

The **ConvertToBW** method converts the bitmap to the black and white color mode.

Parameter	Description
RenderType	Specifies the type of conversion
Intensity	Specifies the intensity of the applied color. This parameter is optional, and its default value is **100**.
Threshold	Specifies the tolerance level for tonal variation for a line art conversion. This parameter is optional, and its default value is **128**.
Halftone	Specifies a halftone screen. This parameter is optional, and its default value is cdrHalftoneSquare (**0**).
HalftoneAngle	Specifies the halftone angle. This parameter is optional, and its default value is **45**.
HalftoneSize	Specifies the lines per unit for a halftone conversion. This parameter is optional, and its default value is **0**.

VBA example

The following VBA example converts all bitmaps on the active page to the black-and-white color mode.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Mode = cdrRGBColorImage Then
    s.Bitmap.ConvertToBW cdrRenderLineArt, , 97
   End If
  End If
 Next s
End Sub
```

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Bitmap.ConvertToDuotone

Sub **ConvertToDuotone(**Duotone As Duotone**)**

Description

Member of Bitmap

The **Duotone** method converts the bitmap to duotone color mode.

Parameter	Description
Duotone	

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Bitmap.ConvertToPaletted

Sub **ConvertToPaletted(**PaletteType As cdrImagePaletteType**, [**DitherType As cdrDitherType = cdrDitherNone**], [**DitherIntensity As Long = 100**], [***Smoothing As Long***], [**NumColors As Long = 256**], [**ColorSensitive As Boolean = False**], [**TargetColor As Long = 16777215**], [**Importance As Long = 500**], [**Lightness As Long = 300**], [**ToleranceA As Long = 300**], [**ToleranceB As Long = 100**], [***Palette As Variant***])**

Description

Member of Bitmap

The **ConvertToPaletted** method converts the bitmap to a specific palette using the specified parameters.

Parameter	Description
PaletteType	Specifies the palette to apply
DitherType	Specifies the type of dithering to apply. This parameter is optional, and its default value is cdrDitherNone (**0**).
DitherIntensity	Specifies the intensity of the dithering. This parameter is optional, and its default value is **100**.
Smoothing	Specifies the type of smoothing to apply. This parameter is optional, and its default value is **0**.
NumColors	Specifies the maximum number of colors in the optimized image. The **PaletteType** must be optimized for this option to be applied. This parameter is optional, and its default value is **256**.
ColorSensitive	Specifies whether color sensitivity is applied. This parameter is optional, and its default value is **False**.
TargetColor	Specifies the color to which the color sensitivity is applied. This parameter is optional, and its default value is **16777215**.
Importance	Specifies the emphasis of the sensitivity of the target color. The TargetColor must be specified for this option to be applied. This parameter is optional, and its default value is **500**.
Lightness	Specifies the lightness of the range sensitivity. This parameter is optional, and its default value is **300**.
ToleranceA	Specifies the tolerance of the range sensitivity on the green-red axis. This parameter is optional, and its default value is **300**.
ToleranceB	Specifies the tolerance of the range sensitivity on the blue-yellow axis. This parameter is optional, and its default value is **100**.
Palette	Specifies a custom palette. This parameter is optional.

VBA example

The following VBA example converts all bitmaps on the active page to the Optimized palette.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Mode <> cdrPalettedImage Then
    s.Bitmap.ConvertToPaletted cdrPaletteOptimized, cdrDitherNone, , 5
   End If
  End If
 Next s
End Sub
```

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Bitmap.ConvertToPaletted2

Sub **ConvertToPaletted2([**Options As StructPaletteOptions**])**

Description

Member of Bitmap

The **ConvertToPaletted2** method converts the bitmap to a specific palette using the specified parameters.

Parameter	Description
Options	Specifies a set of options to apply. This parameter is optional, and its default value is **Nothing**.

VBA example

The following VBA example converts all bitmaps on the active page to a paletted bitmap.

```
Sub Test()
 Dim s As Shape
 Dim Pal As StructPaletteOptions
 Set Pal = CreateStructPaletteOptions
 With Pal
  .DitherType = cdrDitherNone
  .NumColors = 256
  .Smoothing = 5
  .PaletteType = cdrPaletteOptimized
 End With
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Mode <> cdrPalettedImage Then
    s.Bitmap.ConvertToPaletted2 Pal
   End If
  End If
 Next s
End Sub
```

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Bitmap.Crop

Sub **Crop()**

Description

Member of Bitmap

The **Crop** method crops the bitmap to its clipping path.

VBA example

The following VBA example crops all bitmaps which can be cropped.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Cropped Then s.Bitmap.Crop
  End If
 Next s
End Sub
```

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Bitmap.CropEnvelope

Property **CropEnvelope** As Curve

Description

Member of Bitmap

The **CropEnvelope** property returns the bitmap's cropping-path curve.

The **CropEnvelope** property returns a read-only value.

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Bitmap.CropEnvelopeModified

Property **CropEnvelopeModified** As Boolean

Description

Member of Bitmap

The **CropEnvelopeModified** property returns **True** if the cropping path of the bitmap has been modified.

The **CropEnvelopeModified** property returns a read-only value.

VBA example

The following VBA example displays the number of bitmaps that have a modified cropping path.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.CropEnvelopeModified Then n = n + 1
  End If
 Next s
 MsgBox "The crop envelope has been modified on " & n & " bitmaps."
End Sub
```

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Bitmap.Cropped

Property **Cropped** As Boolean

Description

Member of Bitmap

The **Cropped** property returns **True** if the bitmap can be cropped.

The **Cropped** property returns a read-only value.

VBA example

The following VBA example displays the number of bitmaps that can be cropped on the active page.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Cropped Then n = n + 1
  End If
 Next s
 MsgBox "There are " & n & " bitmaps which can be cropped."
End Sub
```

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Bitmap.Duotone

Property **Duotone** As Duotone

Description

Member of Bitmap

The **Duotone** property returns the duotone settings of the bitmap.

The **Duotone** property returns a read-only value.

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Bitmap.Embedded

Property **Embedded** As Boolean

Description

Member of Bitmap

The **Embedded** property returns **True** if the bitmap is embedded.

The **Embedded** property returns a read-only value.

VBA example

The following VBA example displays the number of bitmaps that are embedded in the active page.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Embedded Then n = n + 1
  End If
 Next s
 MsgBox "There are " & n & " embedded bitmaps"
End Sub
```

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Bitmap.ExternalLinkTime

Property **ExternalLinkTime** As Date

Description

Member of Bitmap

The **ExternalLinkTime** returns the time at which the bitmap was externally linked.

The **ExternalLinkTime** returns a read-only value.

VBA example

The following VBA example updates the externally linked bitmap if the original bitmap had been modified since it was linked.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   With s.Bitmap
    If Not .Embedded Then
     If DateDiff("s", FileDateTime(s.Bitmap.LinkFileName), _
       .ExternalLinkTime) <> 0 Then
      .UpdateLink
     End If
    End If
   End With
  End If
 Next s
End Sub
```

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Bitmap.ExternallyLinked

Property **ExternallyLinked** As Boolean

Description

Member of Bitmap

The **ExternallyLinked** property returns **True** if the bitmap is linked to an external file.

The **ExternallyLinked** property returns a read-only value.

VBA example

The following VBA example displays the link status of a bitmap.

```
Sub Link()
 MsgBox ActiveShape.Bitmap.ExternallyLinked
End Sub
```

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Bitmap.Inflate

Sub **Inflate(***Width As Long*, *Height As Long***)**

Description

Member of Bitmap

The **Inflate** method increases the size of a bitmap by adding the specified number of pixels at each side of the bitmap.

Parameter	Description
Width	Specifies the number of pixels to add to each side
Height	Specifies the number of pixels to add to the top and bottom

VBA example

The following VBA example increases the size of the active bitmap to at least 400 × 400 pixels.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   With s.Bitmap
    If .SizeWidth < 400 And .SizeHeight < 400 Then
     .Inflate (400 - .SizeWidth)/2, (400 - .SizeHeight)/2
    End If
   End With
  End If
 Next s
End Sub
```

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Bitmap.IsEPS

Property **IsEPS** As Boolean

Description

Member of Bitmap

The **IsEPS** property returns **True** if the bitmap is a placeholder in the EPS format.

The **IsEPS** property returns a read-only value.

VBA example

The following VBA example searches for a header in the EPS format. If one is found, it is selected.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  s.Selected = False
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.IsEPS Then s.Selected = True
  End If
 Next s
End Sub
```

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Bitmap.LinkFileName

Property **LinkFileName** As String

Description

Member of Bitmap

The **LinkFileName** property returns the filename of an externally linked bitmap.

The **LinkFileName** property returns a read-only value.

VBA example

The following VBA example displays the filenames of all externally linked bitmaps.

```
Sub Test()
 Dim s As Shape
 Dim str As String
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.ExternallyLinked Then str = str & s.Bitmap.LinkFileName & vbCrLf
  End If
 Next s
 MsgBox "The external file names are:" & vbCrLf & str
End Sub
```

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Bitmap.Mode

Property **Mode** As cdrImageType

Description

Member of Bitmap

The **Mode** property returns the color mode of the bitmap.

The **Mode** property returns a read-only value.

VBA example

The following VBA example displays the number of bitmaps in the RGB color mode.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Mode = cdrRGBColorImage Then n = n + 1
  End If
 Next s
 MsgBox "There are " & n & " RGB bitmaps."
End Sub
```

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Bitmap.OPILinked

Property **OPILinked** As Boolean

Description

Member of Bitmap

The **OPILinked** property returns **True** if a bitmap in the TIF format has an Open Prepress Interface (OPI) link to another image.

The **OPILinked** property returns a read-only value.

VBA example

The following VBA example displays the number of bitmaps with OPI links.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.OPILinked Then n = n + 1
  End If
 Next s
 MsgBox "There are " & n & " OPI images."
End Sub
```

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Bitmap.Resample

Sub **Resample([***Width As Long***], [***Height As Long***], [**AntiAlias As Boolean = True**], [***ResolutionX As Double***], [***ResolutionY As Double***])**

Description

Member of Bitmap

The **Resample** method resizes a bitmap.

Parameter	Description
Width	Specifies the width. This parameter is optional, and its default value is **0**.
Height	Specifies the height. This parameter is optional, and its default value is **0**.
AntiAlias	Specifies whether to apply anit-aliasing. This parameter is optional, and its default value is **True**.
ResolutionX	Specifies the horizontal resolution. This parameter is optional, and its default value is **0**.
ResolutionY	Specifies the vertical resolution. This parameter is optional, and its default value is **0**.

VBA example

The following VBA example resizes all bitmaps on the active page. All bitmaps fit in a 150 × 150 pixel frame and maintain their original aspect ratio.

```
Sub Test()
 Dim s As Shape
 Dim Width As Double
 Dim Height As Double
 Dim AspectRatio As Double
 Width = 150
 Height = 150
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   'so if thewidth is greater than theheight
   If s.Bitmap.SizeWidth > s.Bitmap.SizeHeight Then
    'the aspect ratio is Height divided by Width
    AspectRatio = s.Bitmap.SizeHeight / s.Bitmap.SizeWidth
    Height = AspectRatio * Width
   'height is greater than width, so
   ElseIf s.Bitmap.SizeHeight > s.Bitmap.SizeWidth Then
    'the aspect ratio is Width divided by Height
    AspectRatio = s.Bitmap.SizeWidth / s.Bitmap.SizeHeight
    Width = AspectRatio * Height
   End If
   s.Bitmap.Resample CLng(Width), CLng(Height)
  End If
 Next s
End Sub
```

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Bitmap.ResetCropEnvelope

Sub **ResetCropEnvelope()**

Description

Member of Bitmap

The **ResetCropEnvelope** method removes a cropping path from a bitmap.

VBA example

The following VBA example removes the cropping path from all bitmaps on the active page.

```
Sub Test()
 Dim s As Shape
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Cropped Then s.Bitmap.ResetCropEnvelope
  End If
 Next s
End Sub
```

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Bitmap.ResolutionX

Property **ResolutionX** As Long

Description

Member of Bitmap

The **ResolutionX** property returns the horizontal resolution of a bitmap in DPI

The **ResolutionX** property returns a read-only value.

VBA example

The following VBA example displays the horizontal and vertical resolution of a bitmap object.

```
Sub Size()
 With ActiveShape.Bitmap
  MsgBox .ResolutionX
  MsgBox .ResolutionY
 End With
End Sub
```

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Bitmap.ResolutionY

Property **ResolutionY** As Long

Description

Member of Bitmap

The **ResolutionY** property returns the vertical resolution of a bitmap in DPI.

The **ResolutionY** property returns a read-only value.

VBA example

The following VBA example displays the horizontal and vertical resolution of a bitmap object.

```
Sub Size()
 With ActiveShape.Bitmap
  MsgBox .ResolutionX
  MsgBox .ResolutionY
 End With
End Sub
```

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Bitmap.ResolveLink

Sub **ResolveLink()**

Description

Member of Bitmap

The **ResolveLink** method embeds an externally linked bitmap.

VBA example

The following VBA example embeds the active bitmap if it is not already embedded.

```
Sub Embed()
 With ActiveShape.Bitmap
  If .ExternallyLinked = True Then
   .ResolveLink
  End If
 End With
End Sub
```

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Bitmap.SaveAs

Function **SaveAs(***FileName As String***,** Filter As cdrFilter**, [**Compression As cdrCompressionType = cdrCompressionNone**])** As ExportFilter

Description

Member of Bitmap

The **SaveAs** method saves a bitmap to a specified file and returns an ExportFilter object.

Parameter	Description
FileName	Specifies the filename
Filter	Specifies the filter to apply
Compression	Specifies the compression to apply. This parameter is optional, and its default value is cdrCompressionNone (**0**).

VBA example

The following VBA example saves each bitmap on the active page in the BMP format.

```
Sub Test()
 Dim s As Shape
 Dim ex As ExportFilter
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   Set ex = s.Bitmap.SaveAs("c:\" & s.StaticID & ".bmp", cdrBMP, _
     cdrCompressionRLE_LW)
   ex.Finish
  End If
 Next s
End Sub
```

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Bitmap.SizeHeight

Property **SizeHeight** As Long

Description

Member of Bitmap

The **SizeHeight** property returns the vertical size (height) of a bitmap in pixels.

The **SizeHeight** property returns a read-only value.

VBA example

The following VBA example displays the width and height of a bitmap object.

```
Sub Size()
 With ActiveShape.Bitmap
  MsgBox .SizeWidth
  MsgBox .SizeHeight
 End With
End Sub
```

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Bitmap.SizeWidth

Property **SizeWidth** As Long

Description

Member of Bitmap

The **SizeWidth** property returns the horizontal size (width) of a bitmap in pixels.

The **SizeWidth** property returns a read-only value.

VBA example

The following VBA example displays the width and height of a bitmap object.

```
Sub Size()
 With ActiveShape.Bitmap
  MsgBox .SizeWidth
  MsgBox .SizeHeight
 End With
End Sub
```

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Bitmap.Trace

Function **Trace(**TraceType As cdrTraceType**, [**Smoothing As Integer = -1**], [***DetailLevelPercent As Integer***], [**ColorMode As cdrColorType = cdrColorMixed**], [**PaletteID As cdrPaletteID = cdrCustom**], [***ColorCount As Long***], [**DeleteOriginalObject As Boolean = False**], [**RemoveBackground As Boolean = True**], [**RemoveEntireBackColor As Boolean = False**])** As TraceSettings

Description

Member of Bitmap

The **Trace** method converts a bitmap into vector objects.

Parameter	Description
TraceType	Specifies the trace type
Smoothing	Specifies the degree of smoothing. This parameter is optional, and its default value is **-1**.
DetailLevelPercent	Specifies the level of detail. This parameter is optional.
ColorMode	Specifies the color mode. This parameter is optional, and its default value is cdrColorMixed.
PaletteID	Specifies the ID of the palette. This parameter is optional, and its default value is cdrCustom.
ColorCount	Specifies the color count. This parameter is optional.
DeleteOriginalObject	Specifies whether to delete the original object. This parameter is optional, and its default value is **False**.
RemoveBackground	Specifies whether to remove the background. This parameter is optional, and its default value is **True**.
RemoveEntireBackColor	Specifies whether to remove the entire background color. This parameter is optional, and its default value is **False**.

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Bitmap.Transparent

Property **Transparent** As Boolean

Description

Member of Bitmap

The **Transparent** property returns **True** if the bitmap is transparent.

The **Transparent** property returns a read-only value.

VBA example

The following VBA example displays the number of transparent bitmaps that are on the active page.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Transparent Then n = n + 1
  End If
 Next s
 MsgBox "There are " & n & " transparent bitmaps"
End Sub
```

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Bitmap.UpdateLink

Sub **UpdateLink()**

Description

Member of Bitmap

The **UpdateLink** method updates an externally linked bitmap by re-importing the linked file.

VBA example

The following VBA example updates the active bitmap if it is externally linked.

```
Sub Embed()
 With ActiveShape.Bitmap
  If .ExternallyLinked = True Then
   .UpdateLink
  End If
 End With
End Sub
```

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Bitmap.Watermarked

Property **Watermarked** As Boolean

Description

Member of Bitmap

The **Watermarked** property returns **True** if the bitmap has a watermark.

The **Watermarked** property returns a read-only value.

VBA example

The following VBA example displays the number of bitmaps that have a watermark.

```
Sub Test()
 Dim s As Shape, n As Long
 n = 0
 For Each s In ActivePage.Shapes
  If s.Type = cdrBitmapShape Then
   If s.Bitmap.Watermarked Then n = n + 1
  End If
 Next s
 MsgBox "There are " & n & " watermarked bitmaps"
End Sub
```

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BSpline

Properties Methods

Description

The **BSpline** class represents the B-spline settings for a shape (or Shape object).

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BSpline.Closed

Property **Closed** As Boolean

Description

Member of BSpline

The **Closed** property uses a Boolean value to specify whether a B-spline (or BSpline object) is closed.

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BSpline.ControlPoints

Property **ControlPoints** As BSplineControlPoints

Description

Member of BSpline

The **ControlPoints** property returns a BSplineControlPoints object, which represents the control points for a B-spline (or BSpline object).

The **ControlPoints** property returns a read-only value.

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BSpline.CopyAssign

Sub **CopyAssign(**rhs As BSpline**)**

Description

Member of BSpline

The **CopyAssign** method copies the properties of a given B-spline (or BSpline object).

Parameter	Description
rhs	Specifies, through a value of BSpline, the B-spline object from which to copy properties

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BSpline.GetCopy

Function **GetCopy()** As BSpline

Description

Member of BSpline

The **GetCopy** method returns a copy of the specified B-spline (or BSpline object).

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BSpline.InsertControlPoint

Sub **InsertControlPoint(***Index As Long*, *x As Double*, *y As Double*, *Clamped As Boolean***)**

Description

Member of BSpline

The **InsertControlPoint** method inserts one control point before the one at the given index number on a B-spline (or BSpline object).

Parameter	Description
Index	Specifies the index number of the control point before which to insert the new control point
x	Specifies the horizontal position of the new control point
y	Specifies the vertical position of the new control point
Clamped	Specifies whether the B-spline passes through the control point to create a cusp

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BSpline.InsertControlPoints

Sub **InsertControlPoints(***Index As Long*, *HowMany As Long*, *x As Double*, *y As Double*, *Clamped As Boolean***)**

Description

Member of BSpline

The **InsertControlPoints** method inserts the specified number of control points before the one at the given index number on a B-spline (or BSpline object).

Parameter	Description
Index	Specifies the index number of the control point before which to insert the new control points
HowMany	Specifies the number of control points to insert
x	Specifies the horizontal position of the new range of control points
y	Specifies the vertical position of the new range of control points
Clamped	Specifies whether the B-spline passes through the range of control points to create a cusp

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BSplineControlPoint

Properties Methods

Description

The **BSplineControlPoint** class represents a control point for a B-spline (or BSpline object).

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BSplineControlPoint.Clamped

Property **Clamped** As Boolean

Description

Member of BSplineControlPoint

The **Clamped** property specifies whether a control point on a B-spline (or BSpline object) is clamped — that is, whether the B-spline passes through the control point to create a cusp.

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BSplineControlPoint.Delete

Sub **Delete()**

Description

Member of BSplineControlPoint

The **Delete** method deletes a control point on a B-spline (or BSpline object).

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BSplineControlPoint.GetPosition

Sub **GetPosition(***x As Double***,** *y As Double***)**

Description

Member of BSplineControlPoint

The **GetPosition** method retrieves the location of a control point on a B-spline (or BSpline object).

Parameter	Description
x	Specifies the horizontal position of the control point
y	Specifies the vertical position of the control point

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BSplineControlPoint.Index

Property **Index** As Long

Description

Member of BSplineControlPoint

The **Index** property returns the index number that is associated with a control point on a B-spline (or BSpline object).

The **Index** property returns a read-only value.

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BSplineControlPoint.Move

Sub **Move(***x As Double***,** *y As Double***)**

Description

Member of BSplineControlPoint

The **Move** method moves, by a given distance, a control point on a B-spline (or BSpline object).

Parameter	Description
x	Specifies the horizontal distance
y	Specifies the vertical distance

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BSplineControlPoint.SetPosition

Sub **SetPosition(***x As Double***,** *y As Double***)**

Description

Member of BSplineControlPoint

The **SetPosition** method changes the location of a control point on a B-spline (or BSpline object).

Parameter	Description
x	Specifies the new horizontal position of the control point
y	Specifies the new vertical position of the control point

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BSplineControlPoint.SetProperties

Sub **SetProperties(***x As Double***,** *y As Double***,** *Clamped As Boolean***)**

Description

Member of BSplineControlPoint

The **SetProperties** method specifies, in a single method call, all properties for a control point on a B-spline (or BSpline object).

Parameter	Description
x	Specifies the horizontal position of the control point
y	Specifies the vertical position of the control point
Clamped	Specifies whether the B-spline passes through the control point to create a cusp

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BSplineControlPoint.x

Property **x** As Double

Description

Member of BSplineControlPoint

The **x** property specifies the horizontal position of a control point on a B-spline (or BSpline object).

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BSplineControlPoint.y

Property **y** As Double

Description

Member of BSplineControlPoint

The **y** property specifies the vertical position of a control point on a B-spline (or BSpline object).

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BSplineControlPoints

Properties Methods

Description

The **BSplineControlPoints** class represents a collection of control points (or BSplineControlPoint objects) for a B-spline (or BSpline object).

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BSplineControlPoints.Count

Property **Count** As Long

Description

Member of BSplineControlPoints

The **Count** property returns the number of control points in an BSplineControlPoints collection.

The **Count** property returns a read-only value.

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BSplineControlPoints.First

Property **First** As BSplineControlPoint

Description

Member of BSplineControlPoints

The **First** property returns the first control point in a BSplineControlPoints collection.

The **First** property returns a read-only value.

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BSplineControlPoints.Item

Property **Item(***Index As Long***)** As BSplineControlPoint

Description

Member of BSplineControlPoints

The **Item** property returns the index number of a BSplineControlPoint object in an BSplineControlPoints collection. Each control point is identified by an index number represented in base 1. For example, **BSplineControlPoints(5)** refers to the fifth control point in the collection.

Item is the default property of the **BSplineControlPoints** class, and its reference can be omitted when you are accessing items in an **BSplineControlPoints** collection. For example, **BSplineControlPoints.Item(5)** is the same as **BSplineControlPoints(5)** — they both reference the fifth control point in the collection.

The **Item** property returns a read-only value.

Parameter	Description
Index	Specifies the preset placeholder that uniquely identifies each member of the **BSplineControlPoints** collection

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BSplineControlPoints.Last

Property **Last** As BSplineControlPoint

Description

Member of BSplineControlPoints

The **Last** property returns the last control point in a BSplineControlPoints collection.

The **Last** property returns a read-only value.

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BSplineControlPoints.Resize

Sub **Resize(***HowMany As Long***)**

Description

Member of BSplineControlPoints

The **Resize** method changes the number of control points for a B-spline (or BSpline object). New control points, if required, are added at position **(0,0)**.

Parameter	Description
HowMany	Specifies the desired number of control points

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cdr3DMF

The following enumerations have this constant:

- cdrFilter

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cdr3DObjectShape

The following enumerations have this constant:

- cdrShapeType

Copyright 2014 Corel Corporation. All rights reserved.

cdr16ColorsImage

The following enumerations have this constant:

- cdrImageType

Copyright 2014 Corel Corporation. All rights reserved.

cdrAbsoluteSegmentOffset

The following enumerations have this constant:

- cdrSegmentOffsetType

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cdrAddinFilter

The **cdrAddinFilter** enumeration has the following constants:

Constant	Value	Description
cdrAddinFilterNone	**0**	
cdrAddinFilterShapeCreated	**1**	
cdrAddinFilterNew	**2**	
cdrAddinFilterExecute	**4**	

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cdrAddinFilterExecute

The following enumerations have this constant:

- cdrAddinFilter

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cdrAddinFilterNew

The following enumerations have this constant:

- cdrAddinFilter

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cdrAddinFilterNone

The following enumerations have this constant:

- cdrAddinFilter

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cdrAddinFilterShapeCreated

The following enumerations have this constant:

- cdrAddinFilter

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cdrAfrikaans

The following enumerations have this constant:

- cdrTextLanguage

Copyright 2014 Corel Corporation. All rights reserved.

cdrAgate

The following enumerations have this constant:

- cdrUnit

Copyright 2014 Corel Corporation. All rights reserved.

cdrAI

The following enumerations have this constant:

- cdrFilter

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cdrAI9

The following enumerations have this constant:

- cdrFilter

Copyright 2014 Corel Corporation. All rights reserved.

cdrAlbanian

The following enumerations have this constant:

- cdrTextLanguage

Copyright 2014 Corel Corporation. All rights reserved.

cdrAlignBottom

The following enumerations have this constant:

- cdrAlignType

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cdrAlignDistributeH

The **cdrAlignDistributeH** enumeration has the following constants:

Constant	Value	Description
cdrAlignDistributeHNone	**0**	Specifies no horizontal alignment and distribution
cdrAlignDistributeHAlignRight	**1**	Specifies right alignment
cdrAlignDistributeHAlignLeft	**2**	Specifies left alignment
cdrAlignDistributeHAlignCenter	**3**	Specifies center alignment horizontally
cdrAlignDistributeHDistributeRight	**4**	Distributes objects from their right edges
cdrAlignDistributeHDistributeLeft	**5**	Distributes objects from their left edges
cdrAlignDistributeHDistributeCenter	**6**	Distributes objects from their centers
cdrAlignDistributeHDistributeSpacing	**7**	Places equal space between objects

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cdrAlignDistributeHAlignCenter

The following enumerations have this constant:

- cdrAlignDistributeH

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cdrAlignDistributeHAlignLeft

The following enumerations have this constant:

- cdrAlignDistributeH

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cdrAlignDistributeHAlignRight

The following enumerations have this constant:

- cdrAlignDistributeH

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cdrAlignDistributeHDistributeCenter

The following enumerations have this constant:

- cdrAlignDistributeH

Copyright 2014 Corel Corporation. All rights reserved.

cdrAlignDistributeHDistributeLeft

The following enumerations have this constant:

- cdrAlignDistributeH

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cdrAlignDistributeHDistributeRight

The following enumerations have this constant:

- cdrAlignDistributeH

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cdrAlignDistributeHDistributeSpacing

The following enumerations have this constant:

- cdrAlignDistributeH

Copyright 2014 Corel Corporation. All rights reserved.

cdrAlignDistributeHNone

The following enumerations have this constant:

- cdrAlignDistributeH

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cdrAlignDistributeV

The **cdrAlignDistributeV** enumeration has the following constants:

Constant	Value	Description
cdrAlignDistributeVNone	**0**	Specifies no vertical alignment and distribution
cdrAlignDistributeVAlignTop	**1**	Specifies alignment of top edges
cdrAlignDistributeVAlignBottom	**2**	Specifies alignment of bottom edges
cdrAlignDistributeVAlignCenter	**3**	Specifies center alignment vertically
cdrAlignDistributeVDistributeTop	**4**	Distributes objects from their top edges
cdrAlignDistributeVDistributeBottom	**5**	Distributes objects from their bottom edges
cdrAlignDistributeVDistributeCenter	**6**	Distributes objects from their centers vertically
cdrAlignDistributeVDistributeSpacing	**7**	Places equal space between objects vertically

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cdrAlignDistributeVAlignBottom

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVAlignCenter

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVAlignTop

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVDistributeBottom

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVDistributeCenter

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVDistributeSpacing

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVDistributeTop

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignDistributeVNone

The following enumerations have this constant:

- cdrAlignDistributeV

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cdrAlignHCenter

The following enumerations have this constant:

- cdrAlignType

Copyright 2014 Corel Corporation. All rights reserved.

cdrAlignLeft

The following enumerations have this constant:

- cdrAlignType

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cdrAlignment

The **cdrAlignment** enumeration has the following constants:

Constant	Value	Description
cdrNoAlignment	**0**	Specifies no alignment
cdrLeftAlignment	**1**	Specifies left alignment
cdrRightAlignment	**2**	Specifies right alignment
cdrCenterAlignment	**3**	Specifies center alignment
cdrFullJustifyAlignment	**4**	Specifies full justification
cdrForceJustifyAlignment	**5**	Specifies forced justification
cdrMixedAlignment	**6**	Specifies mixed alignment

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cdrAlignRight

The following enumerations have this constant:

- cdrAlignType

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cdrAlignShapesTo

The **cdrAlignShapesTo** enumeration has the following constants:

Constant	Value	Description
cdrAlignShapesToLastSelected	**0**	Aligns shapes with the last selected shape
cdrAlignShapesToEdgeOfPage	**1**	Aligns shapes with the edge of the page
cdrAlignShapesToCenterOfPage	**2**	Aligns shapes with the center of the page
cdrAlignShapesToGrid	**3**	Aligns shapes with the grid
cdrAlignShapesToPoint	**4**	Aligns shapes with a specified point

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cdrAlignShapesToCenterOfPage

The following enumerations have this constant:

- cdrAlignShapesTo

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cdrAlignShapesToEdgeOfPage

The following enumerations have this constant:

- cdrAlignShapesTo

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cdrAlignShapesToGrid

The following enumerations have this constant:

- cdrAlignShapesTo

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cdrAlignShapesToLastSelected

The following enumerations have this constant:

- cdrAlignShapesTo

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cdrAlignShapesToPoint

The following enumerations have this constant:

- cdrAlignShapesTo

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cdrAlignTop

The following enumerations have this constant:

- cdrAlignType

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cdrAlignType

The **cdrAlignType** enumeration has the following constants:

Constant	Value	Description
cdrAlignLeft	**1**	Specifies left alignment
cdrAlignRight	**2**	Specifies right alignment
cdrAlignHCenter	**3**	Specifies horizontal-center alignment
cdrAlignTop	**4**	Specifies top alignment
cdrAlignBottom	**8**	Specifies bottom alignment
cdrAlignVCenter	**12**	Specifies vertical-center alignment

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cdrAlignVCenter

The following enumerations have this constant:

- cdrAlignType

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cdrAllCapsFontCase

The following enumerations have this constant:

- cdrFontCase

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cdrAllFrames

The following enumerations have this constant:

- cdrTextFrames

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cdrAllGuides

The following enumerations have this constant:

- cdrGuideType

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cdrAllPages

The following enumerations have this constant:

- cdrExportRange

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cdrAntiAliasingType

The **cdrAntiAliasingType** enumeration has the following constants:

Constant	Value	Description
cdrNoAntiAliasing	**0**	Specifies no anti-aliasing
cdrNormalAntiAliasing	**1**	Specifies normal anti-aliasing
cdrSupersampling	**2**	Specifies super-sampling

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cdrApplicationClass

The **cdrApplicationClass** enumeration has the following constants:

Constant	Value	Description
cdrCreativeGraphics	**0**	Specifies creative graphics
cdrTechnicalGraphics	**1**	Specifies technical graphics

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cdrApplicationID

The **cdrApplicationID** enumeration has the following constants:

Constant	Value	Description
cdrCorelDRAW	**0**	Specifies CorelDRAW
cdrCorelDESIGNER	**1**	Specifies Corel DESIGNER

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cdrApplyToFill

The following enumerations have this constant:

- cdrTransparencyAppliedTo

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cdrApplyToFillAndOutline

The following enumerations have this constant:

- cdrTransparencyAppliedTo

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cdrApplyToOutline

The following enumerations have this constant:

- cdrTransparencyAppliedTo

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cdrAppStartupMode

The **AppStartupMode** enumeration has the following constants:

Constant	Value	Description
cdrStartupWelcomeScreen	**0**	Specifies to display the **Welcome** screen
cdrStartupNewDocument	**1**	Specifies to start a new document
cdrStartupOpenDocument	**2**	Specifies to open an existing document
cdrStartupLastEditedDocument	**3**	Specifies to open the last-edited document
cdrStartupNewFromTemplate	**4**	Specifies to start a new document from a template
cdrStartupTutorial	**5**	Specifies to start a tutorial
cdrStartupDoNothing	**6**	Specifies to do nothing

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cdrArabic

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicAlgeria

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicBahrain

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicEgypt

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicIraq

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicJordan

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicKuwait

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicLebanon

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicLibya

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicMorocco

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicOman

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicQatar

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicSyria

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicTunisia

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicUAE

The following enumerations have this constant:

- cdrTextLanguage

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cdrArabicYemen

The following enumerations have this constant:

- cdrTextLanguage

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cdrArc

The following enumerations have this constant:

- cdrEllipseType

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cdrArmenian

The following enumerations have this constant:

- cdrTextLanguage

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cdrArtisticFittedText

The following enumerations have this constant:

- cdrTextType

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cdrArtisticMediaGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrArtisticText

The following enumerations have this constant:

- cdrTextType

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cdrAscenderVertPlacement

The following enumerations have this constant:

- cdrFittedVertPlacement

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cdrAssamese

The following enumerations have this constant:

- cdrTextLanguage

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cdrAT1

The following enumerations have this constant:

- cdrFilter

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cdrAutoSense

The following enumerations have this constant:

- cdrFilter

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cdrAVI

The following enumerations have this constant:

- cdrFilter

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cdrAzeriCyrillic

The following enumerations have this constant:

- cdrTextLanguage

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cdrAzeriLatin

The following enumerations have this constant:

- cdrTextLanguage

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cdrBaselineVertPlacement

The following enumerations have this constant:

- cdrFittedVertPlacement

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cdrBasque

The following enumerations have this constant:

- cdrTextLanguage

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cdrBelgianDutch

The following enumerations have this constant:

- cdrTextLanguage

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cdrBelgianFrench

The following enumerations have this constant:

- cdrTextLanguage

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cdrBengali

The following enumerations have this constant:

- cdrTextLanguage

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cdrBevelGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrBitmapPattern

The following enumerations have this constant:

- cdrPatternFillType

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cdrBitmapPatternFillStyle

The following enumerations have this constant:

- cdrFillStyleType

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cdrBitmapShape

The following enumerations have this constant:

- cdrShapeType

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cdrBlackAndWhiteImage

The following enumerations have this constant:

- cdrImageType

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cdrBlend

The following enumerations have this constant:

- cdrEffectType

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cdrBlendGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrBlendMode

The **cdrBlendMode** enumeration has the following constants:

Constant	Value	Description
cdrBlendSteps	**0**	Specifies to use steps
cdrBlendSpacing	**1**	Specifies to use spacing

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cdrBlendSpacing

The following enumerations have this constant:

- cdrBlendMode

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cdrBlendSteps

The following enumerations have this constant:

- cdrBlendMode

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cdrBMP

The following enumerations have this constant:

- cdrFilter

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cdrBoldFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrBoldItalicFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrBottomJustify

The following enumerations have this constant:

- cdrVerticalAlignment

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cdrBottomLeft

The following enumerations have this constant:

- cdrReferencePoint

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cdrBottomLeftPoint

The following enumerations have this constant:

- cdrPresetPoint

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cdrBottomMiddle

The following enumerations have this constant:

- cdrReferencePoint

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cdrBottomPoint

The following enumerations have this constant:

- cdrPresetPoint

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cdrBottomQuadrant

The following enumerations have this constant:

- cdrFittedQuadrant

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cdrBottomRight

The following enumerations have this constant:

- cdrReferencePoint

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cdrBottomRightPoint

The following enumerations have this constant:

- cdrPresetPoint

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cdrBrazilianPortuguese

The following enumerations have this constant:

- cdrTextLanguage

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cdrBulgarian

The following enumerations have this constant:

- cdrTextLanguage

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cdrBurmese

The following enumerations have this constant:

- cdrTextLanguage

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cdrByelorussian

The following enumerations have this constant:

- cdrTextLanguage

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cdrCAL

The following enumerations have this constant:

- cdrFilter

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cdrCascade

The following enumerations have this constant:

- cdrWindowArrangeStyle

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cdrCatalan

The following enumerations have this constant:

- cdrTextLanguage

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cdrCDT

The following enumerations have this constant:

- cdrFilter

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cdrCDX

The following enumerations have this constant:

- cdrFilter

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cdrCenter

The following enumerations have this constant:

- cdrReferencePoint

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cdrCenterAlignment

The following enumerations have this constant:

- cdrAlignment

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cdrCenterJustify

The following enumerations have this constant:

- cdrVerticalAlignment

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cdrCenterPlacement

The following enumerations have this constant:

- cdrFittedPlacement

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cdrCenterPoint

The following enumerations have this constant:

- cdrPresetPoint

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cdrCenterVertPlacement

The following enumerations have this constant:

- cdrFittedVertPlacement

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cdrCentimeter

The following enumerations have this constant:

- cdrUnit

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cdrCGM

The following enumerations have this constant:

- cdrFilter

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cdrCharacterIndexing

The following enumerations have this constant:

- cdrTextIndexingType

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cdrCharSetANSI

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetArabic

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetBaltic

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetChineseBig5

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetDefault

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetEastEurope

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetGreek

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetHangul

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetHebrew

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetJohab

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetMac

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetMixed

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetOEM

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetRussian

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetShiftJIS

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetSymbol

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetThai

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetTurkish

The following enumerations have this constant:

- cdrTextCharSet

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cdrCharSetVietnamese

The following enumerations have this constant:

- cdrTextCharSet

Copyright 2014 Corel Corporation. All rights reserved.

cdrChineseHongKong

The following enumerations have this constant:

- cdrTextLanguage

Copyright 2014 Corel Corporation. All rights reserved.

cdrChineseMacao

The following enumerations have this constant:

- cdrTextLanguage

Copyright 2014 Corel Corporation. All rights reserved.

cdrChineseSingapore

The following enumerations have this constant:

- cdrTextLanguage

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cdrCicero

The following enumerations have this constant:

- cdrUnit

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cdrCLK

The following enumerations have this constant:

- cdrFilter

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cdrCloneAll

The following enumerations have this constant:

- cdrCloneLinkType

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cdrCloneBmpColorMask

The following enumerations have this constant:

- cdrCloneLinkType

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cdrCloneFill

The following enumerations have this constant:

- cdrCloneLinkType

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cdrCloneLinkType

The **cdrCloneLinkType** enumeration has the following constants:

Constant	Value	Description
cdrCloneFill	**1**	
cdrCloneOutline	**2**	
cdrCloneShape	**4**	
cdrCloneTransform	**8**	
cdrCloneBmpColorMask	**16**	
cdrCloneAll	**31**	

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cdrCloneOutline

The following enumerations have this constant:

- cdrCloneLinkType

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cdrCloneShape

The following enumerations have this constant:

- cdrCloneLinkType

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cdrCloneTransform

The following enumerations have this constant:

- cdrCloneLinkType

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cdrCMDCategoryMacros

The following modules have this constant:

- UICategories

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cdrCMDCategoryPlugins

The following modules have this constant:

- UICategories

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cdrCMX5

The following enumerations have this constant:

- cdrFilter

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cdrCMX6

The following enumerations have this constant:

- cdrFilter

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cdrCMX64

The following enumerations have this constant:

- cdrFilter

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cdrCMYKColorImage

The following enumerations have this constant:

- cdrImageType

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cdrCMYKMultiChannelImage

The following enumerations have this constant:

- cdrImageType

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cdrColorBlackAndWhite

The following enumerations have this constant:

- cdrColorType

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cdrColorCMY

The following enumerations have this constant:

- cdrColorType

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cdrColorCMYK

The following enumerations have this constant:

- cdrColorType

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cdrColorGray

The following enumerations have this constant:

- cdrColorType

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cdrColorHLS

The following enumerations have this constant:

- cdrColorType

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cdrColorHSB

The following enumerations have this constant:

- cdrColorType

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cdrColorLab

The following enumerations have this constant:

- cdrColorType

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cdrColorMixed

The following enumerations have this constant:

- cdrColorType

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cdrColorPantone

The following enumerations have this constant:

- cdrColorType

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cdrColorPantoneHex

The following enumerations have this constant:

- cdrColorType

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cdrColorRegistration

The following enumerations have this constant:

- cdrColorType

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cdrColorRGB

The following enumerations have this constant:

- cdrColorType

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cdrColorSpot

The following enumerations have this constant:

- cdrColorType

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cdrColorType

The **cdrColorType** enumeration has the following constants:

Constant	Value	Description
cdrColorPantone	**1**	Specifies PANTONE
cdrColorCMYK	**2**	Specifies CMYK
cdrColorCMY	**4**	Specifies CMY
cdrColorRGB	**5**	Specifies RGB
cdrColorHSB	**6**	Specifies HSB
cdrColorHLS	**7**	Specifies HLS
cdrColorBlackAndWhite	**8**	Specifies black-and-white
cdrColorGray	**9**	Specifies grayscale
cdrColorYIQ	**11**	Specifies YIQ
cdrColorLab	**12**	Specifies LAB
cdrColorPantoneHex	**14**	Specifies PANTONE Hex
cdrColorRegistration	**20**	Specifies registration
cdrColorUserInk	**22**	Specifies user ink
cdrColorSpot	**25**	Specifies spot
cdrColorMultiChannel	**26**	Specifies multi-channel
cdrColorMixed	**99**	Specifies mixed

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cdrColorYIQ

The following enumerations have this constant:

- cdrColorType

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cdrCommandChecked

The following enumerations have this constant:

- cdrCommandCheckState

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cdrCommandCheckState

The **cdrCommandCheckState** enumeration has the following constants:

Constant	Value	Description
cdrCommandUnchecked	**0**	Specifies unchecked
cdrCommandChecked	**1**	Specifies checked

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cdrCommandUnchecked

The following enumerations have this constant:

- cdrCommandCheckState

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cdrCompareAtLeast

The following enumerations have this constant:

- cdrCompareCondition

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cdrCompareAtMost

The following enumerations have this constant:

- cdrCompareCondition

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cdrCompareCondition

The **cdrCompareCondition** enumeration has the following constants:

Constant	Value	Description
cdrCompareEquals	**0**	Specifies “equals”
cdrCompareGreater	**1**	Specifies “greater than”
cdrCompareLess	**2**	Specifies “less than”
cdrCompareAtLeast	**3**	Specifies “at least”
cdrCompareAtMost	**4**	Specifies “at most”

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cdrCompareEquals

The following enumerations have this constant:

- cdrCompareCondition

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cdrCompareFill

The following enumerations have this constant:

- cdrCompareType

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cdrCompareFillType

The following enumerations have this constant:

- cdrCompareType

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cdrCompareGreater

The following enumerations have this constant:

- cdrCompareCondition

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cdrCompareLess

The following enumerations have this constant:

- cdrCompareCondition

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cdrCompareOutline

The following enumerations have this constant:

- cdrCompareType

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cdrCompareOutlineColor

The following enumerations have this constant:

- cdrCompareType

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cdrCompareOutlineWidth

The following enumerations have this constant:

- cdrCompareType

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cdrCompareShapeHeight

The following enumerations have this constant:

- cdrCompareType

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cdrCompareShapeType

The following enumerations have this constant:

- cdrCompareType

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cdrCompareShapeWidth

The following enumerations have this constant:

- cdrCompareType

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cdrCompareType

The **cdrCompareType** enumeration has the following constants:

Constant	Value	Description
cdrCompareShapeType	**1**	Specifies shape type
cdrCompareFillType	**2**	Specifies fill type
cdrCompareFill	**4**	Specifies fill
cdrCompareOutlineWidth	**8**	Specifies outline width
cdrCompareOutlineColor	**16**	Specifies outline color
cdrCompareOutline	**32**	Specifies outline
cdrCompareShapeWidth	**64**	Specifies shape width
cdrCompareShapeHeight	**128**	Specifies shape height

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cdrCompressionCCITT3_1d

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionCCITT3_2d

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionCCITT4

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionHuffman

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionJPEG

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionLZW

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionNone

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionPackBits

The following enumerations have this constant:

- cdrCompressionType

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cdrCompressionType

The **cdrCompressionType** enumeration has the following constants:

Constant	Value	Description
cdrCompressionNone	**0**	Specifies none
cdrCompressionLZW	**1**	Specifies LZW
cdrCompressionPackBits	**2**	
cdrCompressionHuffman	**3**	Specifies Huffman
cdrCompressionCCITT3_1d	**4**	
cdrCompressionCCITT3_2d	**5**	
cdrCompressionCCITT4	**6**	
cdrCompressionZIP	**7**	Specifies ZIP
cdrCompressionJPEG	**8**	Specifies JPEG

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cdrCompressionZIP

The following enumerations have this constant:

- cdrCompressionType

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cdrConicalFountainFill

The following enumerations have this constant:

- cdrFountainFillType

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cdrConnectorBezier

The following enumerations have this constant:

- cdrConnectorType

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cdrConnectorBSpline

The following enumerations have this constant:

- cdrConnectorType

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cdrConnectorObject

The following enumerations have this constant:

- cdrConnectorType

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cdrConnectorRightAngle

The following enumerations have this constant:

- cdrConnectorType

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cdrConnectorShape

The following enumerations have this constant:

- cdrShapeType

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cdrConnectorStraight

The following enumerations have this constant:

- cdrConnectorType

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cdrConnectorType

The **cdrConnectorType** enumeration has the following constants:

Constant	Value	Description
cdrConnectorStraight	**0**	Specifies a straight connector
cdrConnectorRightAngle	**1**	Specifies a right-angled connector
cdrConnectorBSpline	**2**	Specifies a B-spline connector
cdrConnectorBezier	**3**	Specifies a Bézier connector
cdrConnectorObject	**4**	Specifies an object connector

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cdrContour

The following enumerations have this constant:

- cdrEffectType

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cdrContourButtCap

The following enumerations have this constant:

- cdrContourEndCapType

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cdrContourCornerBevel

The following enumerations have this constant:

- cdrContourCornerType

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cdrContourCornerMiteredBevel

The following enumerations have this constant:

- cdrContourCornerType

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cdrContourCornerMiteredOffsetBevel

The following enumerations have this constant:

- cdrContourCornerType

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cdrContourCornerMiteredRound

The following enumerations have this constant:

- cdrContourCornerType

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cdrContourCornerOffsetBevel

The following enumerations have this constant:

- cdrContourCornerType

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cdrContourCornerRound

The following enumerations have this constant:

- cdrContourCornerType

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cdrContourCornerType

The **cdrContourCornerType** enumeration has the following constants:

Constant	Value	Description
cdrContourCornerBevel	**0**	Specifies a beveled contour corner
cdrContourCornerOffsetBevel	**1**	Specifies an offset, beveled contour corner
cdrContourCornerRound	**2**	Specifies a round contour corner
cdrContourCornerMiteredBevel	**3**	Specifies a mitered, beveled contour corner
cdrContourCornerMiteredOffsetBevel	**4**	Specifies a mitered, offset, beveled contour corner
cdrContourCornerMiteredRound	**5**	Specifies a mitered, round contour corner

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cdrContourDirection

The **cdrContourDirection** enumeration has the following constants:

Constant	Value	Description
cdrContourInside	**0**	Specifies an inside contour
cdrContourOutside	**1**	Specifies an outside contour
cdrContourToCenter	**2**	Specifies a contour to the center

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cdrContourEndCapType

The **cdrContourEndCapType** enumeration has the following constants:

Constant	Value	Description
cdrContourButtCap	**0**	Specifies a butted end cap
cdrContourRoundCap	**1**	Specifies a round end cap
cdrContourSquareCap	**2**	Specifies a square end cap

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cdrContourGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrContourInside

The following enumerations have this constant:

- cdrContourDirection

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cdrContourOutside

The following enumerations have this constant:

- cdrContourDirection

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cdrContourRoundCap

The following enumerations have this constant:

- cdrContourEndCapType

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cdrContourSquareCap

The following enumerations have this constant:

- cdrContourEndCapType

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cdrContourToCenter

The following enumerations have this constant:

- cdrContourDirection

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cdrControlPath

The following enumerations have this constant:

- cdrEffectType

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cdrCopyFill

The following enumerations have this constant:

- cdrCopyProperties

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cdrCopyOutlineColor

The following enumerations have this constant:

- cdrCopyProperties

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cdrCopyOutlinePen

The following enumerations have this constant:

- cdrCopyProperties

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cdrCopyProperties

The **cdrCopyProperties** enumeration has the following constants:

Constant	Value	Description
cdrCopyOutlineColor	**1**	Copies the outline color
cdrCopyOutlinePen	**2**	Copies the outline pen
cdrCopyFill	**4**	Copies the fill
cdrCopyTextAttrs	**8**	Copies the text attributes

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cdrCopyTextAttrs

The following enumerations have this constant:

- cdrCopyProperties

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cdrCorelDESIGNER

The following enumerations have this constant:

- cdrApplicationID

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cdrCorelDRAW

The following enumerations have this constant:

- cdrApplicationID

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cdrCornerType

The **cdrCornerType** enumeration has the following constants:

Constant	Value	Description
cdrCornerTypeRound	**0**	Specifies rounded corners
cdrCornerTypeScallop	**1**	Specifies scalloped corners
cdrCornerTypeChamfer	**2**	Specifies chamfered corners

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cdrCornerTypeChamfer

The following enumerations have this constant:

- cdrCornerType

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cdrCornerTypeRound

The following enumerations have this constant:

- cdrCornerType

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cdrCornerTypeScallop

The following enumerations have this constant:

- cdrCornerType

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cdrCPT

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cdrCPT8

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cdrCPT9

The following enumerations have this constant:

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cdrCPT10

The following enumerations have this constant:

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cdrCPT11

The following enumerations have this constant:

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cdrCPX

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- cdrFilter

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cdrCreativeGraphics

The following enumerations have this constant:

- cdrApplicationClass

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cdrCroatian

The following enumerations have this constant:

- cdrTextLanguage

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cdrCSL

The following enumerations have this constant:

- cdrFilter

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cdrCUR

The following enumerations have this constant:

- cdrFilter

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cdrCurrentPage

The following enumerations have this constant:

- cdrExportRange

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cdrCurrentVersion

The following enumerations have this constant:

- cdrFileVersion

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cdrCursorExtPick

The following enumerations have this constant:

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cdrCursorEyeDrop

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorIntersectSingle

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorNodeEdit

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorPenJoin

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorPick

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorPickNone

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorPickOvertarget

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorShape

The **cdrCursorShape** enumeration has the following constants:

Constant	Value	Description
cdrCursorWinAppStarting	**0**	
cdrCursorWinArrow	**1**	
cdrCursorWinCross	**2**	
cdrCursorWinHelp	**3**	
cdrCursorWinIbeam	**4**	
cdrCursorWinNo	**5**	
cdrCursorWinSizeAll	**6**	
cdrCursorWinSizeNeSw	**7**	
cdrCursorWinSizeNs	**8**	
cdrCursorWinSizeNwSe	**9**	
cdrCursorWinSizeWe	**10**	
cdrCursorWinUpArrow	**11**	
cdrCursorWinWait	**12**	
cdrCursorSmallcrosshair	**301**	
cdrCursorPick	**305**	
cdrCursorNodeEdit	**306**	
cdrCursorEyeDrop	**326**	
cdrCursorExtPick	**351**	
cdrCursorPickNone	**305**	
cdrCursorPenJoin	**380**	
cdrCursorPickOvertarget	**396**	
cdrCursorTrimSingle	**428**	
cdrCursorWeldSingle	**430**	
cdrCursorIntersectSingle	**432**	

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cdrCursorSmallcrosshair

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorTrimSingle

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWeldSingle

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinAppStarting

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinArrow

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinCross

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinHelp

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinIbeam

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinNo

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinSizeAll

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinSizeNeSw

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinSizeNs

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinSizeNwSe

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinSizeWe

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinUpArrow

The following enumerations have this constant:

- cdrCursorShape

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cdrCursorWinWait

The following enumerations have this constant:

- cdrCursorShape

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cdrCurveElementFlags

The **cdrCurveElementFlags** enumeration has the following constants:

Constant	Value	Description
cdrFlagSelected	**1**	Specifies selected flags
cdrFlagUser	**4**	Specifies user flags
cdrFlagClosed	**8**	Specifies closed flags
cdrFlagValid	**128**	Specifies valid flags

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cdrCurveElementType

The **cdrCurveElementType** enumeration has the following constants:

Constant	Value	Description
cdrElementStart	**0**	
cdrElementLine	**1**	
cdrElementCurve	**2**	
cdrElementControl	**3**	

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cdrCurveSegment

The following enumerations have this constant:

- cdrSegmentType

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cdrCurveShape

The following enumerations have this constant:

- cdrShapeType

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cdrCuspNode

The following enumerations have this constant:

- cdrNodeType

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cdrCustom

The following enumerations have this constant:

- cdrPaletteID

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cdrCustomEffect

The following enumerations have this constant:

- cdrEffectType

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cdrCustomEffectGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrCustomFountainFillBlend

The following enumerations have this constant:

- cdrFountainFillBlendType

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cdrCustomPalette

The following enumerations have this constant:

- cdrPaletteType

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cdrCustomShape

The following enumerations have this constant:

- cdrShapeType

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cdrCustomVertPlacement

The following enumerations have this constant:

- cdrFittedVertPlacement

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cdrCzech

The following enumerations have this constant:

- cdrTextLanguage

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cdrDanish

The following enumerations have this constant:

- cdrTextLanguage

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cdrDataFormatType

The **cdrDataFormatType** enumeration has the following constants:

Constant	Value	Description
cdrFormatGeneral	**0**	Specifies a general data format
cdrFormatDateTime	**1**	Specifies a date/time data format
cdrFormatLinearAngular	**2**	Specifies a linear/angular data format
cdrFormatNumeric	**3**	Specifies a numeric data format

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cdrDataType

The **cdrDataType** enumeration has the following constants:

Constant	Value	Description
cdrDataTypeString	**0**	Specifies a string
cdrDataTypeNumber	**1**	Specifies a number
cdrDataTypeEvent	**2**	Specifies an event
cdrDataTypeAction	**3**	Specifies an action

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cdrDataTypeAction

The following enumerations have this constant:

- cdrDataType

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cdrDataTypeEvent

The following enumerations have this constant:

- cdrDataType

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cdrDataTypeNumber

The following enumerations have this constant:

- cdrDataType

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cdrDataTypeString

The following enumerations have this constant:

- cdrDataType

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cdrDCS

The following enumerations have this constant:

cdrFilter

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cdrDES

The following enumerations have this constant:

cdrFilter

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cdrDescenderVertPlacement

The following enumerations have this constant:

cdrFittedVertPlacement

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cdrDIC

The following enumerations have this constant:

cdrPaletteID

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cdrDidots

The following enumerations have this constant:

cdrUnit

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cdrDimensionAboveLine

The following enumerations have this constant:

- cdrDimensionPlacement

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cdrDimensionAngular

The following enumerations have this constant:

- cdrDimensionType

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cdrDimensionAngularUnits

The **cdrDimensionAngularUnits** enumeration has the following constants:

Constant	Value	Description
cdrDimensionUnitDegrees	**0**	Specifies degrees for angular dimensions
cdrDimensionUnitRadians	**1**	Specifies radians for angular dimensions
cdrDimensionUnitGradians	**2**	Specifies gradians for angular dimensions
cdrDimensionUnitDegreesSymbol	**3**	Specifies degrees (plus degree symbol) for angular dimensions

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cdrDimensionBelowLine

The following enumerations have this constant:

- cdrDimensionPlacement

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cdrDimensionHorizontal

The following enumerations have this constant:

- cdrLinearDimensionType

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cdrDimensionLinear

The following enumerations have this constant:

- cdrDimensionType

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cdrDimensionLinearUnits

The **cdrDimensionLinearUnits** enumeration has the following constants:

Constant	Value	Description
cdrDimensionUnitInX	**0**	Specifies inches (“ " ”)
cdrDimensionUnitIN	**1**	Specifies inches (“in”)
cdrDimensionUnitInches	**2**	Specifies inches (“inches”)
cdrDimensionUnitFtX	**3**	Specifies feet (“ ' ”)
cdrDimensionUnitFT	**4**	Specifies feet (“ft”)
cdrDimensionUnitMI	**5**	Specifies miles (“mi”)
cdrDimensionUnitMiles	**6**	Specifies miles (“miles”)
cdrDimensionUnitYDS	**7**	Specifies yards (“yds”)
cdrDimensionUnitYards	**8**	Specifies yards (“yards”)
cdrDimensionUnitM	**9**	Specifies meters (“m”)
cdrDimensionUnitMeters	**10**	Specifies meters (“meters”)
cdrDimensionUnitKM	**11**	Specifies kilometers (“km”)
cdrDimensionUnitKilometers	**12**	Specifies kilometers (“kilometers”)
cdrDimensionUnitCM	**13**	Specifies centimeters (“cm”)
cdrDimensionUnitCentimeters	**14**	Specifies centimeters (“centimeters”)
cdrDimensionUnitMM	**15**	Specifies millimeters (“mm”)
cdrDimensionUnitMillimeters	**16**	Specifies millimeters (“millimeters”)
cdrDimensionUnitPicas	**17**	Specifies picas (“picas”)
cdrDimensionUnitPoints	**18**	Specifies points (“points”)
cdrDimensionUnitCiceros	**19**	Specifies ciceros (“ciceros”)
cdrDimensionUnitDidots	**20**	Specifies didots (“didots”)

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cdrDimensionPlacement

The **cdrDimensionPlacement** enumeration has the following constants:

Constant	Value	Description
cdrDimensionAboveLine	**0**	Specifies dimension placement above the line
cdrDimensionWithinLine	**1**	Specifies dimension placement within the line
cdrDimensionBelowLine	**2**	Specifies dimension placement below the line

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cdrDimensionSlanted

The following enumerations have this constant:

- cdrLinearDimensionType

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cdrDimensionStyle

The **cdrDimensionStyle** enumeration has the following constants:

Constant	Value	Description
cdrDimensionStyleDecimal	**0**	Specifies decimal-style dimensions
cdrDimensionStyleFractional	**1**	Specifies fractional-style dimensions
cdrDimensionStyleEngineering	**2**	Specifies engineering-style dimensions
cdrDimensionStyleArchitectural	**3**	Specifies architectural-style dimensions

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cdrDimensionStyleArchitectural

The following enumerations have this constant:

- cdrDimensionStyle

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cdrDimensionStyleDecimal

The following enumerations have this constant:

- cdrDimensionStyle

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cdrDimensionStyleEngineering

The following enumerations have this constant:

- cdrDimensionStyle

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cdrDimensionStyleFractional

The following enumerations have this constant:

- cdrDimensionStyle

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cdrDimensionSymbol

The **cdrDimensionSymbol** enumeration has the following constants:

Constant	Value	Description
cdrDimensionSymbolNone	**0**	Specifies none
cdrDimensionSymbol1After	**1**	
cdrDimensionSymbol1Before	**2**	
cdrDimensionSymbol3After	**3**	
cdrDimensionSymbol3Before	**4**	
cdrDimensionSymbolDAfter	**5**	
cdrDimensionSymbolDBefore	**6**	

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cdrDimensionSymbol1After

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionSymbol1Before

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionSymbol3After

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionSymbol3Before

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionSymbolDAfter

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionSymbolDBefore

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionSymbolNone

The following enumerations have this constant:

- cdrDimensionSymbol

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cdrDimensionType

The **cdrDimensionType** enumeration has the following constants:

Constant	Value	Description
cdrDimensionLinear	**0**	Specifies a linear dimension
cdrDimensionAngular	**1**	Specifies an angular dimension

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cdrDimensionUnitCentimeters

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitCiceros

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitCM

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitDegrees

The following enumerations have this constant:

- cdrDimensionAngularUnits

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cdrDimensionUnitDegreesSymbol

The following enumerations have this constant:

- cdrDimensionAngularUnits

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cdrDimensionUnitDidots

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitFT

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitFtX

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitGradians

The following enumerations have this constant:

- cdrDimensionAngularUnits

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cdrDimensionUnitIN

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitInches

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitInX

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitKilometers

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitKM

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitM

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitMeters

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitMI

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitMiles

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitMillimeters

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitMM

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitPicas

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitPoints

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitRadians

The following enumerations have this constant:

- cdrDimensionAngularUnits

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cdrDimensionUnitYards

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionUnitYDS

The following enumerations have this constant:

- cdrDimensionLinearUnits

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cdrDimensionVertical

The following enumerations have this constant:

- cdrLinearDimensionType

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cdrDimensionWithinLine

The following enumerations have this constant:

- cdrDimensionPlacement

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cdrDirectFountainFillBlend

The following enumerations have this constant:

- cdrFountainFillBlendType

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cdrDirection

The **cdrDirection** enumeration has the following constants:

Constant	Value	Description
cdrLeft	**0**	Specifies left
cdrRight	**1**	Specifies right
cdrUp	**2**	Specifies up
cdrDown	**3**	Specifies down

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cdrDistanceMode

The **cdrDistanceMode** enumeration has the following constants:

Constant	Value	Description
cdrModeNoOffset	0	Specifies no offset
cdrModeOffset	**1**	Specifies distance as offset
cdrModeSpacing	**2**	Specifies distance as spacing

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cdrDistortion

The following enumerations have this constant:

- cdrEffectType

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cdrDistortionCustom

The following enumerations have this constant:

- cdrDistortionType

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cdrDistortionPushPull

The following enumerations have this constant:

- cdrDistortionType

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cdrDistortionTwister

The following enumerations have this constant:

- cdrDistortionType

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cdrDistortionType

The **cdrDistortionType** enumeration has the following constants:

Constant	Value	Description
cdrDistortionPushPull	**0**	Specifies Push-and-pull distortion
cdrDistortionZipper	**1**	Specifies Zipper distortion
cdrDistortionTwister	**2**	Specifies Twister distortion
cdrDistortionCustom	**3**	Specifies custom distortion

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cdrDistortionZipper

The following enumerations have this constant:

- cdrDistortionType

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cdrDistributeArea

The **cdrDistributeArea** enumeration has the following constants:

Constant	Value	Description
cdrDistributeToSelection	**0**	Distributes objects within the bounding box of the selection
cdrDistributeToPage	**1**	Distributes objects over the entire page
cdrDistributeToRect	**2**	Distributes objects within a specified rectangle

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cdrDistributeBottom

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeHCenter

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeHSpacing

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeLeft

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeRight

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeTop

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeToPage

The following enumerations have this constant:

- cdrDistributeArea

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cdrDistributeToRect

The following enumerations have this constant:

- cdrDistributeArea

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cdrDistributeToSelection

The following enumerations have this constant:

- cdrDistributeArea

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cdrDistributeType

The **cdrDistributeType** enumeration has the following constants:

Constant	Value	Description
cdrDistributeLeft	**1**	Specifies left distribution
cdrDistributeRight	**2**	Specifies right distribution
cdrDistributeHCenter	**3**	Specifies horizontal-center distribution
cdrDistributeHSpacing	**4**	Specifies horizontal-spacing distribution
cdrDistributeTop	**8**	Specifies top distribution
cdrDistributeBottom	**16**	Specifies bottom distribution
cdrDistributeVCenter	**24**	Specifies vertical-center distribution
cdrDistributeVSpacing	**32**	Specifies vertical-spacing distribution

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cdrDistributeVCenter

The following enumerations have this constant:

- cdrDistributeType

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cdrDistributeVSpacing

The following enumerations have this constant:

- cdrDistributeType

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cdrDitherFloyd

The following enumerations have this constant:

- cdrDitherType

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cdrDitherJarvis

The following enumerations have this constant:

- cdrDitherType

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cdrDitherNone

The following enumerations have this constant:

- cdrDitherType

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cdrDitherOrdered

The following enumerations have this constant:

- cdrDitherType

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cdrDitherStucki

The following enumerations have this constant:

- cdrDitherType

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cdrDitherType

The **cdrDitherType** enumeration has the following constants:

Constant	Value	Description
cdrDitherNone	**0**	Specifies none
cdrDitherOrdered	**1**	Specifies ordered
cdrDitherJarvis	**2**	Specifies Jarvis
cdrDitherStucki	**3**	Specifies Stucki
cdrDitherFloyd	**4**	Specifies Floyd

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cdrDOC

The following enumerations have this constant:

- cdrFilter

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cdrDocBook

The following enumerations have this constant:

- cdrDocLayout

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cdrDocBooklet

The following enumerations have this constant:

- cdrDocLayout

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cdrDocFullPage

The following enumerations have this constant:

- cdrDocLayout

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cdrDocLayout

The **cdrDocLayout** enumeration has the following constants:

Constant	Value	Description
cdrDocFullPage	**0**	Specifies full-page layout
cdrDocBook	**1**	Specifies a book
cdrDocBooklet	**2**	Specifies a booklet
cdrDocTentCard	**3**	Specifies a tent card
cdrDocSideFoldCard	**4**	Specifies a side-fold card
cdrDocTopFoldCard	**5**	Specifies a top-fold card
cdrDocTriFoldBrochure	**6**	Specifies a tri-fold brochure

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cdrDocSideFoldCard

The following enumerations have this constant:

- cdrDocLayout

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cdrDocTentCard

The following enumerations have this constant:

- cdrDocLayout

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cdrDocTopFoldCard

The following enumerations have this constant:

- cdrDocLayout

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cdrDocTriFoldBrochure

The following enumerations have this constant:

- cdrDocLayout

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cdrDocumentPalette

The following enumerations have this constant:

- cdrPaletteType

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cdrDoubleThinFontLine

The following enumerations have this constant:

- cdrFontLine

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cdrDoubleThinWordFontLine

The following enumerations have this constant:

- cdrFontLine

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cdrDown

The following enumerations have this constant:

- cdrDirection

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cdrDraftView

The following enumerations have this constant:

- cdrViewType

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cdrDropShadow

The following enumerations have this constant:

- cdrEffectType

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cdrDropShadowBottom

The following enumerations have this constant:

- cdrDropShadowType

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cdrDropShadowFlat

The following enumerations have this constant:

- cdrDropShadowType

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cdrDropShadowGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrDropShadowLeft

The following enumerations have this constant:

- cdrDropShadowType

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cdrDropShadowRight

The following enumerations have this constant:

- cdrDropShadowType

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cdrDropShadowTop

The following enumerations have this constant:

- cdrDropShadowType

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cdrDropShadowType

The **cdrDropShadowType** enumeration has the following constants:

Constant	Value	Description
cdrDropShadowFlat	**0**	Specifies flat drop shadow
cdrDropShadowBottom	**1**	Specifies bottom drop shadow
cdrDropShadowTop	**2**	Specifies top drop shadow
cdrDropShadowLeft	**3**	Specifies left drop shadow
cdrDropShadowRight	**4**	Specifies right drop shadow

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cdrDRW

The following enumerations have this constant:

- cdrFilter

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cdrDSF

The following enumerations have this constant:

- cdrFilter

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cdrDuotone

The following enumerations have this constant:

- cdrDuotoneType

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cdrDuotoneImage

The following enumerations have this constant:

- cdrImageType

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cdrDuotoneType

The **cdrDuotoneType** enumeration has the following constants:

Constant	Value	Description
cdrMonotone	**0**	Specifies monotone
cdrDuotone	**1**	Specifies duotone
cdrTritone	**2**	Specifies tritone
cdrQuadtone	**3**	Specifies quadtone

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cdrDutch

The following enumerations have this constant:

- cdrTextLanguage

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cdrDWG

The following enumerations have this constant:

- cdrFilter

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cdrDXF

The following enumerations have this constant:

- cdrFilter

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cdrEdgeFlat

The following enumerations have this constant:

- cdrEdgeType

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cdrEdgeGaussian

The following enumerations have this constant:

- cdrEdgeType

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cdrEdgeInverseSquared

The following enumerations have this constant:

- cdrEdgeType

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cdrEdgeLinear

The following enumerations have this constant:

- cdrEdgeType

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cdrEdgeMesa

The following enumerations have this constant:

- cdrEdgeType

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cdrEdgeSquared

The following enumerations have this constant:

- cdrEdgeType

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cdrEdgeType

The **cdrEdgeType** enumeration has the following constants:

Constant	Value	Description
cdrEdgeLinear	**0**	Specifies linear edge
cdrEdgeSquared	**1**	Specifies squared edge
cdrEdgeFlat	**2**	Specifies flat edge
cdrEdgeInverseSquared	**3**	Specifies inverse-squared edge
cdrEdgeMesa	**4**	Specifies mesa edge
cdrEdgeGaussian	**5**	Specifies Gaussian edge

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cdrEffectType

The **cdrEffectType** enumeration has the following constants:

Constant	Value	Description
cdrBlend	**0**	Specifies a blend
cdrExtrude	**1**	Specifies an extrusion
cdrEnvelope	**2**	Specifies an envelope
cdrTextOnPath	**3**	Specifies text-on-path
cdrControlPath	**4**	Specifies a control path
cdrDropShadow	**5**	Specifies a drop shadow
cdrContour	**6**	Specifies a contour
cdrDistortion	**7**	Specifies a distortion
cdrPerspective	**8**	Specifies a perspective
cdrLens	**9**	Specifies a lens
cdrCustomEffect	**10**	Specifies a custom effect

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cdrElementControl

The following enumerations have this constant:

- cdrCurveElementType

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cdrElementCurve

The following enumerations have this constant:

- cdrCurveElementType

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cdrElementLine

The following enumerations have this constant:

- cdrCurveElementType

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cdrElementStart

The following enumerations have this constant:

- cdrCurveElementType

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cdrEllipse

The following enumerations have this constant:

- cdrEllipseType

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cdrEllipseShape

The following enumerations have this constant:

- cdrShapeType

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cdrEllipseType

The **cdrEllipseType** enumeration has the following constants:

Constant	Value	Description
cdrEllipse	**0**	Specifies an ellipse
cdrArc	**1**	Specifies an arc
cdrPie	**2**	Specifies a pie

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cdrEMF

The following enumerations have this constant:

- cdrFilter

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cdrEnglishAUS

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishBelize

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishCanadian

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishCaribbean

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishIreland

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishJamaica

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishNewZealand

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishPhilippines

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishSouthAfrica

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishTrinidad

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishUK

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishUS

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnglishZimbabwe

The following enumerations have this constant:

- cdrTextLanguage

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cdrEnhancedOutline

The following enumerations have this constant:

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cdrEnhancedView

The following enumerations have this constant:

- cdrViewType

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cdrEnvelope

The following enumerations have this constant:

- cdrEffectType

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cdrEnvelopeCenter

The following enumerations have this constant:

- cdrEnvelopeCopyMode

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cdrEnvelopeCopyMode

The **cdrEnvelopeCopyMode** enumeration has the following constants:

Constant	Value	Description
cdrEnvelopeUseOriginal	**0**	Uses the original envelope
cdrEnvelopeCenter	**1**	Centers the envelope
cdrEnvelopeStretch	**2**	Stretches the envelope
cdrEnvelopeFit	**3**	Fits the envelope

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cdrEnvelopeFit

The following enumerations have this constant:

- cdrEnvelopeCopyMode

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cdrEnvelopeHorizontal

The following enumerations have this constant:

- cdrEnvelopeMode

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cdrEnvelopeMode

The **cdrEnvelopeMode** enumeration has the following constants:

Constant	Value	Description
cdrEnvelopeHorizontal	**0**	Specifies horizontal envelope mapping
cdrEnvelopeOriginal	**1**	Specifies original envelope mapping
cdrEnvelopePutty	**2**	Specifies putty envelope mapping
cdrEnvelopeVertical	**3**	Specifies vertical envelope mapping

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cdrEnvelopeOriginal

The following enumerations have this constant:

- cdrEnvelopeMode

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cdrEnvelopePutty

The following enumerations have this constant:

- cdrEnvelopeMode

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cdrEnvelopeStretch

The following enumerations have this constant:

- cdrEnvelopeCopyMode

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cdrEnvelopeUseOriginal

The following enumerations have this constant:

- cdrEnvelopeCopyMode

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cdrEnvelopeVertical

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- cdrEnvelopeMode

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cdrEPS

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cdrEPSPhotoPaint

The following enumerations have this constant:

- cdrFilter

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cdrEPSShape

The following enumerations have this constant:

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cdrEstonian

The following enumerations have this constant:

- cdrTextLanguage

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cdrEXE

The following enumerations have this constant:

- cdrFilter

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cdrExportRange

The **cdrExportRange** enumeration has the following constants:

Constant	Value	Description
cdrAllPages	**0**	Specifies all pages for export
cdrCurrentPage	**1**	Specifies current page for export
cdrSelection	**2**	Specifies selection for export

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cdrExtraBoldFontStyle

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- cdrFontStyle

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cdrExtraBoldItalicFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrExtraLightFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrExtraLightItalicFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrExtrude

The following enumerations have this constant:

- cdrEffectType

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cdrExtrudeBackParallel

The following enumerations have this constant:

- cdrExtrudeType

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cdrExtrudeBigBack

The following enumerations have this constant:

- cdrExtrudeType

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cdrExtrudeBigFront

The following enumerations have this constant:

- cdrExtrudeType

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cdrExtrudeColorShading

The following enumerations have this constant:

- cdrExtrudeShading

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cdrExtrudeFrontParallel

The following enumerations have this constant:

- cdrExtrudeType

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cdrExtrudeGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrExtrudeLightPosition

The **cdrExtrudeLightPosition** enumeration has the following constants:

Constant	Value	Description
cdrLightFrontTopLeft	**0**	Specifies front, top-left lighting
cdrLightFrontTop	**1**	Specifies front, top lighting
cdrLightFrontTopRight	**2**	Specifies front, top-right lighting
cdrLightFrontLeft	**3**	Specifies front, left lighting
cdrLightFrontCenter	**4**	Specifies front, center lighting
cdrLightFrontRight	**5**	Specifies front, right lighting
cdrLightFrontBottomLeft	**6**	Specifies front, bottom-left lighting
cdrLightFrontBottom	**7**	Specifies front, bottom lighting
cdrLightFrontBottomRight	**8**	Specifies front, bottom-right lighting
cdrLightBackTopLeft	**9**	Specifies back, top-left lighting
cdrLightBackTop	**10**	Specifies back, top lighting
cdrLightBackTopRight	**11**	Specifies back, top-right lighting
cdrLightBackRight	**14**	Specifies back, right lighting
cdrLightBackBottomRight	**17**	Specifies back, bottom-right lighting

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cdrExtrudeObjectFill

The following enumerations have this constant:

- cdrExtrudeShading

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cdrExtrudeShading

The **cdrExtrudeShading** enumeration has the following constants:

Constant	Value	Description
cdrExtrudeObjectFill	**0**	Specifies object-fill shading
cdrExtrudeSolidFill	**1**	Specifies solid-fill shading
cdrExtrudeColorShading	**2**	Specifies color shading

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cdrExtrudeSmallBack

The following enumerations have this constant:

- cdrExtrudeType

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cdrExtrudeSmallFront

The following enumerations have this constant:

- cdrExtrudeType

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cdrExtrudeSolidFill

The following enumerations have this constant:

- cdrExtrudeShading

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cdrExtrudeType

The **cdrExtrudeType** enumeration has the following constants:

Constant	Value	Description
cdrExtrudeSmallBack	**0**	Specifies small, back extrustion
cdrExtrudeSmallFront	**1**	Specifies small, front extrusion
cdrExtrudeBigBack	**2**	Specifies big, back extrusion
cdrExtrudeBigFront	**3**	Specifies big, front extrusion
cdrExtrudeBackParallel	**4**	Specifies back, parallel extrusion
cdrExtrudeFrontParallel	**5**	Specifies front, parallel extrusion

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cdrExtrudeVPType

The **cdrExtrudeVPType** enumeration has the following constants:

Constant	Value	Description
cdrVPLockedToShape	**0**	Specifies locked-to-shape vanishing point
cdrVPLockedToPage	**1**	Specifies locked-to-page vanishing point
cdrVPShared	**2**	Specifies shared vanishing point

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cdrFaeroese

The following enumerations have this constant:

- cdrTextLanguage

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cdrFalse

The following enumerations have this constant:

- cdrTriState

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cdrFarsi

The following enumerations have this constant:

- cdrTextLanguage

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cdrFeatherAverage

The following enumerations have this constant:

- cdrFeatherType

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cdrFeatherInside

The following enumerations have this constant:

- cdrFeatherType

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cdrFeatherMiddle

The following enumerations have this constant:

- cdrFeatherType

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cdrFeatherOutside

The following enumerations have this constant:

- cdrFeatherType

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cdrFeatherType

The **cdrFeatherType** enumeration has the following constants:

Constant	Value	Description
cdrFeatherInside	**0**	Specifies inside feathering
cdrFeatherMiddle	**1**	Specifies middle feathering
cdrFeatherOutside	**2**	Specifies outside feathering
cdrFeatherAverage	**3**	Specifies average feathering

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cdrFH

The following enumerations have this constant:

- cdrFilter

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cdrFileVersion

The **cdrFileVersion** enumeration has the following constants:

Constant	Value	Description
cdrCurrentVersion	**0**	Specifies current version
cdrVersion1	**1**	Specifies version 1
cdrVersion2	**2**	Specifies version 2
cdrVersion3	**3**	Specifies version 3
cdrVersion4	**4**	Specifies version 4
cdrVersion5	**5**	Specifies version 5
cdrVersion6	**6**	Specifies version 6
cdrVersion7	**7**	Specifies version 7
cdrVersion8	**8**	Specifies version 8
cdrVersion9	**9**	Specifies version 9
cdrVersion10	**10**	Specifies version 10
cdrVersion11	**11**	Specifies version 11
cdrVersion12	**12**	Specifies version 12
cdrVersion13	**13**	Specifies version 13
cdrVersion14	**14**	Specifies version 14
cdrVersion15	**15**	Specifies version 15
cdrVersion16	**16**	Specifies version 16
cdrVersion17	**17**	Specifies version 17
cdrVersion105	**105**	Specifies version 10.5

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cdrFillAlternate

The following enumerations have this constant:

- cdrFillMode

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cdrFillMode

The **cdrFillMode** enumeration has the following constants:

Constant	Value	Description
cdrFillAlternate	**0**	Specifies alternate fill mode
cdrFillWinding	**1**	Specifies winding fill mode

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cdrFillStyleType

The **cdrFillStyleType** enumeration has the following constants:

Constant	Value	Description
cdrNoFillStyle	**0**	Specifies no fill
cdrUniformFillStyle	**1**	Specifies a uniform fill
cdrFountainFillStyle	**2**	Specifies a fountain fill
cdrPostscriptFillStyle	**3**	Specifies a PostScript fill
cdrTwoColorPatternFillStyle	**4**	Specifies a two-color-pattern fill
cdrBitmapPatternFillStyle	**6**	Specifies a bitmap-pattern fill
cdrTextureFillStyle	**8**	Specifies a texture fill
cdrFullColorPatternFillStyle	**9**	Specifies a full-color-pattern fill
cdrHatchFillStyle	**10**	Specifies a hatch fill

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cdrFillType

The **cdrFillType** enumeration has the following constants:

Constant	Value	Description
cdrNoFill	**0**	Specifies no fill
cdrUniformFill	**1**	Specifies uniform fill
cdrFountainFill	**2**	Specifies fountain fill
cdrPostscriptFill	**3**	Specifies PostScript fill
cdrTextureFill	**8**	Specifies texture fill
cdrPatternFill	**9**	Specifies pattern fill
cdrHatchFill	**10**	Specifies hatch fill

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cdrFillWinding

The following enumerations have this constant:

- cdrFillMode

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cdrFilter

The **cdrFilter** enumeration has the following constants:

Constant	Value	Description
cdrAutoSense	**0**	Specifies automatic filter detection (Import only)
cdrBMP	**769**	Specifies Windows Bitmap
cdrPCX	**770**	Specifies PaintBrush
cdrTGA	**771**	Specifies Targa Bitmap
cdrTIFF	**772**	Specifies TIFF Bitmap
cdrGIF	**773**	Specifies CompuServe Bitmap
cdrJPEG	**774**	Specifies JPEG Bitmaps
cdrPCD	**775**	Specifies Kodak Photo-CD Image (Import only)
cdrSCT	**776**	Specifies Scitex CT Bitmap
cdrICO	**784**	Specifies Windows 3.x/NT Icon Resource (Import only)
cdrCUR	**785**	Specifies Windows 3.x/NT Cursor Resource (Import only)
cdrEXE	**786**	Specifies Windows 3.x/NT Bitmap Resource (Import only)
cdrIMG	**787**	Specifies GEM Paint File
cdrPSD	**788**	Specifies Adobe Photoshop
cdrPP4	**789**	Specifies Picture Publisher 4 (Import only)
cdrMAC	**791**	Specifies MACPaint Bitmap
cdrOS2BMP	**792**	Specifies OS/2 Bitmap
cdrWI	**793**	Specifies Wavelet Compressed Bitmap
cdrWVL	**793**	Specifies Wavelet Compressed Bitmap
cdrCAL	**800**	Specifies CALS Compressed Bitmap
cdrDCS	**801**	Specifies Desktop Color Separations (Export only)
cdrPNG	**802**	Specifies Portable Network Graphics
cdrPP5	**803**	Specifies Picture Publisher 5.0

cdrEPSPhotoPaint	**804**	Specifies Corel PHOTO-PAINT Encapsulated PostScript
cdrFPX	**806**	Specifies Kodak FlashPix Image
cdrRIFF	**807**	Specifies Corel Painter 5/6 (Import only)
cdrXPM	**809**	Specifies XPixMap Image
cdrXCF	**816**	Specifies Gimp Image (Import only)
cdrJPG2000Import	**818**	Specifies JPG2000 Bitmaps (Import only)
cdrPPF	**819**	Specifies Picture Publisher 10
cdrJP2	**820**	Specifies JPG2000 Standard (Export only)
cdrJPC	**821**	Specifies JPG2000 Codestream (Export only)
cdrPSP	**822**	Specifies Corel Paint Shop Pro Photo
cdrRAW	**823**	Specifies RAW Format
cdrCGM	**1280**	Specifies Computer Graphics Metafile
cdrHPGL	**1281**	Specifies HPGL Plotter File
cdrPLT	**1281**	Specifies HPGL Plotter File
cdrGEM	**1284**	Specifies GEM File
cdrWPG	**1287**	Specifies Corel WordPerfect Graphic
cdrEPS	**1289**	Specifies Encapsulated PostScript
cdrPSEncapsulated	**1289**	Specifies Encapsulated PostScript
cdrPSInterpreted	**1290**	Specifies PostScript Interpreted (Import only)
cdrMET	**1291**	Specifies MET MetaFile (Import only)
cdrNAP	**1292**	Specifies NAP MetaFile (Import only)
cdrPICT	**1293**	Specifies Macintosh PICT
cdrWMF	**1294**	Specifies Windows Metafile
cdrDXF	**1296**	Specifies AutoCAD (DXF)
cdrEMF	**1300**	Specifies Enhanced Windows Metafile
cdrTTF	**1302**	Specifies True Type Font

		(Export only)
cdrAT1	**1303**	Specifies Adobe Type 1 Font (Export only)
cdrAI	**1305**	Specifies Adobe Illustrator
cdrVSD	**1315**	Specifies Visio (Import only)
cdrDWG	**1328**	Specifies AutoCAD (DWG)
cdrFMV	**1329**	Specifies Frame Vector Metafile (Import only)
cdrAI9	**1333**	Specifies Adobe Illustrator 9, Adobe Portable Document Format (Import only)
cdrPDF	**1333**	Specifies Adobe Portable Document Format, Adobe Illustrator 9 (Import only)
cdrPIC	**1334**	Specifies Lotus Pic (Import only)
cdrDRW	**1339**	Specifies Corel/Micrografx Designer (Import only)
cdrDSF	**1339**	Specifies Corel/Micrografx Designer (Import only)
cdrSWF	**1343**	Specifies Macromedia Flash (Export only)
cdrFH	**1344**	Specifies Macromedia Freehand (Import only)
cdrSVG	**1345**	Specifies Scalable Vector Graphics
cdrSVGZ	**1347**	Specifies Compressed SVG
cdrPUB	**1349**	Specifies Microsoft Publisher (Import only)
cdrAVI	**1536**	Specifies Video for Windows
cdrQTM	**1542**	Specifies QuickTime Movie
cdrHTML	**1557**	Specifies HyperText Markup Language (Import only)
cdrGIFAnimation	**1558**	Specifies GIF Animation
cdr3DMF	**1559**	Specifies 3DMF - 3D Model (Import only)
cdrQTVR	**1560**	Specifies QuickTime VR
cdrCMX6	**1793**	Specifies Corel Presentation Exchange

cdrCMX5	**1794**	Specifies Corel Presentation Exchange 5.0
cdrCDR	**1795**	Specifies CorelDRAW
cdrCDX	**1796**	Specifies CorelDRAW Compressed (Import only)
cdrCPX	**1797**	Specifies Corel CMX Compressed (Import only)
cdrCPT8	**1799**	Specifies Corel PHOTO-PAINT 7/8 Image (Export only)
cdrCDT	**1800**	Specifies CorelDRAW Template
cdrPAT	**1801**	Specifies CorelDRAW Pattern File
cdrCLK	**1802**	Specifies Corel R.A.V.E.
cdrDES	**1805**	Specifies Corel DESIGNER
cdrPPT	**1805**	Specifies Microsoft PowerPoint (Import only)
cdrSHW	**1806**	Specifies Corel Presentations (Import only)
cdrCPT	**1808**	Specifies Corel PHOTO-PAINT 11 Image
cdrCPT10	**1808**	Specifies Corel PHOTO-PAINT 9/10 Image
cdrCPT11	**1808**	Specifies Corel PHOTO-PAINT 11 Image
cdrCPT9	**1808**	Specifies Corel PHOTO-PAINT 9/10 Image
cdrCSL	**1810**	Specifies Corel Symbol Library
cdrCMX64	**1811**	Specifies Corel Presentation Exchange 64-bit
cdrTXT	**2048**	Specifies ANSI Text
cdrWord95	**2049**	Specifies MS Word for Windows 6/7
cdrWord2	**2050**	Specifies MS Word for Windows 2.x
cdrWord55	**2051**	Specifies MS Word 3.0, 4.0, 5.0, 5.5
cdrMacWord5	**2052**	Specifies MS Word for Macintosh 4.0, 5.0
cdrRTF	**2053**	Specifies Rich Text Format
cdrWP9	**2055**	Specifies WordPerfect 6 through 9
cdrWPD	**2055**	Specifies WordPerfect 6 and later
cdrWP51	**2056**	Specifies WordPerfect 5.1
cdrWP50	**2057**	Specifies WordPerfect 5.0
cdrWP4	**2058**	Specifies WordPerfect 4.2

cdrWSW	**2059**	Specifies WordStar for Windows 1.x, 2.0
cdrWS7	**2060**	Specifies WordStar 7.0
cdrWS2000	**2061**	Specifies WordStar 2000
cdrWSD	**2061**	Specifies WordStar 2000
cdrXY	**2062**	Specifies XYWrite for Windows 4.0
cdrSAM	**2063**	Specifies Ami Professional (Import only)
cdrXLS	**2064**	Specifies Microsoft Excel (Import only)
cdrWKS	**2066**	Specifies LOTUS 1-2-3 (Import only)
cdrQuattroPro	**2067**	Specifies Corel Quattro Pro (Import only)
cdrDOC	**2068**	Specifies MS Word 97/2000
cdrWord2000	**2068**	Specifies MS Word 97/2000
cdrWS6	**2163**	Specifies WordStar 5.0, 5.5, 6.0 (Import only)
cdrWS4	**2164**	Specifies WordStar 3.3, 3.31, 3.45, 4.0 (Import only)
cdrMIF	**2165**	Specifies FrameMaker 3.0, 4.0, 5.0 (Import only)
cdrWPM	**2166**	Specifies Corel WordPerfect 1.0, 2.x for Macintosh (Import only)

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cdrFinnish

The following enumerations have this constant:

- cdrTextLanguage

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cdrFirstPoint

The following enumerations have this constant:

- cdrPresetPoint

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cdrFittedOrientation

The **cdrFittedOrientation** enumeration has the following constants:

Constant	Value	Description
cdrRotateOrientation	**0**	Specifies rotated orientation
cdrVerticalSkewOrientation	**1**	Specifies vertical skew
cdrHorizontalSkewOrientation	**2**	Specifies horizontal skew
cdrUprightOrientation	**3**	Specifies upright orientation
cdrTransformOrientation	**4**	Specifies transformed orientation

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cdrFittedPlacement

The **cdrFittedPlacement** enumeration has the following constants:

Constant	Value	Description
cdrLeftPlacement	**0**	Specifies left placement
cdrRightPlacement	**1**	Specifies right placement
cdrCenterPlacement	**2**	Specifies center placement

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cdrFittedQuadrant

The **cdrFittedQuadrant** enumeration has the following constants:

Constant	Value	Description
cdrLeftQuadrant	**0**	Specifies the left quadrant
cdrRightQuadrant	**1**	Specifies the right quadrant
cdrTopQuadrant	**2**	Specifies the top quadrant
cdrBottomQuadrant	**3**	Specifies the bottom quadrant

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cdrFittedVertPlacement

The **cdrFittedVertPlacement** enumeration has the following constants:

Constant	Value	Description
cdrCustomVertPlacement	**0**	Specifies custom vertical placement
cdrBaselineVertPlacement	**1**	Specifies bottom vertical placement
cdrAscenderVertPlacement	**2**	Specifies ascender vertical placement
cdrDescenderVertPlacement	**3**	Specifies descender vertical placement
cdrCenterVertPlacement	**4**	Specifies center vertical placement

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cdrFixedPalette

The following enumerations have this constant:

- cdrPaletteType

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cdrFlagClosed

The following enumerations have this constant:

- cdrCurveElementFlags

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cdrFlagSelected

The following enumerations have this constant:

- cdrCurveElementFlags

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cdrFlagUser

The following enumerations have this constant:

- cdrCurveElementFlags

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cdrFlagValid

The following enumerations have this constant:

- cdrCurveElementFlags

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cdrFlipAxes

The **cdrFlipAxes** enumeration has the following constants:

Constant	Value	Description
cdrFlipHorizontal	**1**	Specifies the horizontal axis
cdrFlipVertical	**2**	Specifies the vertical axis
cdrFlipBoth	**3**	Specifies both axes

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cdrFlipBoth

The following enumerations have this constant:

- cdrFlipAxes

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cdrFlipHorizontal

The following enumerations have this constant:

- cdrFlipAxes

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cdrFlipVertical

The following enumerations have this constant:

- cdrFlipAxes

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cdrFMV

The following enumerations have this constant:

- cdrFilter

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cdrFOCOLTONE

The following enumerations have this constant:

- cdrPaletteID

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cdrFontCase

The **cdrFontCase** enumeration has the following constants:

Constant	Value	Description
cdrNormalFontCase	**0**	Specifies normal font case
cdrSmallCapsFontCase	**1**	Specifies small-caps font case
cdrAllCapsFontCase	**2**	Specifies all-caps font case
cdrMixedFontCase	**3**	Specifies mixed font case

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cdrFontLine

The **cdrFontLine** enumeration has the following constants:

Constant	Value	Description
cdrNoFontLine	**0**	Specifies no font line
cdrSingleThinFontLine	**1**	Specifies thin font line
cdrSingleThinWordFontLine	**2**	Specifies thin, word font line
cdrSingleThickFontLine	**3**	Specifies thick font line
cdrSingleThickWordFontLine	**4**	Specifies thin, word font line
cdrDoubleThinFontLine	**5**	Specifies double-thin font line
cdrDoubleThinWordFontLine	**6**	Specifies double-thin, word font line
cdrMixedFontLine	**7**	Specifies mixed font line

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cdrFontPosition

The **cdrFontPosition** enumeration has the following constants:

Constant	Value	Description
cdrNormalFontPosition	**0**	Specifies normal font position
cdrSubscriptFontPosition	**1**	Specifies subscript font position
cdrSuperscriptFontPosition	**2**	Specifies superscipt font position
cdrMixedFontPosition	**3**	Specifies mixed font position

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cdrFontStyle

The **cdrFontStyle** enumeration has the following constants:

Constant	Value	Description
cdrNormalFontStyle	**0**	Specifies normal font style
cdrBoldFontStyle	**1**	Specifies bolded font style
cdrItalicFontStyle	**2**	Specifies italic font style
cdrBoldItalicFontStyle	**3**	Specifies bold, italic font style
cdrThinFontStyle	**4**	Specifies thin font style
cdrThinItalicFontStyle	**5**	Specifies thin, italic font style
cdrExtraLightFontStyle	**6**	Specifies extra-light font style
cdrExtraLightItalicFontStyle	**7**	Specifies extra-light, italic font style
cdrMediumFontStyle	**8**	Specifies medium font style
cdrMediumItalicFontStyle	**9**	Specifies medium, italic font style
cdrSemiBoldFontStyle	**10**	Specifies semi-bold font style
cdrSemiBoldItalicFontStyle	**11**	Specifies semi-bold, italic font style
cdrExtraBoldFontStyle	**12**	Specifies extra-bold font style
cdrExtraBoldItalicFontStyle	**13**	Specifies extra-bold, italic font style
cdrHeavyFontStyle	**14**	Specifies heavy font style
cdrHeavyItalicFontStyle	**15**	Specifies heavy, italic font style
cdrMixedFontStyle	**16**	Specifies mixed font style
cdrLightFontStyle	**17**	Specifies light font style
cdrLightItalicFontStyle	**18**	Specifies light, italic font style

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cdrFoot

The following enumerations have this constant:

- cdrUnit

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cdrForceJustifyAlignment

The following enumerations have this constant:

- cdrAlignment

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cdrFormatDateTime

The following enumerations have this constant:

- cdrDataFormatType

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cdrFormatGeneral

The following enumerations have this constant:

- cdrDataFormatType

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cdrFormatLinearAngular

The following enumerations have this constant:

- cdrDataFormatType

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cdrFormatNumeric

The following enumerations have this constant:

- cdrDataFormatType

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cdrFountainFill

The following enumerations have this constant:

- cdrFillType

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cdrFountainFillBlendType

The **cdrFountainFillBlendType** enumeration has the following constants:

Constant	Value	Description
cdrDirectFountainFillBlend	**0**	Specifies a direct fountain-fill blend
cdrRainbowCWFountainFillBlend	**1**	Specifies a clockwise, rainbow fountain-fill blend
cdrRainbowCCWFountainFillBlend	**2**	Specifies a counterclockwise, rainbow fountain-fill bend
cdrCustomFountainFillBlend	**3**	Specifies a custom fountain-fill blend

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cdrFountainFillSpreadMethod

The **cdrFountainFillSpreadMethod** enumeration has the following constants:

Constant	Value	Description
cdrFountainFillSpreadMethodPad	**0**	Specifies a padded fountain fill
cdrFountainFillSpreadMethodReflect	**1**	Specifies a reflected and repeated fountain fill
cdrFountainFillSpreadMethodRepeat	**2**	Specifies a repeated fountain fill

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cdrFountainFillStyle

The following enumerations have this constant:

- cdrFillStyleType

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cdrFountainFillType

The **cdrFountainFillType** enumeration has the following constants:

Constant	Value	Description
cdrLinearFountainFill	**1**	Specifies a linear fountain fill
cdrRadialFountainFill	**2**	Specifies a radial fountain fill
cdrConicalFountainFill	**3**	Specifies a conical fountain fill
cdrSquareFountainFill	**4**	Specifies a square fountain fill

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cdrFountainTransparency

The following enumerations have this constant:

- cdrTransparencyType

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cdrFPX

The following enumerations have this constant:

- cdrFilter

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cdrFrench

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchCameroon

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchCanadian

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchCotedIvoire

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchLuxembourg

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchMali

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchMonaco

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchReunion

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchSenegal

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchWestIndies

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrenchZaire

The following enumerations have this constant:

- cdrTextLanguage

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cdrFrisianNetherlands

The following enumerations have this constant:

- cdrTextLanguage

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cdrFullColorPattern

The following enumerations have this constant:

- cdrPatternFillType

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cdrFullColorPatternFillStyle

The following enumerations have this constant:

- cdrFillStyleType

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cdrFullJustify

The following enumerations have this constant:

- cdrVerticalAlignment

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cdrFullJustifyAlignment

The following enumerations have this constant:

- cdrAlignment

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cdrGaelicIreland

The following enumerations have this constant:

- cdrTextLanguage

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cdrGaelicScotland

The following enumerations have this constant:

- cdrTextLanguage

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cdrGalician

The following enumerations have this constant:

- cdrTextLanguage

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cdrGEM

The following enumerations have this constant:

- cdrFilter

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cdrGeorgian

The following enumerations have this constant:

- cdrTextLanguage

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cdrGerman

The following enumerations have this constant:

- cdrTextLanguage

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cdrGermanAustria

The following enumerations have this constant:

- cdrTextLanguage

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cdrGermanLiechtenstein

The following enumerations have this constant:

- cdrTextLanguage

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cdrGermanLuxembourg

The following enumerations have this constant:

- cdrTextLanguage

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cdrGIF

The following enumerations have this constant:

- cdrFilter

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cdrGIFAnimation

The following enumerations have this constant:

- cdrFilter

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cdrGrayscaleImage

The following enumerations have this constant:

- cdrImageType

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cdrGreek

The following enumerations have this constant:

- cdrTextLanguage

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cdrGridDot

The following enumerations have this constant:

- cdrGridType

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cdrGridLine

The following enumerations have this constant:

- cdrGridType

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cdrGridType

The **cdrGridType** enumeration has the following constants:

Constant	Value	Description
cdrGridDot	**0**	Specifies dotted grid
cdrGridLine	**1**	Specifies lined grid

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cdrGroupNode

The following enumerations have this constant:

- cdrTreeNodeType

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cdrGroupShape

The following enumerations have this constant:

- cdrShapeType

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cdrGuidelineShape

The following enumerations have this constant:

- cdrShapeType

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cdrGuideType

The **cdrGuideType** enumeration has the following constants:

Constant	Value	Description
cdrAllGuides	**-1**	Specifies all guides
cdrHorizontalGuide	**0**	Specifies horizontal guides
cdrVerticalGuide	**1**	Specifies vertical guides
cdrSlantedGuide	**2**	Specifies slanted guides

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cdrGujarati

The following enumerations have this constant:

- cdrTextLanguage

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cdrHalftoneCross

The following enumerations have this constant:

- cdrHalftoneType

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cdrHalftoneFixed4x4

The following enumerations have this constant:

- cdrHalftoneType

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cdrHalftoneFixed8x8

The following enumerations have this constant:

- cdrHalftoneType

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cdrHalftoneLine

The following enumerations have this constant:

- cdrHalftoneType

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cdrHalftoneRound

The following enumerations have this constant:

- cdrHalftoneType

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cdrHalftoneSquare

The following enumerations have this constant:

- cdrHalftoneType

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cdrHalftoneType

The **cdrHalftoneType** enumeration has the following constants:

Constant	Value	Description
cdrHalftoneSquare	**0**	Specifies square
cdrHalftoneRound	**1**	Specifies round
cdrHalftoneLine	**2**	Specifies line
cdrHalftoneCross	**3**	Specifies cross
cdrHalftoneFixed4x4	**4**	Specifies fixed 4 × 4
cdrHalftoneFixed8x8	**5**	Specifies fixed 8 × 8

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cdrHatchFill

The following enumerations have this constant:

- cdrFillType

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cdrHatchFillStyle

The following enumerations have this constant:

- cdrFillStyleType

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cdrHeavyFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrHeavyItalicFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrHebrew

The following enumerations have this constant:

- cdrTextLanguage

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cdrHindi

The following enumerations have this constant:

- cdrTextLanguage

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cdrHKS

The following enumerations have this constant:

- cdrPaletteID

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cdrHorizontalGuide

The following enumerations have this constant:

- cdrGuideType

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cdrHorizontalSkewOrientation

The following enumerations have this constant:

- cdrFittedOrientation

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cdrHPGL

The following enumerations have this constant:

- cdrFilter

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cdrHTML

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cdrHTMLActiveObjectShape

The following enumerations have this constant:

- cdrShapeType

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cdrHTMLFormObjectShape

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cdrHungarian

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- cdrTextLanguage

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cdrIcelandic

The following enumerations have this constant:

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cdrICO

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- cdrFilter

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cdrcdr

The following enumerations have this constant:

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cdrImageBlackWhite

The following enumerations have this constant:

- cdrImageMode

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cdrImageCMYK

The following enumerations have this constant:

- cdrImageMode

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cdrImageDuotone

The following enumerations have this constant:

- cdrImageMode

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cdrImageGrayscale

The following enumerations have this constant:

- cdrImageMode

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cdrImageGrayscale16

The following enumerations have this constant:

- cdrImageMode

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cdrImageLAB

The following enumerations have this constant:

- cdrImageMode

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cdrImageMode

The **cdrImageMode** enumeration has the following constants:

Constant	Value	Description
cdrImageBlackWhite	**0**	Specifies black-and-white
cdrImageGrayscale16	**1**	Specifies 16-bit grayscale
cdrImageGrayscale	**2**	Specifies grayscale
cdrImagePaletted	**3**	Specifies paletted
cdrImageRGB	**4**	Specifies RGB
cdrImageCMYK	**5**	Specifies CMYK
cdrImageDuotone	**6**	Specifies duotone
cdrImageLAB	**7**	Specifies LAB
cdrImageRGB48	**8**	Specifies 48-bit RGB
cdrImageMultiChannel	**9**	Specifies multi-channel

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cdrImageMultiChannel

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cdrImagePaletted

The following enumerations have this constant:

- cdrImageMode

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cdrImagePaletteType

The **cdrImagePaletteType** enumeration has the following constants:

Constant	Value	Description
cdrPaletteUniform	**0**	Specifies uniform
cdrPaletteStdVGA	**1**	Specifies VGA
cdrPaletteAdaptive	**2**	Specifies adaptive
cdrPaletteOptimized	**3**	Specifies optimized
cdrPaletteBlackBody	**4**	Specifies black body
cdrPaletteGrayscale	**5**	Specifies grayscale
cdrPaletteSystem	**6**	Specifies system
cdrPaletteIE	**7**	Specifies Microsoft Internet Explorer
cdrPaletteNetscape	**8**	Specifies Netscape
cdrPaletteCustom	**9**	Specifies custom

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cdrImageRGB

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cdrImageRGB48

The following enumerations have this constant:

- cdrImageMode

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cdrImageType

The **cdrImageType** enumeration has the following constants:

Constant	Value	Description
cdrBlackAndWhiteImage	**0**	Specifies black-and-white
cdr16ColorsImage	**1**	Specifies 16-color
cdrGrayscaleImage	**2**	Specifies grayscale
cdrPalettedImage	**3**	Specifies paletted
cdrRGBColorImage	**4**	Specifies RGB
cdrCMYKColorImage	**5**	Specifies CMYK
cdrDuotoneImage	**6**	Specifies duotone
cdrLABImage	**7**	Specifies LAB
cdrCMYKMultiChannelImage	**8**	Specifies multi-channel CMYK
cdrRGBMultiChannelImage	**9**	Specifies multi-channel RGB
cdrSpotMultiChannelImage	**10**	Specifies multi-channel spot

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cdrIMG

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cdrImportCrop

The following enumerations have this constant:

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cdrImportFormattingOnly

The following enumerations have this constant:

- cdrImportTextFormatting

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cdrImportFull

The following enumerations have this constant:

- cdrImportMode

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cdrImportMode

The **cdrImportMode** enumeration has the following constants:

Constant	Value	Description
cdrImportFull	**0**	Specifies full import
cdrImportCrop	**1**	Specifies cropped import
cdrImportResample	**2**	Specifies resampled import

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cdrImportNoFormatting

The following enumerations have this constant:

- cdrImportTextFormatting

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cdrImportResample

The following enumerations have this constant:

- cdrImportMode

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cdrImportTextFontAndFormatting

The following enumerations have this constant:

- cdrImportTextFormatting

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cdrImportTextFormatting

The **cdrImportTextFormatting** enumeration has the following constants:

Constant	Value	Description
cdrImportTextFontAndFormatting	**0**	Specifies import of text font and formatting
cdrImportNoFormatting	**1**	Specifies import of text without formatting
cdrImportFormattingOnly	**2**	Specifies import of text formatting only

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cdrInch

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cdrIndonesian

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cdrInsideShape

The following enumerations have this constant:

- cdrPositionOfPointOverShape

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cdrItalian

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- cdrTextLanguage

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cdrItalicFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrJapanese

The following enumerations have this constant:

- cdrTextLanguage

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cdrJP2

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cdrJPC

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cdrJPEG

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cdrJPG2000Import

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cdrKannada

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cdrKashmiri

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cdrKazakh

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cdrKhmer

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cdrKilometer

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cdrKirghiz

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cdrKonkani

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cdrKorean

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cdrLab

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- cdrPaletteID

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cdrLABImage

The following enumerations have this constant:

- cdrImageType

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cdrLandscape

The following enumerations have this constant:

- cdrPageOrientation

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cdrLanguageMixed

The following enumerations have this constant:

- cdrTextLanguage

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cdrLanguageNone

The following enumerations have this constant:

- cdrTextLanguage

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cdrLao

The following enumerations have this constant:

- cdrTextLanguage

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cdrLastPoint

The following enumerations have this constant:

- cdrPresetPoint

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cdrLatvian

The following enumerations have this constant:

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cdrLayerNode

The following enumerations have this constant:

- cdrTreeNodeType

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cdrLeft

The following enumerations have this constant:

- cdrDirection

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cdrLeftAlignment

The following enumerations have this constant:

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cdrLeftPlacement

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cdrLeftPoint

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cdrLeftQuadrant

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cdrLens

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- cdrEffectType

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cdrLensBrighten

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cdrLensColorAdd

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cdrLensColorLimit

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cdrLensCustomColorMap

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cdrLensFishEye

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cdrLensHeatMap

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cdrLensInvert

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cdrLensMagnify

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cdrLensTintedGrayscale

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cdrLensTransparency

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cdrLensType

The **cdrLensType** enumeration has the following constants:

Constant	Value	Description
cdrLensMagnify	**0**	Specifies **Magnify** lens
cdrLensFishEye	**1**	Specifies **Fish Eye** lens
cdrLensWireframe	**2**	Specifies **Wireframe** lens
cdrLensColorLimit	**3**	Specifies **Color Limit** lens
cdrLensColorAdd	**4**	Specifies **Color Add** lens
cdrLensInvert	**5**	Specifies **Invert** lens
cdrLensBrighten	**6**	Specifies **Brighten** lens
cdrLensTintedGrayscale	**7**	Specifies **Tinted Grayscale** lens
cdrLensHeatMap	**8**	Specifies **Heat Map** lens
cdrLensTransparency	**9**	Specifies **Transparency** lens
cdrLensCustomColorMap	**10**	Specifies **Custom Color Map** lens

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cdrLensWireframe

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cdrLevelContainer

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cdrLevelDocument

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cdrLevelGroup

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cdrLevelLayer

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cdrLevelPage

The following enumerations have this constant:

- cdrShapeLevel

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cdrLightBackBottomRight

The following enumerations have this constant:

- cdrExtrudeLightPosition

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cdrLightBackRight

The following enumerations have this constant:

- cdrExtrudeLightPosition

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cdrLightBackTop

The following enumerations have this constant:

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cdrLightBackTopLeft

The following enumerations have this constant:

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cdrLightBackTopRight

The following enumerations have this constant:

- cdrExtrudeLightPosition

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cdrLightFontStyle

The following enumerations have this constant:

- cdrFontStyle

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cdrLightFrontBottom

The following enumerations have this constant:

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cdrLightFrontBottomLeft

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cdrLightFrontBottomRight

The following enumerations have this constant:

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cdrLightFrontCenter

The following enumerations have this constant:

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cdrLightFrontLeft

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cdrLightFrontRight

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cdrLightFrontTop

The following enumerations have this constant:

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cdrLightFrontTopLeft

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cdrLightFrontTopRight

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- cdrExtrudeLightPosition

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cdrLightItalicFontStyle

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cdrLinearDimensionShape

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cdrLinearDimensionType

The **cdrLinearDimensionType** enumeration has the following constants:

Constant	Value	Description
cdrDimensionHorizontal	**0**	Specifies horizontal linear dimension
cdrDimensionVertical	**1**	Specifies vertical linear dimension
cdrDimensionSlanted	**2**	Specifies slanted linear dimension

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cdrLinearFountainFill

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cdrLineSegment

The following enumerations have this constant:

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cdrLineSpacingType

The **cdrLineSpacingType** enumeration has the following constants:

Constant	Value	Description
cdrPercentOfCharacterHeightLineSpacing	**0**	Specifies percentage of character height
cdrPointLineSpacing	**1**	Specifies point
cdrPercentOfPointSizeLineSpacing	**2**	Specifies percentage of point size
cdrMixedLineSpacing	**3**	Specifies mixed

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cdrLinkAllConnections

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cdrLinkAllDependentClones

The following enumerations have this constant:

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cdrLinkAllDependents

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cdrLinkAllDependentsButClones

The following enumerations have this constant:

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cdrLinkDirectControls

The following enumerations have this constant:

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cdrLinkDirectControlsButClones

The following enumerations have this constant:

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cdrLinkDirectDependentClones

The following enumerations have this constant:

- cdrShapeLinkType

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cdrLinkDirectDependents

The following enumerations have this constant:

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cdrLinkDirectDependentsButClones

The following enumerations have this constant:

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